1.1. The project summary

Project Number [1]	720770	Project Acronym [2]	DAFIA

One form per project
General information

Project title [3]	Biomacromolecules from municipal solid bio-waste fractions and fish waste for high added value applications.
Starting date [4]	01/01/2017
Duration in months [5]	48
Call (part) identifier [6]	H2020-NMBP-BIO-2016
Topic	BIOTEC-02-2016 Bioconversion of non-agricultural waste into biomolecules for industrial applications
Fixed EC Keywords	
Free keywords	Municipal solids waste, Marine rest, fermentation, microbial, dicarboxylic acid, diamines, polyamides, nucleic acid, flame retardant, coating, gelatine, green chemistry.

Abstract [7]

Municipal solids waste (MSW) are collected by municipalities and represents more than 500 kg/capita (EU-27 average), 300 million tonnes overall every year in the EU-32. Currently, approximately 50% of this volume is landfilled. More than 1.3 million tonnes of Marine rest raw material (MRRM) are generated in Europe each year. Some countries, such as Norway and Denmark, have traditionally for animal feed. It will therefore be a challenge for the industry to develop methods to turn fish viscera and skin, currently considered as undesirable raw materials for hydrolysis and human consumption, into profitable products.
DAFIA will exploit MSW and MRRM as feedstocks for high value products. The parallel exploitation of the two feedstocks may create synergies. This expertise will be utilised in process development from MSW, while at the same time, new added-value products may be identified from both feed stocks.
The main objective of the DAFIA project is to explore the conversion routes of municipal solid waste (MSW), and marine rest raw-materials (MRRM) from the fish processing industries, to obtain high added value products, i.e. flame retardants, edible/barrier coatings and chemical building blocks (dicarboxylic acids and diamine) to produce polyamides and polyesters for a wide range industrial applications.
Different value-chains and products will be selected and explored based on the potential commercial value and the technical feasibility including new microbial strains and processes for conversion of major feedstock fractions, enzymatic and chemical modifications of components isolated from the feedstock or produced in microbial processes. Up to four cost-effective molecule groups suitable for the final selected applications will be targeted (nucleic acids, dicarboxylic acids, diamines and gelatine), & two value-chains (MSW & MRRM) will be evaluated at pilot scale to reach TRL5.

1.2. List of Beneficiaries

Project Number [1]	720770	Project Acronym [2]	DAFIA

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS	AIMPLAS	Spain	1	48
2	POLITECNICO DI TORINO	POLITO	Italy	1	48
3	SINTEF FISKERI OG HAVBRUK AS	SINTEF-FA	Norway	1	48
4	STIFTELSEN SINTEF	SINTEF-MC	Norway	1	48
5	DANMARKS TEKNISKE UNIVERSITET	DTU	Denmark	1	48
6	CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS	CNRS	France	1	48
7	NUTRIMAR AS	NUTRIMAR	Norway	1	48
8	INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL	IRIS	Spain	1	48
9	BIOTREND - INOVACAO E ENGENHARIA EM BIOTECNOLOGIA SA	BIOTREND	Portugal	1	48
10	DAREN LABORATORIES & SCIENTIFIC CONSULTANTS LTD	DLABS	Israel	1	48
11	MINE PLASTIK ENDUSTRIYEL URUNLER VE SERT MADEN KIMYA LIMITED SIRKETI	MINE	Turkey	1	48
12	BIO BASE EUROPE PILOT PLANT VZW	BBEPP	Belgium	1	48
13	BIOPOLIS SL	BIOPOLIS	Spain	1	48
14	ARKEMA FRANCE	ARKEMA	France	1	48
15	The National Non-Food Crops Centre	NNFCC	United Kingdom	1	48

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	Personmonths[11]	Start month[12]	End month[13]
WP1	Characterization, handling and preprocessing of feedstocks.	13 - BIOPOLIS		1	30
WP2	Isolation and purification of products from pre-processed feedstocks	3 - SINTEF-FA		2	22
WP3	Microbial conversion of feedstocks into value chemicals	5 - DTU		6	36
WP4	Flame retardants from nucleic acids from fish RRM and microbial biomass.	10 - DLABS		8	24
WP5	Fish gelatine-derived barrier coating for packaging and edible coatings for food application	8 - IRIS		10	34
WP6	Green chemical routes to produce polymers from developed building blocks.	1 - AIMPLAS		6	36
WP7	Product validation by industrial partners	12 - BBEPP		30	46
WP8	Environmental, regulatory & economic analyses	9 - BIOTREND		1	48
WP9	Communication, Dissemination, Exploitation, Business plan and Technology Transfer.	11 - MINE		1	48
WP10	Project Management	1 - AIMPLAS		1	48
WP11	Ethics requirements	1 - AIMPLAS		1	48
		Total			

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3. IMPLEMENTATION

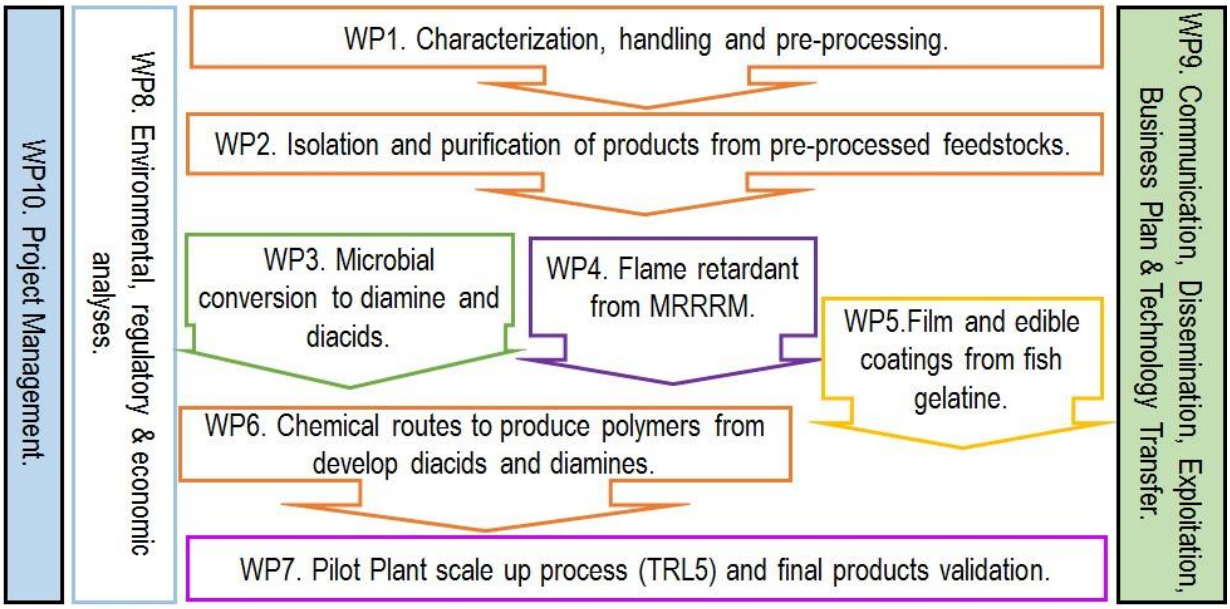
3.1 WORK PLAN - WORK PACKAGES, DELIVERABLES AND MILESTONES

3.1.1. Overall structure of the work plan

This covers all the planned research activities from MRRM and MSW pre-treatment and extraction of specified components (WP1 and WP2), optimization to obtain the most useful intermediates for the selected final products (diamines and diacids monomers in WP3, flame retardants in WP4 and gelatine-based products in WP5). WP6 aims to produce new polyamides from WP3 monomers and flame retardant polyamides using the additives of WP4. WP7 will scale up (TRL5) the developments proposed in WP1 to WP6. The environmental and economic viability of the new products will be evaluated in WP8. Furthermore, WP9 activities are related to communication, dissemination, exploitation, business plan and technology transfer activities. WP10 will provide the full management of the project in order to make all the proposed developments in line with defined DAFIA objectives.

WP7: Pilot Plant Scale-up process (TRL5) and final products validation.	**WP Leader:** BBEU
Carry out the optimization of the fermentative production of target molecules together with the validation of the developed DSP methods at 100 L scale. Finally, the overall biorefinery concept will be demonstrated by producing 1 target molecule (diacid or diamine) on pilot fermentation scale (1000 L). The products studied at lab scale in WP4 and WP5 will be scaled up (TRL5) in the pilot lines of the industrial partners. IRIS, AIMPLAS and NUTRIMAR will upscale and validate the edible and barrier coatings developed in WP5 in fish packaging applications.	
WP8: Environmental, regulatory & economic analyses.	**WP Leader:** BIOTREND
Assess the environmental and economic impact of the full process proposed in DAFIA throughout each stage of development, to ensure compliance with legislation, to maximise market acceptance and facilitate increasing optimisation of these processes and products.	
WP9: Communication, Dissemination, Exploitation, Business plan &Technology Transfer.	**WP Leader:** MINE.
Complete IPR strategy, dissemination and technology transfer activities to ensure the full exploitation of the innovations and scientific results will be carried out. The active involvement of companies in the project, representing the different sectors active in the value chain will ensure wide exploitation with realistic objectives and plans. Communication, Dissemination and Exploitation Plans will be developed, including the PEDR.	
WP10: **Project Management.**	**WP Leader** AIMPLAS
Project activities of coordination, efficient grant management (resources optimization), communication with the EC, risk management	

3.1.2. Graphical presentation of the components showing how they inter-relate (Pert chart).



.1.3. Timing of the different work packages and their components (Gantt chart).

Work Packages and Related Tasks		Partners Num.	Deliverables	MONTHS																							
				2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
WP1	**Characterization, handling & pre-processing of feedstocks.**	**3, 7, 9, (13)**	**D1.1-D1.5**									**H**	**H**					**H**									
T1.1	Characterization and standardization of raw materials	3,7,(13)	D1.1				□																				
T1.2	Initial fractionation of lipids, protein and other water-soluble compounds	3,7,(13)	D1.3-1.4									□	□														
T1.3	Pre-processing, fractionation and standardisation of MSW for use as a fermentation substrate	3,7,9,(13)	D1.5															□									
T1.4	Definition, compilation and summary of end-user products requirements developed in DAFIA project	(1),7,8,10,12,14	D1.2					□																			
WP2	**Isolation & purification of products from pre-processed.**	**2,(3),4,5,7,8**	**D2.1-D2.5**									**H**		**H**													
T2.1.	Isolation and purification of gelatine.	(3),7,8	D2.1									□															
T2.2	Isolation and purification of nucleic acids.	2,(4)	D2.2									□															
T2.3	Utilisation of protein-rich residues.	(3),4,5,7	D2.3											□													
WP3	**Microbial conversion of feedstocks into value chemicals.**	**4,(5),9,12,14**	**D3.1-D3.4**															**H**	**H**	**H**	**H**						
T3.1	Microbial production 1,5-pentanediamine from waste feedstock.	(4),9,12	D3.1																□								
T3.2	Microbial production of C6 dicarboxylic acids from waste feed.	(5),9,12	D3.2-3.3															□	□								
T3.3	Fermentation optimization & downstream process development for recovery of chemicals from fermentation broth.	4,5,(9),12,14	D3.4																		□						
WP4	**Flame retardants from nucleic acids from fish RRM & mic.**	**2,(10)**	**D4.1-D4.3**											**H**	**H**												
T4.1	Bonding of lignin/amines to nucleic acid or protein backbone.	2,(10)	D4.1											□													
T4.2	Compatibilization of the flame retardants with polyamides.	2,(10)	D4.2									□															
T4.3	Finding optimal combinations of flame retardants.	(2),1	D4.3												□												

WP5	**Film and edible coatings from fish gelatine.**	**1,7,(8)**	**D5.1-D5.3**													**H**					**H**						
T5.1	Edible and barrier coatings development.	7,(8),1	D5.1																								
T5.2	Active fibres and packaging development.	7,(8),1																									
T5.3	Upscaling of the coating and lamination process.	(1),8	D5.2																								
WP6	**Green chemical routes to produce polymers from building**	**(1),2,10,11,14**	**D6.1-D6.4**																	**H**	**H**	**H**					
T6.1	Production of polyamide 56 at lab level.	(1),14	D6.1																								
T6.2	Muconic acid as an adhesion precursor in polyamides.	6,(14)	D6.2																								
T6.3	Flame retardant bio-polyamides.	(1),10,11,14	D6.3																								

Associated with document Ref. Ares(2016)6406795 - 14/11/2016

Work Packages and Related Tasks		Partners Num.	Delivera-bles	MONTHS																							
				2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
T6.4	Flame retardants biopolyamides characterization.	(1),2,10,11,14	D5.6																		♦						
WP7	**Pilot Plant Scale-up process (TRL5) & product validation**	**All (12)**	**D7.1-D7.7**																				H	H	H	H	
T7.1	Demonstration of the overall biorefinery concept.	9,(12), 13	D7.1-7.3																				♦	♦	♦		
T7.2	Scale up of Flame Retardant Synthesis.	2,(10)	D7.4																							♦	
T7.3	Scale up of edible and barrier films.	(1),7,8	D7.5																							♦	
T7.4	Scale up of production of polyamides & adhesion promoters.	1,6,(14)	D7.6																						♦		
T7.5	Scale up of Flame Retardant polyamide compounds.	1,2,(11),14	D7.7																							♦	
WP8	**Environmental, regulatory & economic analyses.**	**All(9)**	**D8.1-D8.4**																						H	H	H
T8.1	Building a data repository.	All(9)																									
T8.2	Evaluation of synergies with other projects involved in MSW & animal waste valorisation.	3-5,7-9,12-(13),14,15																									
T8.3	Economic analysis & mass and energy balances.	All(9)	D8.1-8.4												♦											♦	
T8.4	Product life-cycle-assessment.	All(1)	D8.2																							♦	
T8.5	Regulatory and safety analysis.	1,(7),8,15	D8.3																								♦
T8.6	Evaluation of Economic and LCA risk.	All(14)																									
WP9	**Dissemination, Exploitation And Technology Transfer**	**All(11)**	**D9.1-9.9**		H							H												H			H
T9.1	Communication	All(11)	D9.1-D9.3		♦	♦						♦									♦						♦
T9.2	Dissemination plan.	All(11)																									
T9.3	IPR Management	All(11)	D9.4-D9.5			♦																					♦
T9.4	Exploitation plan	All(11)																									
T9.5	Business plans for key results	All(16)	D9.6																								♦
T9.6	Technology transfer	All(8)	D9.7-9.9																♦				♦			♦	
WP10	**Project Management**	**All(1)**	**D10.1-10.2**		H																						
T10.1	Management of the resources.	All(1)		╬		▲			▲			╬			▲			▲			╬			▲			╬
T10.2	Risk management. TMC's members	(1)																									
T10.3	Communication with EC.	(1)	D10.1-10.2					♦									♦										

*EB: Executive Board, TMC: Technical management Committee

(1): (Responsible), (X) Leader Legend: ♦ Deliverable Milestone ▲ Meeting ╬ Kick-Off, Mid Term, Final Meeting

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1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	MSW/MRRM composition and factors affecting variability	WP1	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D1.2	Report on main requirements and standards related to the DAFIA products.	WP1	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D1.3	Methods for MSW pre-processing and initial fractionation, and optimized strategy selected	WP1	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D1.4	Methods for MRRM pre-processing and initial fractionation, and optimized strategy selected	WP1	3 - SINTEF-FA	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D1.5	Pre-treatment of MSW for use as fermentation substrate and selected strategies, including experiences from fermentation and downstream processingWP3	WP1	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D2.1	Laboratory-scale evaluation of methods for isolation of gelatine	WP2	3 - SINTEF-FA	Report	Confidential, only for members of the consortium (including the Commission Services)	18

D2.2	Laboratory-scale evaluation of methods for isolation of nucleic acids	WP2	4 - SINTEF-MC	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.3	Characterisation and possible usage of residual fractions after extraction of gelatine and nucleic acids	WP2	3 - SINTEF-FA	Report	Confidential, only for members of the consortium (including the Commission Services)	22
D3.1	Lab-scale process for microbial production of PDA from pre-treated MSW and MRRM.	WP3	4 - SINTEF-MC	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D3.2	Lab-scale process for microbial production of CCM from pre-treated MSW.	WP3	5 - DTU	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D3.3	Lab-scale process for microbial production of AA from pre-treated MSW.	WP3	5 - DTU	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D3.4	Process for extracting muconic acid, adipic acid and 1.5pentanediamine from complex fermentation broth.	WP3	9 - BIOTREND	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.1	Modification of nucleic acid/protein backbone with lignin/melamine.	WP4	10 - DLABS	Report	Confidential, only for members of the consortium (including the Commission Services)	22

D4.2	Compatibilization of FR with selected polyamides	WP4	10 - DLABS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D4.3	Characterization of the obtained flame retardants	WP4	2 - POLITO	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.1	Development and characterisation of gelatine edible and barrier coatings including bioactive compounds	WP5	8 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	24

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D5.2	Upscaling of coated films and laminates with gelatine barrier coatings including bioactive compounds	WP5	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D6.1	Polymerization optimization of biobased PA56.	WP6	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.2	Sample of produced mercaptoadipic acid/ester and their associated report	WP6	6 - CNRS	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.3	Report of processing conditions of biobased PA56 compounds including data-sheet.	WP6	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	34

D7.1	Demonstration scale of MSW pretreatment and fractioning	WP7	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	40
D7.2	Report and tech transfer on scale-up of the fermentation of 2 target molecules at 150 L scale.	WP7	9 - BIOTREND	Report	Confidential, only for members of the consortium (including the Commission Services)	40
D7.3	Report on 150 L-scale recovery of diacid and/or diamine from complex fermentation broth.	WP7	12 - BBEPP	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D7.4	Report on the pilotscale production (fermentation and recovery) of the target molecule from raw feedstock.	WP7	12 - BBEPP	Report	Confidential, only for members of the consortium (including the Commission Services)	44

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D7.5	Pilot-scale production of flame retardant synthesis.	WP7	10 - DLABS	Report	Confidential, only for members of the consortium (including the Commission Services)	44
D7.6	Pilot-scale production of barrier packaging film.	WP7	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	44
D7.7	Validation of the different fish packaging solutions	WP7	8 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	46

D7.8	Pilot-scale production of FR polyamide compounds.	WP7	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	44
D7.9	Pilot-scale production of polyamides for coatings.	WP7	14 - ARKEMA	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D8.1	Report on economic analysis.	WP8	9 - BIOTREND	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D8.2	Report on Life Cycle Assessment (LCA).	WP8	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D8.3	Report on regulatory analysis and safety aspects.	WP8	7 - NUTRIMAR	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D8.4	Final report on simulation of the	WP8	9 - BIOTREND	Report	Confidential, only for members	46

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
	system and M&E balances				of the consortium (including the Commission Services)	
D9.1	Project web site	WP9	1 - AIMPLAS	Websites, patents filling, etc.	Confidential, only for members of the consortium (including the Commission Services)	3

D9.2	Initial Communication and dissemination Plan	WP9	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	16
D9.3	Report on performed communication activities and dissemination Plan updates	WP9	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D9.4	IPR policy in line with the Exploitation Plan	WP9	15 - NNFCC	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D9.5	Report of the IPR management activities	WP9	15 - NNFCC	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D9.6	Final Exploitation Plan & Business Plans.	WP9	15 - NNFCC	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D9.7	Technology Transfer Activities, including training to SMEs, other activities and workshops	WP9	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D9.8	Guide to explore the conversion routes of MSW, MRRM to	WP9	11 - MINE	Report	Confidential, only for members of the consortium (including the	46

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
	obtain high added value products.				Commission Services)	

D9.9	Final version of the PEDR.	WP9	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D10.1	1st Interim Progress Report including Risk indicators and Risk assessment	WP10	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D10.2	2nd Interim Progress Report including Risk indicators and Risk assessment	WP10	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	27
D11.1	EPQ - Requirement No. 2	WP11	1 - AIMPLAS	Ethics	Confidential, only for members of the consortium (including the Commission Services)	3
D11.2	POPD - Requirement No. 3	WP11	1 - AIMPLAS	Ethics	Confidential, only for members of the consortium (including the Commission Services)	3

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	13 - BIOPOLIS
Work package title	Characterization, handling and pre-processing of feedstocks.		
Start month	1	**End month**	30

Objectives

Objectives: The techno-economic feasibility of biorefinery approaches to valorise complex, residual feedstocks has been shown to be highly dependent on the efficiency of the selected sorting and pretreatment operations and their integration with further purification and bioprocessing stages. The general objective of this WP will be to ensure availability and compatibility of the selected raw materials and their fractions in an integrated approach with WP2 and WP3. In particular, the objectives of WP1 will be:
To characterize the composition of the selected raw materials and to determine their geographical, seasonal and stockassociated variability.
To evaluate and select preservation and pre-processing techniques in order to promote subsequent fractionation and biotransformation.
To generate batches of selected materials and their fractions for further analysis and processing.

Description of work and role of partners

WP1 - Characterization, handling and pre-processing of feedstocks. [Months: 1-30]
BIOPOLIS, AIMPLAS, SINTEF-FA, SINTEF-MC, NUTRIMAR, IRIS, BIOTREND, DLABS, MINE, ARKEMA
Task 1.1. Characterization and standardization of raw materials. Task Leader: BIOPOLIS; Other partners: SFA & NUTRIMAR.
Selected batches of MSW and marine rest raw materials (MRRM) will be sourced and sorted. The organic fraction of MSW and MRRM will be analysed following standardized NREL methodologies based on acid hydrolysis and HPLC/biochemical analysis and complimentary compositional analysis by differential extraction and gravimetry or specific determinations. The relative content of starch, cellulose, hemicellulose (NREL methodology), lipids (chloroform and chloroform/methanol extraction, gas chromatography), water extractives, pectin (EDTA extraction), lignin (hypochlorite-acetic acid extraction), protein (total nitrogen Kjeldahl, spectrophotometry) and ashes (calcination) will be determined. Other relevant components that may interfere with further processing (e.g. complex polysaccharides, polycyclic aromatic hydrocarbons, heavy metals) will be quantified attending to WP2 and WP3 results. More specific analyses of the target components will be part of the respective WPs.
Between 20-30 batches will be analysed in order to account for regional, seasonal and stock-associated (e.g. mixed MSW and agricultural wastes, pre-processed/sorted MSW, sorted MRRM) variability. Stability will be evaluated up to 3 months to define logistics and supply. Thermal and chemical pre-treatments (e.g. hygienization by pressure and temperature, low strength alkali or acid addition and/or steaming) will be evaluated to ensure preservation and avoid undesired microbial spoilage. Standard batches will be selected and distributed for further development.

Task 1.2. Initial fractionation of lipids, protein and other water-soluble compounds. Task Leader: BIOPOLIS; Other partners: SFA & NUTRIMAR.
The complexity and diversity of the two feedstocks hinders the efficient recovery of individual components with valorisation potential. In order to facilitate fractionation of lipids, protein and polysaccharides, pre-processing methods will be developed taking advantage of established technologies from the fish processing industries and from previous and on-going projects on use of MSW for biofuel production. Synergies between the two feedstocks will be utilised. For both feedstocks, the generated fractions will be characterized (e.g. carbohydrate content, lipid and amino acid profiles) and quantified in order to formulate the corresponding mass balance and select the optimum set-up. Amount of free amino acids, formation of biogenic amines, and total volatile nitrogen will be used as quality indicator for the proteinrich fractions, and amount of free fatty acids, oxidative status (peroxide and anisidine values) for the lipid fractions.

Subtask 1.2.1 Initial fractionation of MRRM (NUTRIMAR, SFA) Processes for fractionation of lipids and proteins rich fractions are well established in the fish processing industries, but adaptations and optimizations to the raw material (viscera) and targeted products (nucleic acids and proteins) are needed. Thermal processing and/or proteolytic enzymes will be applied to break protein-lipid bonds, and for further hydrolysis of the protein fraction, if required, before a mechanical separation into a lipid phase, water phase and solid (sediment) phase. Use of cheap, commercial enzymes and utilisation of the endogenous enzymes of the raw material will be studied. The sediments-fraction from fish viscera will be applied for isolation of nucleic acids and gelatine (WP2), while the liquid phase can be combined with other protein residues generated in Task 2.2, for further processing in Tasks 2.3 and 5.2.

Subtask 1.2.2 Initial fractionation of MSW (BIOPOLIS) Similar principles as for MRRM will be applied for MSW, but due to a higher content of polysaccharides, including cellulose and also some lignin, and lower content of protein and lipids, other enzymes and process conditions may be required.

Pretreatments combining biotechnological (proteases, cellulases) and physic-chemical processes (milling, steam explosion, organosolv, mild chemical catalysis) will be developed in order to facilitate the direct isolation of components (aim of Task 1.2) and the use of the organic fraction of MSW as fermentation substrate (aim of Task 1.3).

For protein isolation, if the water content of the MSW sample is very high, concentration by centrifugation or filtration will be applied as a first stage. In order to increase the yield of the enzymatic hydrolysis, some pretreatment processes will be studied: extraction with ethanol to remove color, previous lipid extraction, milling, heating, additional enzymes. Test conditions will be selected in terms of solid loads and other variables. The enzymatic processes will be evaluated on these pre-treated samples. Enzymes working at neutral or basic pH will be preferred, such as papain (pH 7.0), pancreatin (pH7.5), Alcalase® 2.5L (pH 8) and Flavourzyme®1000L (pH 7). Working temperature will be around 50-60ºC, and 3-4 hours of incubation. Optimal dosage described in literature will be tested. Finally, free aminoacids will be quantified by a colorimetric test based on nihidrin reaction and its aminogram will be analyzed by HPLC. The enzymatic approach will be the preferred one, although, combination with mild acid hydrolysis will be evaluated in order to maximize yield and minimize processing time. The protein hydrolyzate will be employed as a Nitrogen source for fermentation procedures in WP3.

For lipid isolation a similar approach as for MRRM (subtask 1.2.1) will be applied, based on thermal processing and/or proteolytic enzymes to break protein-lipid bonds, and mechanical separation. Lipid separation using organic solvents such as hexane (avoiding toxic solvents and implementing recycling systems) will be taken into account as a second option. The lipid fraction will be analyzed in order to know the content in acylglycerols (by GC-FID). Analysis of fatty acid composition will be performed by GC-FID or GC-MS of methanolized samples.

The liquid phase for MSW will contain the soluble carbohydrates (mono-, di, oligosaccharides, and starch), while insoluble (e.g. cellulose) will follow the sludge fraction. Depending on the further applications (Task 1.3 and 1.4), other enzymes may be introduced before the separation. For instance, for use as a fermentation substrate, polysaccharide hydrolysing enzymes can be introduced before the separation, yielding most of the carbohydrates in the liquid (soluble) fraction.

Task 1.3. Pre-processing, fractionation and standardisation of MSW for use as a fermentation substrate. Task Leader: BIOPOLIS; Other partners: SMC, SFA, BIOTREND & NUTRIMAR.
MSW is rich in starch, cellulose and pectin. For use as a fermentation substrate (e.g. carbon source), the carbohydrate fraction can be isolated and processed, or the complete lipid-free fractions from Task 1.3 can be applied, also serving as a nitrogen and nutrient source. The first approach would be more expensive, but will facilitate standardising the composition and ensure reproducibility.
The polysaccharides need to be solubilised and hydrolysed to monomers. Commercial, polysaccharide-hydrolysing enzymes (amylases, cellulase cocktails) can be added before the separation step in Task 1.3, or to the sludge fraction from Task 1.3. Depending on the structure and availability of the cellulose, additional thermal (autoclave, steam explosion), and/or chemical (acid/base) treatments may be required. Solubilising unfractionated MSW, without removal of lipids, will be evaluated for production of adipic acid by the lipolytic yeast Yarrowia lipolytica (WP3, Task 3.2). The remaining nitrogen-containing fractions will be evaluated as a source of nitrogen and growth factors.
Possible content of toxic, or growth inhibiting, substances in fractions prepared by different methods will be evaluated in small-scale growth experiments with relevant industrial microorganisms. This Task should allow iterations with the fermentation and DSP (WP3).

Task 1.4 Definition, compilation and summary of end-user products requirements developed in DAFIA project. Task leader: AIMPLAS with the support of end-users companies (NUTRIMAR, IRIS, FARMAK, ARKEMA, DLABS and MINE).
Based on a review of the actual material properties, technical specification of the demonstrator parts, further technical and environmental requirements, and especially economic limitations, a list of specific requirements will be defined for each application (fish packaging, polyamide 56 with flame retardant properties for automotive aplications and polyamides with improved adhesion to metal and glass, ..) and also to define the purity and reactivity of the intermediate chemicals (adipic acid, muconic acid and pentadiamine) to produce the polyamides, the protein-nucleic acid intermediates to produce the flame retardants and the gelatine and peptides for the fish packaging coatings.
Beside European and international technical standards, material and product properties that are required from certain standards for each application, are important factors in material development. Standards to be review comprise e.g. flame retardancy, antibacterial and antifungal properties, chemical and thermal resistance, barrier properties, adhesion, purity and corresponding mechanical testing standard, end of life and recycling demands and specific tests of the individual applications.
In a further evaluation step these requirements will be summarised to specific requirement-lists that serve as a basis for future development. The aim here is to group requirements so that more general target areas can be defined that allow the use of material developments in more than one application.

Participation per Partner
Partner number and short name
1 - AIMPLAS
3 - SINTEF-FA
4 - SINTEF-MC
7 - NUTRIMAR
8 - IRIS
9 - BIOTREND
10 - DLABS
11 - MINE

13 - BIOPOLIS
14 - ARKEMA

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	MSW/MRRM composition and factors affecting variability	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D1.2	Report on main requirements and standards related to the DAFIA products.	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D1.3	Methods for MSW pre-processing and initial fractionation, and optimized strategy selected	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D1.4	Methods for MRRM pre-processing and initial fractionation, and optimized strategy selected	3 - SINTEF-FA	Report	Confidential, only for members of the consortium (including the Commission Services)	18

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.5	Pre-treatment of MSW for use as fermentation substrate and selected strategies, including experiences from fermentation and downstream processingWP3	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	30

Description of deliverables

D1.1. Report on MSW/MRRM composition and factors affecting variability.

D1.2. Report on main requirements and standards related to the DAFIA products.

D1.3. Report on methods for MSW pre-processing and initial fractionation, and optimized strategy selected.

D1.4. Report on methods for MRRM pre-processing and initial fractionation, and optimized strategy selected.

D1.5. Report on pre-treatment of MSW for use as fermentation substrate and selected strategies, including experiences from fermentation and downstream processing WP3.

D1.1 : MSW/MRRM composition and factors affecting variability [8]
MSW/MRRM composition and factors affecting variability

D1.2 : Report on main requirements and standards related to the DAFIA products. [10] Main requirements and standards related to the DAFIA products.

D1.3 : Methods for MSW pre-processing and initial fractionation, and optimized strategy selected [18]
Methods for MSW pre-processing and initial fractionation, and optimized strategy selected

D1.4 : Methods for MRRM pre-processing and initial fractionation, and optimized strategy selected [18]
Methods for MRRM pre-processing and initial fractionation, and optimized strategy selected

D1.5 : Pre-treatment of MSW for use as fermentation substrate and selected strategies, including experiences from fermentation and downstream processingWP3 [30]
Pre-treatment of MSW for use as fermentation substrate and selected strategies, including experiences from fermentation and downstream processingWP3

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP2	**Lead beneficiary [10]**	3 - SINTEF-FA
Work package title	Isolation and purification of products from pre-processed feedstocks		
Start month	2	**End month**	22

Objectives

The aim of WP2 is to develop cost-efficient isolation and purification methods for nucleic acids and gelatine from the pre-processed feedstocks prepared in WP1. The processes should enable utilisation of all residues, either as increased value-products or as fermentation substrates (WP3).

Description of work and role of partners

WP2 - Isolation and purification of products from pre-processed feedstocks [Months: 2-22]
SINTEF-FA, POLITO, SINTEF-MC, DTU, NUTRIMAR, IRIS
Task 2.1 Isolation and purification of gelatine. Task Leader: SFA; Other partners: NUTRIMAR & IRIS.
Gelatine will be isolated from sorted and unprocessed fractions of fish RRM (skin, backbones) and from the insoluble fractions obtained after fractionation/hydrolysis of fish viscera into oil, stick water/hydrolysate and sediments (within WP1). Several extraction parameters (like extraction temperature and time, pH, fractionation of extracted gelatine,) for improving gelatine extraction technologies with the focus on yield and textural properties will be tested. In order to replace mammalian gelatines in food industry and employ them in the production of coatings, mixing with several other ingredients (see WP5) for improvement of textural characteristics will be evaluated. Molecular weight distribution, amino acid composition as well as viscosity, film forming capacity and gel strength will be analysed on extracted gelatines. The isolation process of the selected fraction, with feedback from WP5, will be scaled up in WP7 to obtain gelatine in the range of several kg. Recommendations for the post-project implementation of the process by NUTRIMAR will be provided.

Task 2.2 Isolation and purification of nucleic acids. Task Leader: SMC; Other partners: POLITO

Nucleic acids will be isolated and purified after fractionation of fish viscera into oil, stickwater and solids (WP1), and from microbial biomass (WP3). Due to the variation of the starting material, different strategies (mechanical, chemical and enzymatic) or their combinations will be applied during initial isolation steps to disrupt cells and release nucleic acids. Removal of contaminating proteins (including nucleases) and lipids will be performed by cost-effective protocols using proteases and/or based on the "salting out method" i.e., employing NaCl and detergents as reported in literature, and precipitation of nucleic acids will be done using isopropanol. Precipitated nucleic acids will be washed with 70% ethanol, dried, reconstituted in appropriate buffer and analysed for nucleic acid yield and purity. The minimum required purity is A260/A280>1.6.

Task 2.3 Utilisation of protein-rich residues (after extraction of gelatine and nucleic acids). Task Leader: SFA; Other partners: SMC, DTU & NUTRIMAR
To have sustainable and profitable utilisation of rest raw materials, all the obtained fractions should be used in the most optimal way. Residual raw material remaining after extraction of gelatine and nucleic acids will be characterised in order to define their possible utilisation. In particular, the antioxidant properties of selected fractions and bioactive compounds (protein hydrolysates) will be evaluated in respect of their use in WP5. Chemical composition, microbiological quality and technological properties of the residual fractions will be evaluated, and the quality assessed by the same methods as in Task 1.3, in relation to their use as food/feed/fertiliser products. Different technological solutions will be integrated to upgrade the obtained products by increasing the yield and quality of relevant products. Methods for preservation and stabilisation of residual fractions, as well as concentration and removal of undesired components will be set. Bioactive compounds will be provided to WP5

Participation per Partner
Partner number and short name
2 - POLITO
3 - SINTEF-FA
4 - SINTEF-MC
Partner number and short name
5 - DTU
7 - NUTRIMAR
8 - IRIS

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.1	Laboratory-scale evaluation of methods for isolation of gelatine	3 - SINTEF-FA	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.2	Laboratory-scale evaluation of methods for isolation of nucleic acids	4 - SINTEF-MC	Report	Confidential, only for members of the consortium (including the Commission Services)	18

D2.3	Characterisation and possible usage of residual fractions after extraction of gelatine and nucleic acids	3 - SINTEF-FA	Report	Confidential, only for members of the consortium (including the Commission Services)	22

Description of deliverables

D2.1. Report on laboratory-scale evaluation of methods for isolation of gelatine.

D2.2. Report on laboratory-scale evaluation of methods for isolation of nucleic acids.

D2.3. Report on characterisation and possible usage of residual fractions after extraction of gelatine and nucleic acids

D2.1 : Laboratory-scale evaluation of methods for isolation of gelatine [18]
Laboratory-scale evaluation of methods for isolation of gelatine

D2.2 : Laboratory-scale evaluation of methods for isolation of nucleic acids [18]
Laboratory-scale evaluation of methods for isolation of nucleic acids

D2.3 : Characterisation and possible usage of residual fractions after extraction of gelatine and nucleic acids [22]
Characterisation and possible usage of residual fractions after extraction of gelatine and nucleic acids

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP3	**Lead beneficiary** [10]	5 - DTU
Work package title	Microbial conversion of feedstocks into value chemicals		
Start month	6	**End month**	36

Objectives

The main objectives proposed in WP3 will be: i) To develop strains and fermentation process for conversion of pretreated MSW and MRRM into diamine, ii) To develop strains and fermentation process for conversion of pre-treated MSW into C6 dicarboxylic acids: cis,cis-muconic acid and adipic acid and iii) To develop downstream processing for recovery of 1,5 pentanediamine (PDA), cis,cis-muconic acid (CCM) and adipic acid (AA) from fermented broths.

Description of work and role of partners

WP3 - Microbial conversion of feedstocks into value chemicals [Months: 6-36]
DTU, SINTEF-MC, BIOTREND, BBEPP, BIOPOLIS, ARKEMA
Task 3.1 Microbial production 1,5 pentanediamine from waste feedstocks. Task Leader: SMC, Other partners: BIOTREND, BBEU.
In this task, efficient microbial processes for conversion of MSW into PDA will be developed, using MRRM hydrolysates as nitrogen and nutrient source. Taking advantage of previously established and in-house strains belonging to Brevibacterium and Corynebacterium, specifically selected for high L-lysine production levels (the precursor for PDA), SMC will apply and genetically modify these strains for conversion of L-lysine into cadaverine. Multiple variants of the key enzyme lysine decarboxylase and multiple verified and putative PDA exporters will be selected based on the literature and by using sequence databases. The genes encoding enzyme variants will be synthesized in codonoptimized versions and heterologously expressed alone, and in combinations (operon), in the above-mentioned L-lysine overproducing strains. The strains will be screened for PDA production by cultivation on commonly used defined growth media containing glucose, and by applying MSW-derived carbon sources alone and in combination with MRRM-derived nitrogen and nutrient sources.
In parallel to the activities above, the L-lysine overproducing strains will be characterized for growth on complex substrates containing MSW-derived carbon sources and MRRM-derived nutrient sources. Consumption of both substrates will be analysed, in addition to the L-lysine production level. If necessary, the best-suited PDA-producing strains will be adapted towards improved production rates while retaining high 1,5 pentanediamine production levels

on complex, mixed substrates using serial culture passages (adapted evolution). Selected optimized PDA-producing strains will be cultivated in controlled lab-scale (1L) fermentors utilizing MSW- and MRRM-derived carbon and nitrogen sources, respectively, to evaluate their performance. The strains will be transferred to Task 3.3 for optimization of the fermentation process and for development of downstream processing.

Task 3.2 Microbial production of C6 dicarboxylic acids from waste feedstocks. Task Leader: DTU, Other partners: BIOTREND, BBEU, BIOPOLIS

In task 3.2 microbial fermentation process for conversion of municipal solid waste MSW hydrolysate into C6 dicarboxylic acids, cis,cis-muconic acid (CCM) and adipic acid (AA), will be developed. The task will start with the development of suitable microbial chassis for utilization of MSW hydrolysate by adaptive laboratory evolution. MSW hydrolysate contains fats, proteins and C6 and C5 carbohydrates; to achieve optimal conversion of the substrate into the product, it is desired to completely utilize all the components of the substrate. DTU has previously developed advanced genetic tools that allow rapid modification of yeast S. cerevisiae and Y. lipolytica100.These yeast hosts will be engineered for production of the target dicarboxylic acids and for optimal utilization of the MSW feedstock. One host is a robust industrial baker's yeast Saccharomyces cerevisiae strain that DTU has engineered for xylose utilization in the previous work101. Another host is a naturally hardy Yarrowia lipolytica strain, also genetically engineered for xylose utilization (unpublished results).

BIOPOLIS has different strains of Thermobifida fusca, which is a native producer of adipic acid102. The various candidate strains will be adapted using serial culture passages to be able to grow efficiently on MSW hydrolysate. The resulting strains will be tested in laboratory-scale (1L) fermentations to analyse the utilization of MSW components over time. The adapted strains will be also re-sequenced and the discovered genetic changes will be reverse engineered into the original strains to identify the causal mutations for improved utilization of MSW hydrolysate. The most robust strain capable of growing in MSW hydrolysate will be selected as the host strain for further modifications.

In the next step the two MSW-utilizing strains of S. cerevisiae and Y. lipolytica will be genetically engineered for production of the corresponding CCM and AA. We have previously engineered S. cerevisiae for production of CCM

via the aromatic amino acids pathway (Skjoedt et al., Nature Chem Biol, accepted) and DTU will now implement the optimal biosynthetic pathway towards CCM into the MSW-utilizing S. cerevisiae strain. In order to implement adipic acid production in Y. lipolytica-oxidase and block the degradation of fatty acyl-CoAs in peroxysomes at C6 chain length, this will prevent the degradation of adipic acid and will cause conversion of lipids from the waste into adipic acid. In the case of T. fusca, the gene cluster of the reverse adipate degradation pathway (β-ketothioase, 3-hydroxyacyl-CoA, 3hydroxyadipyl-CoA dehydrogenase, 5-carboxy-2-pentenoyl-CoA, and succinyl-CoA synthetase) will be improved by adding new enzymes or by testing other combinations of enzymes using more efficient genes from other microorganisms. In addition, the possibility to modulate the expression levels of the genes by acting on the promoters will be tested in order to reduce the toxicity of no-balanced intermediates and to increase carbon flux toward the desired product. Further genetic engineering to obtain specific deletion mutants or new recombinant bacteria will be oriented by system biology predictions. In order to increase the scalability of the fermentation process a knock- out mutant in the ssgB (gen conding the sporulation specific cell division protein) will be attempted.
In order to enable de novo biosynthesis of adipic acid from other components of the waste, such as glucose and xylose, DTU will introduce reverse-oxidation process in the cytosol of Y. lipolytica and envisage blocking it at the C6 step. The resulting CCM and AA producing strains will be evaluated in controlled laboratory-scale bioreactors to follow the time course of conversion of the substrate into the acids and by-products. For both diacid platforms, the strain design will also target the minimisation of the co-production of mono-carboxylic acids. A metabolic model of the best performing T. fusca strain will be implemented using omics data in order to detect the bottlenecks of the production process and to guide the implementation of specific knockouts or the cloning of new genes. As a starting point, the existing genome metabolism model of F. fusca will be employed103. Further, the gene cluster of the reverse adipate degradation pathway (β-ketothioase, 3-hydroxyacyl-CoA, 3-hydroxyadipyl-CoA dehydrogenase, 5-carboxy-2-pentenoyl-CoA, and succinylCoA synthetase) will be improved by adding new enzymes or, if required, by testing other combinations of enzymes using more efficient genes from other microorganisms. In addition, the possibility to modulate the expression levels of the genes by acting on the promoters will be tested in order to reduce the toxicity of non-balanced intermediates and to increase carbon flux toward the desired product. The genetic engineering to obtain specific deletion mutants or new recombinant bacteria will be oriented by system biology predictions. Further, a knock-out mutant in the ssgB (gen conding the sporulation specific cell division protein) will be attempted. The two best-performing strains will be transferred to BIOTREND for optimization of the fermentation process and for the development of downstream processing (Task 3.3).

Task 3.3 Fermentation optimization and downstream process development for recovery of chemicals from fermentation broth. Task Leader: BIOTREND, Other partners: BBEU, SMC, DTU & ARKEMA.
In this task we will optimize fermentation conditions and develop downstream strategies for recovery of chemicals (CCM) from the fermentation broth based on the waste stream feedstocks. Expected impurities are proteins, lipids, non-fermented carbohydrates, aromatic compounds, furfural, hydroxymethylfurfural, and solids, depending on how the medium is prepared and the fermentation process. Further some fermentation by-products such as other organic acids or alcohols may be present as well.
The task will start with development of recovery and purification methods using simulated broths while the optimised strains from Tasks 3.1 and 3.2 are not made available. These simulated broths will be prepared by fermenting the waste stream with the working strains able to grow on the specific raw material and adding the expected concentrations of the product of interest. Later work will refocus on optimizing separation processes using whole fermentation broths using novel strains. Various purification methods of the diacids will be tested (ex. precipitation, adsorption, reactive extraction, and crystallization (cool, evaporative, anti-solvent crystallization)), in order to reach the required specifications for product use or polymerization. The separation of any mono-carboxylic acids that might be present in the broth will be a key parameter, since these would act as polymerisation terminators if present in the diacid end-product. For 1,5pentanediamine, state-of-the-art technologies will be evaluated to obtain an efficient and cost-effective recovery method of this diamine from the complex broth. This typically includes salt precipitation, solvent/repulsive extraction and ionexchange chromatography.
Fermentation process optimisation for increased productivities and the assessment of the robustness of the process will be performed, once the producing strains become available from SMC and DTU. The fermentation process will be implemented and characterised in a battery of fully controlled 2L bioreactors, which will allow capturing the dynamics of the process and implement robust and advanced feeding strategies for process intensification. The conditions of the most successful process will be established at 10L scale and the processing protocol transferred to WP7. The implementation of in situ-separation strategies will also be investigated in synergy with task T3.1 and T3.2, aiming at reducing product inhibition phenomena and the consumption of neutralising agents during the fermentation. The protocols and operational parameters obtained in this task will be used as starting point for pilot-scale optimization and validation of the recovery methods in WP7

Partner number and short name
4 - SINTEF-MC
5 - DTU
9 - BIOTREND
12 - BBEPP
13 - BIOPOLIS
14 - ARKEMA

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Lab-scale process for microbial production of PDA from pre-treated MSW and MRRM.	4 - SINTEF-MC	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D3.2	Lab-scale process for microbial production of CCM from pre-treated MSW.	5 - DTU	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D3.3	Lab-scale process for microbial production of AA from pre-treated MSW.	5 - DTU	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D3.4	Process for extracting muconic acid, adipic acid and 1.5-pentanediamine from complex fermentation broth.	9 - BIOTREND	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D3.1. Report on lab-scale process for microbial production of PDA from pre-treated

D3.2. Report on lab-scale process for microbial production of CCM from pre-treated MSW.

D3.3. Report on lab-scale process for microbial production of AA from pre-treated

D3.4. Report on process specifications for extracting muconic acid, adipic acid and 1.5-pentanediamine from complex fermentation broth, meeting end-user specifications.

D3.1 : Lab-scale process for microbial production of PDA from pre-treated MSW and MRRM. [32] Lab-scale process for microbial production of PDA from pre-treated MSW and MRRM.

D3.2 : Lab-scale process for microbial production of CCM from pre-treated MSW. [30] Lab-scale process for microbial production of CCM from pre-treated MSW.

D3.3 : Lab-scale process for microbial production of AA from pre-treated MSW. [32] Lab-scale process for microbial production of AA from pre-treated MSW.

D3.4 : Process for extracting muconic acid, adipic acid and 1.5-pentanediamine from complex fermentation broth. [36]
Process for extracting muconic acid, adipic acid and 1.5-pentanediamine from complex fermentation broth.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP4	**Lead beneficiary** [10]	10 - DLABS
Work package title	Flame retardants from nucleic acids from fish RRM and microbial biomass.		
Start month	8	**End month**	24

Objectives

The main objective is the development of novel flame retardants based on nucleic acid and proteins extracted from MRRM. Specific objectives:
1.- Improvement of the FR action of nucleic acid and proteins from MRRM by Chemical modification and connection with other FR compounds (lignosulphonates, phosphates, carbonates, borates, silicates, etc..).
2.- Improve the compatibilization of nucleic acid-based FRs to bio-polyamides.

Description of work and role of partners

WP4 - Flame retardants from nucleic acids from fish RRM and microbial biomass. [Months: 8-24]
DLABS, AIMPLAS, POLITO
Task 4.1: Bonding of lignin and/or melamine (or other amines) to the nucleic acid and/or protein backbone. Task Leader: DLABS, Other partners: POLITO
Strategies will be based on an initial exhaustive literature and patent survey on the modification of the extracted biomacromolecules. Usually, the concurrent presence of three well-known non-halogen fire-protecting elements (i.e. P, N & S) can provide the polymer with enhanced flame retardant features, favoring the formation of a protective and stable carbonaceous residue, called char.
In this context, the modifications of nucleic acid and/or protein backbones will be carried out, possibly exploiting synergistic effects, which will be evaluated according to the synergistic effectiveness parameter. In particular, the presence of such reactive groups as amines or carboxylic acids in the recovered nucleic acid and/or proteins will be exploited for covalently linking P-based groups to the biomacromolecule backbone or by ionic connections between the acidic phosphate salt and basic amines. As an example, phosphorylation reactions involving phosphoric acid or acidic phosphate esters will enable us to covalently link phosphate groups onto the protein/nucleic acid backbone, hence possibly increasing the efficiency of the flame retardant biomacromolecule. Furthermore, lignin will be exploited as a char promoter covalently linked to the biomacromolecule backbone. To this aim, a Mannich type reaction between the lignin's aromatic rings, aldehydes and amines from the DNA and proteins will be performed. A similar approach will be utilized for bonding melamine to the nucleic acid/proteins.
An important issue here is the analytical characterization of initial materials and products. This will give information concerning the concentrations of active groups (amine, OH, aromatic, carboxyl) suitable for connections to other molecules. This is very important for the effective development of FRs. These data can be obtained by elemental analysis, ICP, titrations, UV, FTIR, NMR and HPLC. Information about the molecular weight of the polymeric materials can be found by GPC, SEC and gel electrophoresis. The thermal and fire properties of the synthesized material will be analysed by DSC and TGA (in nitrogen and air), LOI, UL94 and cone calorimetry tests. More specifically, DLABS will determine the concentrations of reactive groups on the recovered nucleic acid and proteins by using FTIR, elemental analysis, ICP and titrations and will decide with POLITO, the reactions required for the modification of the biomacromolecules (i.e. phosphorylation and covalent bonding with lignin and/or melamine). The modified products will be tested as far as their thermal stability is considered: DLABS will perform DSC and TGA (in nitrogen and air) measurements; POLITO will help in the thermal characterization by performing TGA tests coupled with FTIR analysis, in order to assess the type of products derived from the pyrolysis and oxidation of the modified biomacromolecules.
Adjustment of the products will be performed according to feedback from AIMPLAS and POLITO on the mechanical and flammability performances (WP6). For these latter, POLITO will perform UL94, LOI and cone calorimetry tests. A continuous exchange of information on the results obtained between DLABS, POLITOI and AIMPLAS will advance the development of the project, possibly correcting some modification reactions toward a high FR effect of the recovered nucleic acid and proteins.

Task 4.2: Compatibilization of the FRs with polyamides. Task Leader: DLABS, Other partners: POLITO.
The nature of the target polymer will determined the choice of the connecting bonds. The connection can be performed by covalent and/or ionic bond and/or by making common salts with multivalent metal cations and carbonates, silicates, borates and anions. By this method not only compatibilization may be achieved but the combination of chemically connected different FRs may give synergistic effects. Different target polyamides may require different chemical moieties for compatibilization. Compatibilization will be qualitatively assessed through SEM-EDX analyses.

Furthermore, AIMPLAS and POLITO will perform mechanical tests on standard dumb-bell specimens, aiming at assessing the effect of the modified FR biomacromolecules on the mechanical behavior of the obtained compounds.

Task 4.3 Finding optimal combinations of flame retardants. Task Leader: POLITO, Other partners: AIMPLAS.
In this task, FRs from previous task will be combined and a library of FRs suitable for specific target polymers will be constructed according to experimental data and feedback from AIMPLAS and POLITO. Optimal combinations should be chosen, on the basis of synergistic effects, mechanical properties of the obtained compounds and also of costeffectiveness of the proposed FR solutions. To this aim, DLABS, POLITO and AIMPLAS will work together in order to achieve the envisaged target.

Participation per Partner
Partner number and short name
1 - AIMPLAS
2 - POLITO
10 - DLABS

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Modification of nucleic acid/protein backbone with lignin/melamine.	10 - DLABS	Report	Confidential, only for members of the consortium (including the Commission Services)	22
D4.2	Compatibilization of FR with selected polyamides	10 - DLABS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D4.3	Characterization of the obtained flame retardants	2 - POLITO	Report	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D4.1. Report on the modification of nucleic acid/protein backbone with lignin/melamine.

D4.2. Report on the compatibilization of flame retardants with selected polyamides.

D4.3. Report on the thermal, mechanical and fire characterization of the obtained green flame retardants.

D4.1 : Modification of nucleic acid/protein backbone with lignin/melamine. [22]
Modification of nucleic acid/protein backbone with lignin/melamine.

D4.2 : Compatibilization of FR with selected polyamides [24]
Compatibilization of FR with selected polyamides

D4.3 : Characterization of the obtained flame retardants [24]

Characterization of the obtained flame retardants

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP5	**Lead beneficiary** [10]	8 - IRIS
Work package title	Fish gelatine-derived barrier coating for packaging and edible coatings for food application		
Start month	10	**End month**	34

Objectives

This WP will study the potential of MRRM in the food and food packaging sectors according to the following specific objectives:
To develop barrier coating for packaging and edible coatings solutions from fish gelatine
To enhance the packaging and edible coatings solutions adding encapsulated active ingredients
To apply the developed coating on packaging films at pilot scale

Description of work and role of partners

WP5 - Fish gelatine-derived barrier coating for packaging and edible coatings for food application [Months: 10-34]
IRIS, AIMPLAS, SINTEF-FA, NUTRIMAR
Task 5.1 Edible and barrier coatings development. Task Leader: IRIS, Other partners: SFA, NUTRIMAR
Selected fish gelatine proteins extracted from fish RRM in task 2.1 have potential to be valorized as gas barrier biopolymers coating for packaging of oxidation sensitive food (e.g. meat, fish, cheese, etc.) to substitute conventionally used fossil barrier polymers (generally applied as interim barrier layer in multilayer film configuration) or directly applied on the food itself by dipping (a technology currently mostly used for fruits).
Aqueous gelatine protein solution at different pH and/or temperature treatment will be prepared. When needed, natural plasticizers and cross linkers will be selected in accordance to the list of substances approved for food contact or as food additive respectively. The formulation and process influence will be investigated at lab scale applying the coating on plastic and as standalone films. These coatings and films will be characterized in terms of optical properties, stability and water barrier. For promising solutions, oxygen barrier will be measured at different relative humidity as well as light barrier. To allow starting this work early, until fractions from WP2 are available (final fractions will be available at the latest by M18 with D2.1 although earlier ones will be evaluated to provide feedback for further optimization), commercial compounds will be used. Depending on the results, iterative feedback will be given to NUTRIMAR and SINTEF to optimize the available fish by-product.

Task 5.2 Active coatings and packaging development. Task Leader: IRIS, Other partners: SFA, NUTRIMAR
Bioactive peptides (derived from fish RRM in tasks 1.2.1 and 2.3), or other bioactive compounds, such peptides and polyphenols, will be selected (taking into account their later organoleptic acceptability by the consumer in the target application to be incorporated into active packaging and edible coating solutions in comparison with synthetic antioxidants such as BHT.
The expected role of each of the listed substances are summarised as follow:

Type of action over the packed food : Type of additive to extract from MRRM
Antimicrobial/Antioxidant : Polyphenols and Bioactive peptides
Crosslinker : Polyphenols
Plasticizer : Bioactive peptides

First of all, the antioxidant capacity (tested in WP2 for selected compounds) and antimicrobial activity of the bioactive compounds will be studied, as well as their temperature resistance. The suitable concentrations of the active coating solution will be studied in vitro in terms of antimicrobial activity against microorganisms relevant for food conservation/ spoilage (Gram-positive and Gram-negative bacteria and moulds), as well as in terms of antioxidant activity (antioxidant assays like Folin–Ciocalteu reagent (FCR), DPPH (1,1-Diphenyl-2-picryl-hydrazyl), FRAP (Ferric reducing ability of plasma)).
In order to obtain a more progressive controlled release, alternative encapsulation routes of the bioactive compounds will be considered for direct application on the food surface or dispersed in the coatings from the previous task. For instance, those will be mixed with gelatine in a dilute phosphate buffer saline water-ethanol solution to obtain sub-micro fibres by electrospinning which will be optionally crosslinked using selected bioactive substances. Different encapsulated bioactive concentrations will be produced and the release rate will be studied in different liquid food simulants to select the optimal formulations. Trials with coated plastics and standalone films will be undertaken as in the previous task and similar characterizations will be repeated.

Task 5.3 Upscaling of the coating and lamination process. Task Leader: AIMPLAS, Other partners: IRIS

Based on the lab scale inputs from IRIS in tasks 5.1 and 5.2, and selecting respectively one barrier coating and one active coating formulation of those targeted at packagingaccording the need of each food product defined in WP1, AIMPLAS will upscale the application on a 250 mm width commercial plastic films (eg. Polyethylene tereftalate (PET) or polyolephins). Initial processing tests will be done in a semi-automatic multicoater using meter bars allowing coating thickness in the range of 0.5-1.5 microns. The thickness homogeneity and coating weight of these samples will be evaluated by IRIS and AIMPLAS will be in charge of water vapour, UV and oxygen barrier evaluations.
The coating formulations will be coated and optionally laminated at pilot scale. Visual and barrier (UV, humidity, Oxygen) properties as well as adhesion of the coating on the substrate will be tested using normalized and calibrated scratch equipament. Iterative optimisation will be performed until the coating is qualitatively and quantitatively suitable (reaching same OTR but also bioactive properties as for lab samples measured as task 5.2 described). The knowledge will fuel into further upscaling in AIMPLAS in WP7.

Participation per Partner

Partner number and short name
1 - AIMPLAS
3 - SINTEF-FA
7 - NUTRIMAR
8 - IRIS
Total

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D5.1	Development and characterisation of gelatine edible and barrier coatings including bioactive compounds	8 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.2	Upscaling of coated films and laminates with gelatine barrier coatings including bioactive compounds	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	34

Description of deliverables

D5.1. Report on the development and characterisation of gelatine edible and barrier coatings optionally including bioactive compounds

D5.2. Report of the upscaling of coated films and laminates with gelatine barrier coatings optionally including bioactive compounds

D5.1 : Development and characterisation of gelatine edible and barrier coatings including bioactive compounds [24]
Development and characterisation of gelatine edible and barrier coatings including bioactive compounds

D5.2 : Upscaling of coated films and laminates with gelatine barrier coatings including bioactive compounds [34]
Upscaling of coated films and laminates with gelatine barrier coatings including bioactive compounds

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP6	**Lead beneficiary** [10]	1 - AIMPLAS
Work package title	Green chemical routes to produce polymers from developed building blocks.		
Start month	6	**End month**	36

Objectives

Objectives: The objective of WP6 is: i) to synthesize new 100% renewable biopolyamide 5.6 from the diamine 1.5pentanediamine (PDA) and adipic acid (AA) obtained in WP3, ii) to evaluate muconic acid (CCM) modification to graft SH groups that promote adhesion on metallic surfaces of long chain polyamides and to evaluate muconic acid ammoniation to lead to hexamethylenediamine (HMDA) and iii) develop a 100% renewable flame-retarded polyamides using polyamide 56 and other biobased polyamides and the flame retardants (FR) obtained in WP4.

Description of work and role of partners

WP6 - Green chemical routes to produce polymers from developed building blocks. [Months: 6-36] **AIMPLAS**, POLITO, CNRS, DLABS, MINE, ARKEMA

Task 6.1 Production of polyamide 5.6 at lab level. Task Leader: AIMPLAS. Others partners: ARKEMA.

During this task, new 100% renewable bio-polyamide 5.6 (PA5.6) from PDA and AA obtained in WP3 will be synthetized. One of the main challenges in the polymerization reaction of polyamides is the presence of impurities as other diamines or monofunctional monomers. In order to overcome it and minimize the cost of downstream process. AIMPLAS will develop a two-stage melt polycondensation method based on the development of a di-ester intermediate as is sketched below. The diester approach is inspired by the polyethylenfuranoate and polybutylenfuranoate chemistry developed since 2011 by several research groups104-105-106 and some companies as Avantium107, in this context AIMPLAS has developed similar procedures under the framework of Biorefine 2G project. (FP7-KBBE).

The first reaction of figure 3.1.a will involve the synthesis of diethyladipate or dimethyladipate, once the conversion reaction reach the appropriate value (95-99%) in a second step the intermediate will be polycondensated in the presence of 1,5-pentaenediamine. Both reactions will be optimized in terms of catalyst, stirring systems, temperature, vacuum and reaction time in a full monitored 0.5L steel reactor.

Figure 3.1.a Scheme of biopolyamide 5,6 synthesis-step 1.

Due to the expected limitations of supply initially during the two first years of the project, initial polymerization trials of polyamides will be carried employing commercial grade starting materials which will be “artificially” contaminated with some monofunctional compounds in order to simulate as close as possible to the expected bio-starting materials. The polyamide obtained directly from the reactor will be purified using high vacuum system in order to remove residual volatiles and water/ethanol from the reaction. AIMPLAS will obtain a 3-5 kilos of these polyamides to be used in task 6.3.

The PA56 will be characterized employing several techniques as Differential scanning calorimetry (DSC), thermogravimetry (TGA), Gel permeation chromatography (GPC) and rheological tests, which are currently available at AIMPLAS. In addition, the molecular weight of the new PA56 will be analysed employing a new NMR method based in the correlation of diffusion coefficients acquired by proton NMR spectroscopy and the MW. The method allows to explore and analyse with just some milligrams of samples properties as Molecular Weight and polydispersity index. Furthermore, it is copolymerisation of ester and diamine, impurities like N-alkylation will be evaluated in order to prevent polymer degradation.

Task 6.2 Muconic acid as monomer and an adhesion precursor in polyamides. Task Leader: ARKEMA. Other partners: CNRS.

Due to the expected cost of production of muconic acid and its added functionality versus adipic acid (i.e. 2 unsaturations) ARKEMA targets chemical conversion where these are either an advantage or where they would not increase the cost of conversion. In a first reaction, muconic acid/ester is reacted with H2S to graft one (or 2) thiol function on the adipic acid backbone. The mercaptoadipic acid/ester will be initially prepared in similar conditions than mercaptosuccinic acid (thiomalic acid), a well-known, intermediate for the synthesis of the insecticide malathion. Catalysts and process conditions will be screened to identify the best conditions to add one or two thiols groups. The reaction will be monitored for the optimum of selectivity, since it is not necessary to achieve a full conversion as long as diacids are still produced. The new monomers will be evaluated in polyamide (such as Platamid® www.arkema.com/en/products/product-finder/range-viewer/Platamid-Copolyamides/) and/or polyester applications for improved surface adhesion (in coating resins). In these applications, the mercapto compounds will be tested as an additive, i.e. diluted in other monomers and the assessment will be made by the observation of the adhesion on metallic and glass surfaces.

In a second application, muconic acid ammonium salt will be converted to the corresponding diamide (HMDA), then to the dinitrile. Reaction conditions used classically for the conversion of the C10 linear diacid (sebacic acid) will be used, i.e. using a catalysed batch process. When the dinitrile quality is satisfactory (conversion >99 %, monofunctional<0.1 %), the obtained dinitrile will be further hydrogenated to obtain the HMDA. The quality of the diamine has to match monomer quality, for which the main criteria will be the monofunctional content which has to be below 0.1 % to reach the highest molecular weight.

Task 6.3 Flame retardant bio-polyamides. Task Leader: AIMPLAS, Other partners: ARKEMA, MINE, DLABS.
Due to the expected limitations of supply initial during the two first years of the project, compounding trials of the polyamides and nucleic based FR will be done in two stages under commercial bio-polyamide grades (PA56 & long chain polyamides of ARKEMA) and in a second stage over the bio-PA 56 synthesized in the project. Final FR-biopolyamide formulation will be developed at pilot scale/relevant environment (TRL5) in WP7.
Sub-Task 6.3.1. Flame retardants biopolyamide 56. In this sub-task, flame retardant additives and commercial PA56 will be fed in a twin-screw extruder in order to obtain a FR-polyamide that fulfils the main automotive standards against fire. The different parameters to consider in order to maximize the dispersion, avoiding flame retardant damaging, will be: i) screw and barrel design, ii) feeding protocol (form and place), iii) screw speed, iv) throughput, v) type and percentage of flame retardant, vi) type and percentage of other additives to improve formulation.
Sub-Task 6.3.2. Flame retardants long chain biopolyamides. In this sub-task, flame retardant additives will be tested in comparison with existing halogen containing or phosphorous containing flame retarded long chain bio-based polyamide formulations (PA11 or PA10-10). The PA matrices will be compounded with different concentrations of additives in labscale twin-screw extruders and the fire resistance properties of moulded tests bars such as Limited Oxygen Index and UL 94 normalized FR tests will be measured. According to first results with pure additives, new optimized formulations with possible combinations between the new additives and other components will then be developed to meet some precise application FR requirements in transportation industry. From the BRIGIT project, we have defined in detail these requirements and have a closed contact with companies such as SOLARIS and FIAT.
Sub-Task 6.3.3. Additivation of synthesized bio-polyamide. In this sub-task the previously developed PA56 (Task 6.1), will be modified with the proper additives to obtain a polyamide grade that could be processed directly by extrusion and injection moulding. The additive selection will be guided by the desired final properties (e.g. mechanical, rheological, weathering, etc.), driven by the standards and requirements of the automotive sector. In this matter the following additives could be used: thermal stabilizers, chain extenders/anti-hydrolysis, UV filters, lubricants, impact modifiers, pigments, fillers, etc. to achieve the suitable viscosity for each selected process. Biobased flame retardants will be added to the bio-PA56 using as a start the formulation developed in subtask 6.3.1.

Task 6.4. Characterization of the Flame retarded biopolyamides. Task Leader: AIMPLAS, Other partners: ARKEMA, POLITO, DLABS, MINE.
In this task some specimens will be produced to check the properties of the developed polyamide and effectiveness of the flame retardant. The following sub-task will be carried out:
Sub-Task 6.4.1.Test specimen manufacture. Both methods, injection and compression moulding are adequate processing ways to obtain standard test specimens suitable for characterization. However the effect of processing conditions is crucial, especially for this new flame retardant; an important role not only to fill the cavity properly, but also keeping the flame retardant properties and avoiding the degradation and re-agglomeration of the FR and other additives during the process. This sub-task will give important information about the processability of the compounds obtained in task 6.3 and will run parallel with compounding task, giving guidelines to find the right formulation in terms of flame retardants composition and processability in standard plastic equipment.
Sub-Task 6.4.2.Characterization. Following the regulation derived from the automotive industry, flame retardancy tests will be carried out. These tests will be also done simultaneously with the compounding and processing trials in order to have feedback and optimize the formulations. The flame characterization will be done under the following standards: Federal Motor Vehicle Safety Standards (FMVSS), DIN 75200 and ISO 3795, in all cases aiming to a flame propagation speed up to 100 mm/min (according to the limit set by each car manufacturer).
Mechanical (tensile, impact, flexural properties), physical and thermal (density, HDT, etc.), and morphological properties of the developed compounds will also be tested using the suitable EN/ISO standards.

[redacted]

Partner number and short name
1 - AIMPLAS
2 - POLITO

6 - CNRS
10 - DLABS
11 - MINE
14 - ARKEMA

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Polymerization optimization of bio-based PA56.	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.2	Sample of produced mercaptoadipic acid/ ester and their associated report	6 - CNRS	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.3	Report of processing conditions of biobased PA56 compounds including data-sheet.	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	34

Description of deliverables

D6.1. Report of polymerization optimization of bio-polyamide 56.

D6.2. Sample of produced mercaptoadipic acid/ester and their associated report.

D6.3. Report of processing conditions of bio-based PA56 compounds including data-sheet.

D6.4. Report of the processing, and main properties of developed FR polyamides.

D6.1 : Polymerization optimization of bio-based PA56. [32]
Polymerization optimization of bio-based PA56.

D6.2 : Sample of produced mercaptoadipic acid/ester and their associated report [32]
Sample of produced mercaptoadipic acid/ester and their associated report

D6.3 : Report of processing conditions of bio-based PA56 compounds including data-sheet. [34]
Report of processing conditions of bio-based PA56 compounds including data-sheet.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP7	**Lead beneficiary** [10]	12 - BBEPP
Work package title	Product validation by industrial partners		
Start month	30	**End month**	46

Objectives

Objectives: The main objective of this WP is to achieve a TRL5 in most products/technologies developed in previous WPs (fermentation and further DSP, Flame retardants, biopolyamides and packaging barrier films).
Optimization of the fermentation process to obtain a robust and efficient production system at 150L scale (100L working volume).
Validation and optimization of the developed downstream processing methods.
Scale-up of the most promising fermentation process at 1,500L scale (1,000L working volume).
Scale-up the production of the gelatine-based coating to allow validating the new packaging demonstrators in terms of optimal food (fish) conservation

Description of work and role of partners

WP7 - Product validation by industrial partners [Months: 30-46]
BBEPP, AIMPLAS, POLITO, SINTEF-FA, CNRS, NUTRIMAR, IRIS, BIOTREND, DLABS, MINE, BIOPOLIS, ARKEMA
Task 7.1 Demonstration of the overall biorefinery concept. Task Leader: BBEU, Other partners: BIOTREND, BIOPOLIS.
Subtask 7.1.1 Scale up of MSW pretreatment and conditioning as a fermentation substrate. In order to demonstrate the overall biorefinery concept taking into account all the stages of the process, substrate preparation will be optimized and scaled-up, focusing in the preparation of MSW as substrate for the target molecule fermentation in WP3. All the steps developed in WP1 will be moved to the necessary scale to feed with enough substrate the fermentative production in subtask 7.1.4. For instance, if necessary, steam explosion will be performed at 400-L volume (up to 21 bar), and hydrolysis up to 300-L, with auxiliary equipment [solid biomass storage (35 m3), mills (200 kg/h), extractor with vapour (3 m3), solid/liquid separator (15 m3/h)].

Subtask 7.1.2 Optimization of the fermentative production of two target molecules. For the 2 best performing strains developed in WP3, an optimized fermentation process will be developed. Both strains will be first validated and then further improved in 10-200 L fermentors at BIOTREND. As these scales tend to allow better oxygen transfer rates, also higher productivities can be achieved. Besides production rates, titters and conversion yields will be improved to reduce production costs by fine-tuning the medium composition and feeding rates. Finally, the applied fermentation strategy will also take into account the next downstream processing steps to allow a simpler and cost-effective recovery of the target molecules. A simplification of the control strategy of the fermentation implemented at 200L will be sought in order to allow for a robust and straightforward upward scale transition in T7.3.

Sub Task 7.1.3 Validation of the developed DSP methods at 100 L scale. As the DSP contributes often to a major share of the final product cost, the developed DSP methods on the simulated broth in WP3 will now be validated and improved with the final fermentation broths from Task 7.1. For each target molecule, 2 x 100 L broth will be processed and evaluated at BBEU. In particular, the scalability of the unit operations, their transferability to industrial context and the compliance of the final product with the target specifications will be assessed.

Sub Task 7.1.4 Fermentative production of 1 target molecule at pilot scale. The most efficient production strain and related DSP method will be selected to demonstrate the fermentative production of a high value chemical from a waste feedstock at a pilot-scale. To this end, the process developed at 200 L will be implemented and validated at 1000 L scale (1500 L reactor volume) to assess its robustness. Subsequent pilot-scale purification of the target molecule will result in a ready to use product meeting the required specifications. The obtained data from the 2 x 1000 L fermentations will be used to perform a first techno-economical assessment in WP8.

Task 7.2: Scale up of Flame Retardant Synthesis. Optimization of flame retardancy, mechanical properties, cost and ecological impact on large scale (20L vessels). Task Leader: DLABS, Other partners: NUTRIMAR. POLITO, AIMPLAS.
Based on the process set up in WP2, and using their existing equipment for concentrating and hydrolysing proteins, NUTRIMAR will upscale the production of nucleic acid and proteins (in the range of 10 kg batch) according the results of WP4.

The scale up of flame retardants polyols will be addressed by DLABS (up to 10L) and AIMPLAS (up to 60 L). Using a stirred heated stainless steel reactor with a bottom tap wide enough to allow a slurry to exit, the chemical routes optimized in task 4.2 to obtain the new flame retardant will be scale up to obtain at least 50 kg to be validated in task 7.4. The processing temperature will be up 150ºC and the obtained material will be filtered and dried in suitable ovens to avoid further reactivity problems.

Task 7.3: Scale up of edible and barrier films. Task Leader: AIMPLAS, Other partners: NUTRIMAR, SFA & IRIS. Sub-Task 7.3.1 Upscaling of gelatine-based coatings production. Based on the process set up in WP2, and using their existing equipment for concentrating and hydrolysing proteins, NUTRIMAR will upscale the production of the suitable grades of gelatine (in the range of 10 kg batch) and bioactive compounds that worked best in WP5.

Sub-Task 7.3.2 Upscaling of the coating and lamination process. Coating formulations that show the best functional and barrier properties will be applied onto monolayer and multilayer PET/polyolefins based films and sheets producing currently at AIMPLAS facilities using a semi-industrial roll-to-roll coating equipment featuring standard coating technologies (such as rotogravure or flexography) at AIMPLAS pilot plant facilities. In the range of 50-100 linear meters of films produced at at low speed (less than 100 m/min) to produce a small series of packaging for next subtask. The best coating thickness obtained in task 5.3 will be used as an start point for final coating thickness optimization that fulfil the requirements defined in WP1.
Resulting properties will be characterized by AIMPLAS and compared with those obtained in WP5 (water, oxygen and UV barrier and coating-substrate adhesion) and an iterative optimisation will be performed to reach properties that suit the food packaging market.

Sub-Task 7.3.3 Validation trials of the new packaging solutions. With the technologies previously developed number of edible coatings and packaging configurations can be used separately or combined as will be demonstrated in the present task packing fresh and frozen fish fillets. Case scenarios for the validation of the different fish gelatine derived edible coatings and packaging solutions are detailed in the following table:

Type of Packaging

Unpacked: Fish with gelatine edible coating (Frozen fillet)

Monolayer PET film: Fish with gelatine edible coating (Frozen fillet) and Fish with gelatine edible active coating (Fresh fillet)
Monolayer PET film with gelatine barrier coating: Uncoated Fish (Fresh fillet) and Fish with gelatine edible active coating (Fresh fillet)
Multilayer PET/EVOH/PE film with active coating: Uncoated Fish (Fresh fillet)

Fresh fish fillets will be acquired through the supply chain of NUTRIMAR to be tested at IRIS. Some samples will be dipped into edible coatings optionally incorporating active ingredients as defined in the previous table. Part of them will be frozen. Monolayer as well as multilayer films produced in the previous task will be used as packaging solutions in the different planned configuration. In each case, a benchmark representing the currently used packaging material for fish will be used for comparison. Since it does not apply to active packaging, where the release of quantum satis amount of bioactive compounds is allowed, the unwanted migration of substances from coatings on multilayer films will be measured only for the passive packaging solutions. With the exception of not being performed in automatic packing equipment/lines, the results will be representative of the larger scale validation requirements of the target fish packaging application.
The fish shelf life (via physico-chemical, microbiological and organoleptic studies) will be tested. At several storage periods, analyses of different parameters (appearance, colour, odour, pH, weight loss, lipid-oxidation and concentration of the different bioactive compounds from task 5.2, etc.) will be carried out. Recommendations for further upscaling will be provided in case of the edible coatings so it could be trasnferred by eg. dipping or curtain coating in contiuous processing lines to the fish industry whereas in case of the packaging solutions, FARMAMAK will give further recommendations to reach full scale production (eg. coating and lamination at>200m/min). Based on the properties obtained and market experience, AIMPLAS will also suggest other foodstuff for which the new material should be suitable.

Task 7.4: Scale up of production of bio-based polyamides and polyamides with CCM adhesion promoters. Task Leader: ARKEMA, Other partners: CNRS, AIMPLAS.
The scale up of the production of polyamide 56 will be done by AIMPLAS (from 0.5L up to 10 L). Using a stirred heated stainless steel reactor with a bottom tap wide enough to allow a slurry to exit, the chemical routes and reactor conditions optimized in task 6.1 to obtain the biopolyamide from the selected diacid and diamine will be scaled up to obtain at least 20 kg to be validated in task 7.3 and 7.5. The polyamide will be purified by high vacuum.
Adhesion promoters will be evaluated in polymerisation reaction. Polymerization will be initially tested in disposable glass tubes to identify the appropriate loading of co-monomers. Dosing from 0 to 15 wt. % will be investigated, in order to keep the overall cost of production in the acceptable range. Since the mercapto-promoters are expected to be obtained initially as diesters, the polymerization is expected to take place in 2 stages, with a first step which is an hydrolysis of the ester, combined with a removal of the alcohol, and then a copolymerization of the obtained diacid. If the dosing of the mercapto-promoter is sufficiently low, the polycondensation reaction (diacid+diamine) would release enough water by itself to avoid the preliminary hydrolysis step, nevertheless it needs to be studied. The obtained polyamide properties will be evaluated by GPC, DSC and NMR. Mercapto diacids will be used in formulations, to prepare polyamides at kg to 10 kg scale.

Task 7.5: Scale up of production of Flame Retardant compounds on selected polyamides (industrial facilities). Task Leader: MINE, Other partners: AIMPLAS, POLITO & ARKEMA.
The compound manufacturing process developed at pilot plant level (WP6) will be scaled-up to MINE and ARKEMA industrial facilities. Machine parameters such as screw design, components feeding, coupling agents, formulation and pressure and temperature profile will be analysed in order to obtain an optimized industrial (from 10 kg/h lab scale up to 100 kg/h MINE and ARKEMA facilities. The two selected polyamides produced in task 7.6 and containing the FR additive from 7.4 will be evaluated at industrial level using the standards detailed in task 6.4

Participation per Partner
Partner number and short name
1 - AIMPLAS
2 - POLITO
3 - SINTEF-FA
6 - CNRS

7 - NUTRIMAR
8 - IRIS
9 - BIOTREND
10 - DLABS
11 - MINE
12 - BBEPP
13 - BIOPOLIS
14 - ARKEMA

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	Demonstration scale of MSW pretreatment and fractioning	13 - BIOPOLIS	Report	Confidential, only for members of the consortium (including the Commission Services)	40
D7.2	Report and tech transfer on scale-up of the fermentation of 2 target molecules at 150 L scale.	9 - BIOTREND	Report	Confidential, only for members of the consortium (including	40

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	
D7.3	Report on 150 L-scale recovery of diacid and/ or diamine from complex fermentation broth.	12 - BBEPP	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D7.4	Report on the pilot-scale production (fermentation and recovery) of the target molecule from raw feedstock.	12 - BBEPP	Report	Confidential, only for members of the consortium (including the Commission Services)	44
D7.5	Pilot-scale production of flame retardant synthesis.	10 - DLABS	Report	Confidential, only for members of the consortium (including the Commission Services)	44

D7.6	Pilot-scale production of barrier packaging film.	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	44
D7.7	Validation of the different fish packaging solutions	8 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D7.8	Pilot-scale production of FR polyamide compounds.	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	44
D7.9	Pilot-scale production of polyamides for coatings.	14 - ARKEMA	Report	Confidential, only for members of the consortium (including the Commission Services)	46

Description of deliverables

D7.1 Report and tech transfer on scale-up of MSW as fermentative substrate, or fraction recovery for pilot-scale fermentation.

D7.2. Report and tech transfer on scale-up of the fermentation of 2 target molecules at 10-200 L scale.

D7.3. Report on 150 L-scale recovery of two target molecules (diacids and/or diamine) from complex fermentation broth.

D7.4. Report on the pilot-scale production (fermentation and recovery) of the final target molecule from raw feedstock.

D7.5. Report on the pilot-scale production of flame retardant synthesis. D7.6.

Report on the pilot-scale production of barrier packaging film.

D7.7. Report on the validation of the different fish packaging solutions

D7.8. Report on the pilot-scale production of flame retardant polyamide compounds.

D7.9. Report on the pilot-scale production of polyamides for coatings.

D7.1 : Demonstration scale of MSW pretreatment and fractioning [40]
Demonstration scale of MSW pretreatment and fractioning

D7.2 : Report and tech transfer on scale-up of the fermentation of 2 target molecules at 150 L scale. [40] Report and tech transfer on scale-up of the fermentation of 2 target molecules at 150 L scale.

D7.3 : Report on 150 L-scale recovery of diacid and/or diamine from complex fermentation broth. [42] Report on 150 L-scale recovery of diacid and/or diamine from complex fermentation broth.

D7.4 : Report on the pilot-scale production (fermentation and recovery) of the target molecule from raw feedstock. [44]
Report on the pilot-scale production (fermentation and recovery) of the target molecule from raw feedstock.

D7.5 : Pilot-scale production of flame retardant synthesis. [44] Pilot-scale production of flame retardant synthesis.

D7.6 : Pilot-scale production of barrier packaging film. [44] Pilot-scale production of barrier packaging film.

D7.7 : Validation of the different fish packaging solutions [46]
Validation of the different fish packaging solutions

D7.8 : Pilot-scale production of FR polyamide compounds. [44] Pilot-scale production of FR polyamide compounds.

D7.9 : Pilot-scale production of polyamides for coatings. [46] Pilot-scale production of polyamides for coatings.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP8	**Lead beneficiary [10]**	9 - BIOTREND
Work package title	Environmental, regulatory & economic analyses		
Start month	1	**End month**	48

Objectives

Study the environmental suitability of the processes by a Life Cycle Assessment (LCA).
Assess the economic viability each the product stream through the development of a technical and economic evaluation to guide the decision points throughout the project and assess economic viability.
Ensure that the new material and their processing technology complies with the relevant regulatory landscape.

Description of work and role of partners

WP8 - Environmental, regulatory & economic analyses [Months: 1-48]
BIOTREND, AIMPLAS, POLITO, SINTEF-FA, SINTEF-MC, DTU, CNRS, NUTRIMAR, IRIS, DLABS, MINE, BBEPP, BIOPOLIS, ARKEMA, NNFCC
Task 8.1 Building a data repository. Task Leader: BIOTREND, with the support of all partners.
The economical and life cycle assessment rely on putting together all process relevant information regarding process flowsheets, material and energy balances, specifications of equipment to be used in the larger scale. Although the

complete set of information will only be available at the end of the process, every 6 months the information will be collected from partners, continuously refining it, integrating new layers of detail, and timely identifying information that might be lacking. Further to process information, market information will also be obtained from publicly available sources and the partners, mainly the end users (market volumes and segmentation, prices) and the raw material processors (availability, costs). The gathered information will be continuously shared with T8.2 & T8.3.

Task 8.2 Evaluation of synergies with other projects involved in MSW & animal RRM valorisation. Task Leader: BIOPOLIS, Other partners:SFA, SMC, DTU, NUTRIMAR, IRIS, BIOTREND, ARKEMA, BBEU & NNFCC.
There are several projects aiming to make biofuels and/or bio-based products from MSW or animal RRM (FP6, FP7 and H2020). Some of them feed the waste to microorganism; some others first go via a hydrothermal liquefaction process or a gasification. In the second case, it is important to check that no chemical by-products will affect the downstream microorganism process, as for example happened at Ineos Bio's process (in Florida) to produce ethanol, where they faced problems of contamination with HCN. (same if such a step is introduced in this project). Synergy with current or past fuel related projects will be evaluated using the CORDIS database, personal contacts and information from the different partners will be employed. A stakeholder board, or an advisory board will be proposed in the first months of the project that include mainly companies and research centres from the biofuel sector (such as IMECAL, Lanzatech, Ineos, Enerkem, French institute of Petroleum and New Energies, BP, Shell, UBE or TOTAL). In terms of Board members, we plan to involve the European association of the chemical industry and The European Polymer manufacturer association.

Task 8.3 Economic Analysis & Mass and energy balances. Task Leader: BIOTREND, with the support of all partners.
Throughout the project, a constant technical and economic analysis monitoring of the developments and improvement in the production methods, particularly those of WP1 to WP7, will be undertaken by BIOTREND with the help of the other project members, particularly the leaders of WP1 to WP7. Constantly feeding back information and data to the process development activities helping on decision taking and choosing the best between several possible processing alternatives. Particular attention will be given to the different integration opportunities between the fermentation and downstream operations with reference to the established specifications of the produced material and, when possible, to raw-material recycling and energy integration.
To evaluate the impact of the project in terms of energy and mass streams, including the actual CO2 balance which aims to be positive, virtual simulations of the some DAFIA process at the process scale and/or larger scale (as a case study) will be run in order to evaluate the different balances.
Software packages such as SuperPro designer will be used to swiftly model the different processing options. In addition to the final economical assessment, an intermediate report on cost calculation will be delivered at M24 which will be based on a simple conceptual design for the various processing streams considered within DAFIA. This will allow prioritizing activities in workpackages WP1 to WP7, either by contributing to the selection of the processing streams to be adopted in higher scale development or by identifying critical processing steps that require further optimization and by establishing minimum performance targets to be achieved.

Task 8.4. Product Life-Cycle-Assessment. Leader AIMPLAS, with the support of all partners.

Life Cycle Assessment considers the entire life cycle of a product to assess the environmental impacts associated with all the stages of the product's life from-cradle-to-grave (i.e. from raw material extraction through material processing, manufacture, distribution, use, repair and maintenance, and disposal or recycling). LCA gives for instance a measure of the water footprint and greenhouse gas emissions. .

To ensure the consistency of the LCA studies performed in DAFIA, Task 8.1 will build upon existing and ongoing LCA activities on bio-based products and processes, and will take into account the guidelines of the ILCD handbook project108, which has developed guidelines to make LCA studies more coherent and comparable. Furthermore, the LCA approach will be used to compare bio-based and non-biobased products within the same LCA framework, this comparative approach generates results which are more robust than a reliance on absolute values and will help in the marketing claims of the advantage of the new solutions vs. conventional synthetic counterparts.

The LCAs will be performed using dedicated software (e.g. Umberto NXT software) and the Ecoinvent database for a selected set of raw material/production process/product, taking into account use and disposal or re-use, i.e. 'from cradle-to-grave'. The LCA will follow the ILCD guidelines and ISO 14040 methodologies Definition of goal and scope, including system boundaries and functional units. It is likely that four products will be studied: 1) Flame retardant, 2) Diamines and diacids 3) Biopolymers and 4) Coatings.

Life Cycle Inventory: detailed data on the processes and flows will be obtained from the partners and will be completed by data from existing and ongoing LCA activities on bio-based products and processes, as well as relevant metadata.

Life Cycle Impact Assessment: the impact of the developed technologies and products will be assessed through a set of indicators (e.g. global warming potential, energy demand from non-renewable resources, water consumption)

Interpretation of the results: special care will be taken to ensure that the results are reasonable and realistic. The results of the LCA will then be compared to those of equivalent non bio-based products

A dynamic approach with feedback to the process development activities will be implemented, using the outputs of the LCA analysis. The pilot tests and demonstrators will likewise provide feedback to the Life Cycle Inventory, with up-to-date field data.

The LCA will serve both as a characterisation tool for the performance of the developed processes, and as a means of identifying areas where process optimisation can make the largest environmental savings. This information will be fed back to the RTD development and scale-up activities so that effective process optimisations may be realised.

Task 8.5. Regulatory and safety analysis. Task Leader: NUTRIMAR Other participants: ARKEMA, AIMPLAS, IRIS, MINE.

The verification that the products comply with the existing regulatory landscape for the use of the rest raw materials and for the use of the final products on the envisaged applications and fulfil EC Directives and specific national laws in the EU28, regulatory compliance will be regularly checked throughout the project. For example, Regulation EC 10/2011 on plastics intended to come into contact with food and Regulation EC 450/2009 on active and intelligent materials and articles intended to come into contact with food.

The assessment should be performed before the end of the project in order to enable any processing/product specification changes. Input from final users required. A regulatory compliance report will be written to make sure that the products and technologies also fulfil all standards and legislation in force with regards to safety criteria, the developments will be evaluated according to these standards.

Task 8.6 Evaluation of Economic and LCA risk. Task Leader: ARKEMA, with the support of all partners.

In this task, a sensitivity analysis will be performed on the economic and LCA analysis, to deal with uncertainties associated to the deterministic results from Tasks 8.2 and 8.3. Indeed, the scenarios in these tasks will be produced with a unique set of data and a probabilistic approach will be required in which a set of data is used as input (for example historical raw material costs), yielding as output a statistical distribution of possible product costs. The final result is, for example, a probability of the production cost to be below a given figure. These approaches helps providing relevant information for decision-taking procedures of adopting specific innovative technologies, quantifying risks and benefits (ex. production cost, reduced carbon footprint, etc). Uncertainties will be assessed for the input values of the deterministic model and then the data processed using a Monte-Carlo simulation. The sophisticated models will take into account that the statistical distribution is not necessarily Gaussian, but can have different shapes. For example, in the future the crude oil price has a low probability to be at 40 $/bl, a fair probability to be at 100 $/bl and a low probability to be at 200 $/bl. The shape of the statistical distribution can be taken from different sources (ex. for the case of crude oil price, historical data over the last 10 years, or predictions of the International Energy Agency…) for all data taken into account. The models will be established with the most adequate tool for each task (ex. Excel, SuperPro Designer and/or Aspen) and interfaced with the CrystalBall software for the probabilistic simulations109. The result of this task will fuel into a scenario for post-project upscaling to greater TRL that allow future commercialization of viable products as inputs for task 9.4.

Participation per Partner

Partner number and short name
1 - AIMPLAS
2 - POLITO
3 - SINTEF-FA
4 - SINTEF-MC
5 - DTU
6 - CNRS
7 - NUTRIMAR
8 - IRIS
9 - BIOTREND
10 - DLABS
11 - MINE
12 - BBEPP
13 - BIOPOLIS
14 - ARKEMA
15 - NNFCC

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	Report on economic analysis.	9 - BIOTREND	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D8.2	Report on Life Cycle Assessment (LCA).	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D8.3	Report on regulatory analysis and safety aspects.	7 - NUTRIMAR	Report	Confidential, only for members of the consortium (including the Commission Services)	48

D8.4	Final report on simulation of the system and M&E balances	9 - BIOTREND	Report	Confidential, only for members of the consortium (including the Commission Services)	46

Description of deliverables

D8.1 Report on economic analysis (Preliminary month 24 and Final Month 46).

D8.2 Report on Life Cycle Assessment (LCA).

D8.3 Report on regulatory analysis and safety aspects.

D8.1 : Report on economic analysis. [24] Report on economic analysis.

D8.2 : Report on Life Cycle Assessment (LCA). [46] Report on Life Cycle Assessment (LCA).

D8.3 : Report on regulatory analysis and safety aspects. [48] Report on regulatory analysis and safety aspects.

D8.4 : Final report on simulation of the system and M&E balances [46] Simulation of the system and M&E balances

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**

Work package number [9]	WP9	**Lead beneficiary** [10]	11 - MINE
Work package title	Communication, Dissemination, Exploitation, Business plan and Technology Transfer.		
Start month	1	**End month**	48

Objectives

This Workpakage encompasses the development and delivery of the project Plan for Exploitation of the Results and Dissemination (PERD). The work package will be led by MINE and draw upon the expertise and market knowledge of all partners. WP9 comprises of six interlinked actions designed to ensure successful exploitation of the projects technical innovations. The six tasks are 1) Communication, 2) Dissemination, 3) Exploitation, 4) Business Models and Plans, 5) Intellectual Property Management and 6) Technology Transfer.

Description of work and role of partners

WP9 - Communication, Dissemination, Exploitation, Business plan and Technology Transfer. [Months: 1-48]
MINE, AIMPLAS, POLITO, SINTEF-FA, SINTEF-MC, DTU, CNRS, NUTRIMAR, IRIS, BIOTREND, DLABS, BBEPP, BIOPOLIS, ARKEMA, NNFCC
Task 9.1 Communication. Task Leader: MINE with the support of all partners.
The purpose of the communication plan is to maximise awareness of project aims and outputs amongst a broad stakeholder base. Communication activities complement dissemination actions as they add public value to the achievements of the project by transforming the sometimes-complex scientific and technological results into media resources. We will develop clear and credible key messages and communications tools to be used by the project partners to communicate, educate, engage and assess consumers' needs around the different applications that the project undertakes.
The communication plan will identify channels for tailored, target-specific communication and will include guidelines for using and disseminating the knowledge gained during the project lifetime. This plan will identify the relevant trade media (e.g. bio-based chemicals and materials, sustainability, industrial biotechnology as well as packaging and the relevant specialty chemicals publications).
Stakeholder mapping - At the start of the project, the Consortium will review and expand the list (email pool) of groups, projects, experts and other stakeholders, identified during the proposal phase, and develop a stakeholder mapping to use in its communication activities. Stakeholder groups will include technical and non-technical business managers (segregated by value chain position), policy makers, politicians, NGOs and the public.
Messaging - WP9 will work on defining the key messages that need to be used in the communications materials. The messages, based on scientific information, will be concise, clear and easy to understand, and tailored to the target stakeholder group. The project communications tools will include:
Project logo and marketing communications tools - The first step to a successful project communications strategy and implementation is the visual identity of the project itself. A DAFIA logo along with the project's visual identity will be designed and produced. The logo and the project's visual identity will be rolled out to the different communications tools that will be produced to engage consumers.
Project website and communication channels - At the beginning of the project, WP9 will set up the DAFIA website. The website, in English, will be developed to provide an overview of the project objectives and programme of work, to track its progress and present relevant results of individual work packages in near real time. Project deliverables will be available on the website and project workshops, consultations and activities will be promoted. The website will be publicly available and will serve as a useful source of information for the general public/consumers and other relevant stakeholders. Another channel for dissemination of the DAFIA objectives, organisation and results is through social media. WP9 will set-up and maintain a presence in social media networks, including Facebook, Twitter and YouTube. The DAFIA social media presence will broaden the reach to the public and experts interested in following the project and offering feedback and suggestions. The presence in social media, along with the website, will allow the consortium to make a rough estimate on how innovative, substantiated and valuable views are expressed by consortium members and outside stakeholders, e.g. by the number of hits/visits, 'likes' etc.

Task 9.2 Dissemination. Task Leader: MINE with the support of all partners.

The purpose of the dissemination plan is to support project exploitation through technical dissemination and industrial outreach. The DAFIA Consortium will use a variety of platforms tailored, when possible, to reach particular industry and research groups and maintain communication with them. Press releases will be drafted and disseminated and opportunities to publish through the trade media will be compiled. The consortium will also carefully select events organised by others to participate in and present the project results towards the end of it.

Based on the stakeholder mapping done at the beginning of the project (Task 9.1), a dissemination plan (a Working Document among the partners) with tailored means to reach main target audiences will be developed. The plan will designate specific points of contact for value chain participants, research communities, press, the general public and the European bioeconomy process industry that will be maintained during the course of the project, so that the DAFIA consortium can always be contacted by relevant actors in the field and interested persons for any query on content and/ or findings but also as opportunities for possible collaborations.

Task 9.3 Exploitation. Task Leader: NNFCC with the support of all partners.The Exploitation Plan will be designed at the beginning of the project as a working document and updated constantly throughout the project. The plan will be reviewed yearly with progress and identified issues recorded and exploitations actions updated. At the end of the project, a detailed exploitation plan will be finalised. Exploitable outputs will be identified and discussed at regular exploitation workshops;

Exploitation workshops - The whole consortium will be involved in exploitation workshops that will be led by NNFCC. The objective of the workshops will be the characterisation of the exploitable results and identification of new results as the project progresses. The workshops will serve as screening mechanism to identify the key exploitable results to be taken through to business model and plan development (Task 9.4). The workshops will include update and discussion between consortium partners about competition (existing or emerging alternative solutions) and an initial identification and assessment of the risks and barriers to be overcome in order to enable the exploitation of the results and design a mitigation plan. The ultimate goals of the final exploitation workshop will be: 1. Fine-tuning and validation of the exploitation plan, in particular the management of IP and the business model(s) for each of the retained results; and 2. Exploiting new knowledge which will imply interactive activities, led by the exploitation experts of NNFCC and involving all consortium partners. The exploitation workshops will be organised on a yearly basis, and whenever possible just before/after the General Assemblies. The outcomes of these workshops will feed into the Exploitation Plan.

The Exploitation Plan will address the following aspects: 1) List of project results and identification of key exploitable results (KER i.e. most promising results). The final exploitation plan will incorporate the information and knowledge developed in Task 9.4 and also an exploitation risk analysis.

Exploitation risk analysis will lead to a risk management plan, taking into account the internal and external factors which may prevent the successful exploitation of the project results. This task follows three steps: 1) the identification of exploitation risks. They may include technical and non-technical risks e.g. human, economic, legal, IPR, safety; the risk register will draw from the information and knowledge gathered in all work packages and task, and, 2) the assessment of these exploitation risks along two dimensions: probability of occurrence and potential impact. A matrix combining these two dimensions then allows ranking the different risks and 3) the design of a risk management plan detailing the actions to put any type of exploitation risk under proper control. The risk management will be implemented during the project with recommendations for the pre-exploitation phase in the final exploitation plan.

Task 9.4 Business Models and Plans for KER. NNFCC with the support of all partners.
Development of business models: NNFCC will develop business models to support the continued exploitation and eventual commercialisation of each key project result (please see Impact section for more details). This task will be led in coordination with the dissemination/communication (Tasks9.1/9.2) activities to ensure the coherence of the project's messages towards the outside world.
The task will involve outlining, refining and defining the business models through two actions.
Action 1. Business Model Development. The development of a successful business model for an innovation requires an understanding of its potential value and the costs involved in its realisation. A two-step process will be used to build an initial plan.
Step 1) Understand, refine and define the innovations value proposition through the analysis of, and discussion with consortium partners and through literature review considering 1) Policy/Environmental/Social/Technological (PEST) issues, 2) competitive intensity analysis, 3) the need for alliances and collaboration and finally a Strengths/Weaknesses/ Opportunities/Threats (SWOT) analysis. NNFCC with the support of the whole consortium will analyse the potential for innovations to address new emerging market opportunities (white space) or whether the innovations address current business problems and whether these problems are acknowledged (blatant and critical) or unacknowledged (latent and aspirational). The nature of the problem the innovation addresses is important for understanding the time and effort that may be required to bring the innovation to market.
Step 2) Build economic models describing the cost and revenue projections for the innovations, this allows profit margins to be calculated and business sensitivities to be identified. This action will draw on information produced in Task8.2. Estimating revenue potential draws on the knowledge gathered in the development of the value proposition. The economic model will allow understanding of future cash flows and highlight the need for additional funding or external investment to reach the first sales.

Action 2. Business Model Validation. To ensure the validity of the business model it will be tested and refined through engagement and discussion with target customer segments and commercial partner networks. A two-step process will be used to validate the business model.
Step 1) To ensure valuable feedback, careful stakeholder identification and recruitment is required. In collaboration with project partners, a targeted list based on stakeholder profiles will be drawn-up. The target list will ensure appropriate value chain coverage with an emphasis on the potential to become early innovation adopters. Stakeholders will be recruited through an initial contact, introduction of the value proposition and request for initial feedback, recruitment will draw on the stakeholder database developed for dissemination activities. Step 2) Feedback on the business model will be captured through interviews, questionnaires and, if relevant, visits to the experimental demonstration sites, depending on the targets' profiles. The feedback analysis will lead to a review and update of the value proposition and revenue/ cost models within the business models.
Business Plan: The developed business models will be detailed in a business plan that will be the central part of the exploitation plan. Business Plan will be developed as a sustainability strategy outlining what should happen to the project outputs at the end of the project, and to explore how they can be sustained. The business plan will include client definition, a strategy for marketing, sales and operations financial models and go-to market strategies.

Task 9.5 Intellectual Property Management. Task leader: IRIS with the support of all partners.
This task will undertake a regular review and assessment of the consortiums 'Freedom to Operate' in the areas of exploitable outputs. Reviewing 'Freedom to Operate' throughout the project will ensure that the consortium is safe to produce a product or use a production technology without infringing or violating third party intellectual property rights or highlight the need for mitigating actions by the consortium (eg. acquisition of license, which would impact the exploitation).

Furthermore, IPR management will include the mapping of existing patents and potentially overlapping IPR; and when the developing partner identifies possible novelty in a KER being developed, an assessment of the new knowledge generated during the project (outputs and other knowledge such as data); a proposition of optimal IPR protection options (e.g. patent, copyright, trademark, confidentiality), in line with the selected business model options (Task 9.4) and taking into account possible co-ownership; the proper implementation of the IPR protection measures during the project. The findings and recommendations for IPR protections will be included in the final exploitation plan (D9.3). IRIS will lead the IPR management by providing guidance to the partners for their FTO and patentability search but each partner will be responsible of the right management of his result. When necessary, after the initial identification of a promising (patentable) idea, external professional support from a patent agent will be sought.

Task 9.6 Technology Transfer. Task Leader: MINE with the support of all partners.
In this task, the technical skills required to ensure the wide take up and exploitation of the materials and processes developed in the project will be gathered and the' know how' disseminated. Technology transfer will target the wide chemical building blocks (dicarboxylic acids and diamine) to produce polyamides and polyesters for a wide range industrial applications. This will ensure a high level of scientific and technological knowledge transferred to the industrial companies at different levels according to the needs of each sector: scientists, engineers, technicians, workers etc. The activities planned to achieve a successful technology transfer are:
Workshops/seminars oriented to different sectors (green chemistry, etc.). The two workshops will be organised in M30 and M48, in two different European countries (TBD), in cooperation with Task 9.2. The subject will be selected at M18 and M30, depending on the intermediate results.
Training at companies' facilities. Technology transfer will be facilitated through the arrangement of 'short stay' training courses designed for scientists/Technicians to take place at partner companies' facilities. These courses will be organised in order to foster a quick transfer of the knowledge and technologies developed in the project.
Technical Guidance Document: A 'Guide to the production and use of green chemical intermediates and resins using MSW and MRRM as a main raw material' will be prepared and published after completion of the project, in order to contribute to the technology transfer, dissemination and training. According to best industrial and environmental practices, all partners will co-operate in the elaboration of this Guide, which will contribute to an optimal profit of the new chemical products advantages. This guide will be developed by all Partners (D9.8) and will include the safety, regulatory, economic and environmental issues identified in WP8, and will be made public in the second Project Workshop (M48). This guide will not contain any information conflicting with Partner's exploitation interests and will be made available also under Open Access approach.
Publishing scientific articles in journals and conferences (subject to the consortium's IP protection publication policy) and detailed in table 2.2.c. (section 2).
Development of policy briefs and case studies targeting policy makers responsible for drafting EU or national legislation, regulations and guidelines affecting AFPW valorisation, including sustainability guidelines and standards. This will also be fuelled by the results of task 9.5 in terms of influence of current policies on the market deployment of the project results (eg. barrier or gaps).

Participation per Partner

Partner number and short name
1 - AIMPLAS
2 - POLITO
3 - SINTEF-FA
4 - SINTEF-MC
5 - DTU
6 - CNRS
7 - NUTRIMAR
8 - IRIS
9 - BIOTREND

10 - DLABS
11 - MINE
12 - BBEPP
13 - BIOPOLIS
14 - ARKEMA
15 - NNFCC

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D9.1	Project web site	1 - AIMPLAS	Websites, patents filling, etc.	Confidential, only for members of the consortium (including the Commission Services)	3
D9.2	Initial Communication and dissemination Plan	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	16
D9.3	Report on performed communication activities and dissemination Plan updates	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D9.4	IPR policy in line with the Exploitation Plan	15 - NNFCC	Report	Confidential, only for members of the consortium (including	6

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	
D9.5	Report of the IPR management activities	15 - NNFCC	Report	Confidential, only for members of the consortium (including the Commission Services)	48

D9.6	Final Exploitation Plan & Business Plans.	15 - NNFCC	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D9.7	Technology Transfer Activities, including training to SMEs, other activities and workshops	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D9.8	Guide to explore the conversion routes of MSW, MRRM to obtain high added value products.	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D9.9	Final version of the PEDR.	11 - MINE	Report	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D9.1 Project web site

D9.2 Initial Communication and dissemination Plan

D9.3 Report on performed communication activities and dissemination Plan updates

D9.4 IPR policy implementation in line with the Exploitation Plan

D9.5 Report of the IPR management activities

D9.6 Final Exploitation Plan including Business Plans of key results

MINE D9.7 Report of Technology Transfer Activities, including training to SMEs, other activities and workshops M30/48

D9.8 Guide to explore the conversion routes of MSW, and marine rest raw-materials (MRRM) from the fish processing industries, to obtain high added value products.

D9.9. Final version of the PEDR.

D9.1 : Project web site [3]

Project web site

D9.2 : Initial Communication and dissemination Plan [16]

Initial Communication and dissemination Plan

D9.3 : Report on performed communication activities and dissemination Plan updates [18]
Report on performed communication activities and dissemination Plan updates

D9.4 : IPR policy in line with the Exploitation Plan [6]
IPR policy in line with the Exploitation Plan

D9.5 : Report of the IPR management activities [48]
Report of the IPR management activities

D9.6 : Final Exploitation Plan & Business Plans. [48] Final
Exploitation Plan & Business Plans.

D9.7 : Technology Transfer Activities, including training to SMEs, other activities and workshops [30]
Technology Transfer Activities, including training to SMEs, other activities and workshops

D9.8 : Guide to explore the conversion routes of MSW, MRRM to obtain high added value products. [46] Guide to explore the conversion routes of MSW, MRRM to obtain high added value products.

D9.9 : Final version of the PEDR. [48] Final
version of the PEDR.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**

Work package number [9]	WP10	**Lead beneficiary** [10]	1 - AIMPLAS
Work package title	Project Management		
Start month	1	**End month**	48

Objectives

Overall project co-ordination and management. To optimize the resources and to ensure that all EC requirements for communication and reporting are met. Specific objectives:
Setting up of Project Management and Administration: Definition of Management Procedures; development of procedures, rules and templates for the consortium reporting activity, management platform through web-based project intranet.
Management Co-ordination of the Project: Administrative and financial project management; legal representation of the project.
Project Administration, control and support activities: Co-ordination and development activity reporting; Requesting and processing cost statements and cost certificates; Follow-up the progress of the project; Provide generic support at project level.
Project Communication: Define, provide, set-up and maintain the necessary communication procedures and tools between all the partners of the project to ensure an efficient communication flow. Communication with the EC representatives.
Risk management: to carry out a continuous (ongoing) assessment of the progress of the project. To review the progress of the work in relation to the defined success measures and to report on this progress. Where appropriate, contingency plans will be identified and implemented

Description of work and role of partners

WP10 - Project Management [Months: 1-48]
AIMPLAS, POLITO, SINTEF-FA, SINTEF-MC, DTU, CNRS, NUTRIMAR, IRIS, BIOTREND, DLABS, MINE, BBEPP, BIOPOLIS, ARKEMA, NNFCC
Task 10.1 Management of the Resources. Task Leader: AIMPLAS Involved partners: All WP leaders.
The project will be managed on a daily basis by the lead partner AIMPLAS. Full consortium project management meetings will be held every quarter to discuss technical progress and contractual issues (conference calls). Project management duties will include a dedicated project intranet which will act as a project management and communication tool and will include a password protected area for the sharing of meeting arrangements, minutes, reports and financial statements.
The Project Coordinator has the responsibility of managing the project cost effectively, including day to day management aspects, distribution of work within project partners, assessment of results, risk management, etc. Due to the number of partners involved (15) and the number of Member States (10), the management effort required should not be underestimated, requiring of an efficient and well-designed management structure. The project management structure, i.e. Project Coordinator, General Assembly (GA) and Executive Board (EB) is fully described in section 3.2. The General Assembly, representing the voice of the whole consortium, will form the highest decision making body. The EB will form the decision-making and arbitration body (on behalf of the GA) and will be responsible for the overall strategic direction of the project, in order to ensure that the project progresses properly and the risks are properly identified and managed. Intellectual/Industrial Property Rights (IPR) are managed by the IPR & Exploitation Board (IPREB), whose responsibilities are also described in section 3.2. In all these boards, the membership ensures that there is a strong industrial leadership in all Project activities. The TMC, formed by the WP leaders, who are in turn responsible for the coordination of all the activities performed by each participant within the work package and for on time delivery of their specific tasks and the subsequent report of problems or special issues, will assess the technical progress of the Project reporting to the EB and the GA.
Management Co-ordination. The Project Co-ordinator (AIMPLAS) will assume the administrative and financial management of the project, assuring the project's progress. The Coordinator supports the Project Consortium in legal, administrative and financial issues. This co-ordination will lead the strategic, political, economic and operational of the project, articulated through the different Boards. AIMPLAS will thus be in charge of the preparation, organization, administration, minuting and follow up of the main project meetings (GA, EB, IPREB, TMC) and will represent the Project to the EC. The tasks are fully detailed in section 3.2.2.
Project administration. Progress of each task will be assessed every three months to closely control the development activities of the partners in order to ensure that there is no deviation from the original plan or that corrective actions are

taken in case of deviations. Financial control will be performed for the official reporting periods (requesting and processing the cost statements and the necessary cost certificates) as well as periodic Intermediate internal financial controls. It is foreseen to send to the EC, two official progress reports (M18, M36) and a final report in M48. Details about the methods for monitoring and reporting progress, and documenting results, are provided in Section 3.2. An internal project web site will be created at the beginning of the project, in order to facilitate the monitoring and reporting of progress and documenting of results (see WP9). The website will also allow interactive internal communication between the partners involved in the different tasks. An overall description of all the Committees, their membership and procedures is provided in section 3.2.

Task 10.2 Risk management. Task Leader: AIMPLAS. Involved partners: TMC Members
Based on the initial risk identification reported in table 3.1.c., at the beginning of the project and for each work package, the risk indicators will be defined taking into account the overall project progress, and will be reported in M9.1 and D9.1 Each WP leader will collect the risk indicators and will monitor technical progress at WP level along the project implementation. Progress of each task will be assessed every 3 months to control the developed activities of the partners in order to ensure that there is no deviation from the original plan or that corrective actions are taken when appropriate. The TMC and EB will review and assess the progress of each WP in relation to the identified risk indicators and assist the WP leaders. When appropriate, contingency plans and countermeasures will be developed and when approved by the EB, will be discussed with the EC Scientific Officer. If approved, they will be implemented. The progress will be reported periodically, in the contractual reports (month 18, 36, 48) and in the internal intermediate progress reports, where deviations from the work programme objectives (Deliverables and Milestones) are identified, they will be addressed and corrective actions/plans and countermeasures implemented to ensure that the Project reaches its planned outcomes.

Task 10.3. Communication with EC. Task Leader: AIMPLAS.
The co-ordinator is the sole point of contact between the EC and the project consortium. AIMPLAS will ensure that all milestones are met and that all reports are prepared in accordance with the contractual requirements. A Milestone report will be issued and sent to the EC describing the Milestone completion and its implication in the project progress. The structure for monitoring and reporting progress will consist of a series of reports and meetings. Technical reports will be sent to the EC to meet the reporting pattern expected. Documentation will be in English and in Microsoft Word format. All reports will be assembled by the Project Co-ordinator in a master file, which will be uploaded to the project intranet. Review procedures will be agreed and included in the Consortium Agreement to ensure that all the project deliverables reach a satisfactory quality standard (e.g. technical deliverables will be assessed by the TMC or certain appointed members and the Project Coordinator prior to submitting them to the EC).

Participation per Partner
Partner number and short name
1 - AIMPLAS
2 - POLITO
3 - SINTEF-FA
4 - SINTEF-MC
5 - DTU
6 - CNRS
7 - NUTRIMAR
8 - IRIS
9 - BIOTREND
10 - DLABS
11 - MINE

12 - BBEPP
13 - BIOPOLIS
14 - ARKEMA
15 - NNFCC

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D10.1	1st Interim Progress Report including Risk indicators and Risk assessment	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D10.2	2nd Interim Progress Report including Risk indicators and Risk assessment	1 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	27

Description of deliverables

D10.1 1st Interim Progress Report including Risk indicators and Risk assessment M09

D10.2 2nd Interim Progress Report including Risk indicators and Risk assessment M27

D10.1 : 1st Interim Progress Report including Risk indicators and Risk assessment [9]
1st Interim Progress Report including Risk indicators and Risk assessment

D10.2 : 2nd Interim Progress Report including Risk indicators and Risk assessment [27]
2nd Interim Progress Report including Risk indicators and Risk assessment

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP11	**Lead beneficiary** [10]	1 - AIMPLAS
Work package title	Ethics requirements		
Start month	1	**End month**	48

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP11 - Ethics requirements [Months: 1-48]
AIMPLAS
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D11.1	EPQ - Requirement No. 2	1 - AIMPLAS	Ethics	Confidential, only for members of the consortium (including the Commission Services)	3
D11.2	POPD - Requirement No. 3	1 - AIMPLAS	Ethics	Confidential, only for members of the consortium (including the Commission Services)	3

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D11.1 : EPQ - Requirement No. 2 [3]

The applicant must provide further information about the possible harm to the environment caused by the research and state the measures that will be taken to mitigate the risks. If relevant, copies of facilities authorisations must be provided (e.g. security classification of laboratory, GMO authorisation). The applicant must ensure that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

D11.2 : POPD - Requirement No. 3 [3]

Detailed information must be provided on the procedures that will be implemented for data collection, storage, protection, retention and destruction and confirmation that they comply with national and EU legislation. The applicant must explicitly confirm that the data used are publicly available. In case data is not publicly available, relevant authorisations must be provided.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Composition of MSW/ MRRM available		13 - BIOPOLIS	8	Comparison with reported values in literature for similar feedstocks
MS2	Samples of preprocessed & fractionated MSW available for Task 1.3		13 - BIOPOLIS	20	Sufficient amounts of different fractions for lab tests within WP2 & WP3.
MS3	Samples of preprocessed & fractionated MRRM available for WP2		3 - SINTEF-FA	30	Sufficient amounts of different fractions for lab tests within WP2.
MS4	Technological solution for extraction of gelatine from fish rest raw materials tested.		3 - SINTEF-FA	18	Required purity verified by analyses. Delivery 100 g. to WP5.
MS5	Process for extraction of nucleic acids tested.		4 - SINTEF-MC	22	Required purity verified by analyses. Delivery 125 g. to WP4
MS6	At least 20 g/kg MSW of 1.5-pentanediamine produced in laboratoryscale fermentation (1L)		13 - BIOPOLIS	30	Verified by chemical analyses
MS7	At least 5 g/kg MSW of CCM and AA produced in laboratory-scale fermentation (1L).		5 - DTU	32	Verified by HPLC analysis
MS8	At least 5 g/kg MSW of AA produced in laboratory-scale fermentation (1L).		13 - BIOPOLIS	34	Verified by HPLC analysis.
MS9	DSP method has been validated to separate and purify PDA, CCM & AA from complex fermentation broth, while meeting end-user specifications.		9 - BIOTREND	36	Samples of each product have been produced meeting the specifications required by the end-users.
MS10	Samples of compatible FR (150 g scale) ready to be employed in WP6.		10 - DLABS	22	Compatibility verified by SEM-EDX analysis.

MS11	Nucleic acid based FR with a weight loss < 2% at 300ºC as assessed by TGA.		2 - POLITO	24	Verified by TGA analyses in nitrogen and air.

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS12	Edible, active and barrier coatings available at lab scale		8 - IRIS	24	Developed coatings have good film forming properties, as well as respectively good barrier to oxygen and/or antimicrobial properties.
MS13	Packaging samples produced at pilot scale		8 - IRIS	34	Developed Coating is processable using AIMPLAS' lamination set.
MS14	Bio-polyamide 5.6 compounds with adequate properties and processability to be used in automotive and packaging applications		1 - AIMPLAS	32	Rheological and mechanical tests using suitable EN/ISO standards.
MS15	Production of Monomers derived from CCM so that polymer tests can be engaged.		14 - ARKEMA	34	Production of samples with required purity for polymer tests.
MS16	Production of samples with required purity for polymer tests.		1 - AIMPLAS	36	Full characterized fire-proof polyamides using suitable FMVSS and DIN standards.
MS17	Demonstration scale of MSW pretreatment and fractioning		13 - BIOPOLIS	40	Process successfully implemented at 300L with at least the yields and concentrations achieved at laboratory process
MS18	Scale-up of fermentative production of 2 target molecules at minimal 100 L scale.		9 - BIOTREND	40	Process successfully implemented at 100L with at least the productivity achieved at 2L process.
MS19	DSP for 2 target molecules has been implemented to treat 100 L broth yielding product conform specifications.		12 - BBEPP	42	Process of M3.3 implemented at 100L for 2 target molecules, with comparable yield and meeting the specifications required by the end-users.

MS20	Demonstration of the fermentation and purification at a 1000 L scale to yield 5 kg of 99 % pure product.		12 - BBEPP	44	5 kg of target product meeting the specifications required by the end-users has been produced.
MS21	DAFIA solutions are validated as suitable packaging solutions		1 - AIMPLAS	46	Full characterized package including water & oxygen barrier and migration tests.

Milestone number[18]	**Milestone title**	**WP number[9]**	**Lead beneficiary**	**Due Date (in months)[17]**	**Means of verification**
MS22	Fireproof polyamides fulfil the fire, thermal and mechanical standards of the inner plastic parts for the automotive industry		11 - MINE	46	Full characterized fire-proof polyamides using suitable FMVSS and DIN standards.
MS23	Polyamide 56 with at least 85% of properties of current market PA56 (with just 45% biobased). New polyamide with adhesion promoters with improved technical properties.		14 - ARKEMA	46	Full characterized polyamides using suitable EN/ISO standards.
MS24	Environmental performance of the new materials and processes according to standards and EU legislation in force.		1 - AIMPLAS	48	Check the environmental regulatory assessments that affect the product
MS25	Final cost of intermediates and products at least 15% cheaper than current biobased market analogues.		9 - BIOTREND	44	Final costs of the products taking into account all process steps.
MS26	The new biobased products and intermediates fulfil the related regulatory requirements.		1 - AIMPLAS	46	Check the regulatory assessments that affect the product.
MS27	Project website running		1 - AIMPLAS	3	Website uploaded content & running

MS28	Advanced exploitation strategy defined		15 - NNFCC	18	Previous to issue the draft PEDR, the exploitable results are completely agreed, as well as co-ownership and preliminary business plans for each result outlined.
MS29	Final Business/ Exploitation & Communication Plans agreed		15 - NNFCC	42	Agreement on final Business and Communication Plans
MS30	Up to 15 peer-reviewed papers and five patents ready to be submitted.		11 - MINE	48	Paper submitted to different journals and patents prepared to be submitted.
MS31	Two workshops successfully organized		11 - MINE	48	Assistance of 50 firms in each one.

Milestone number[18]	**Milestone title**	**WP number**[9]	**Lead beneficiary**	**Due Date (in months)**[17]	**Means of verification**
MS32	WPs risk indicators identified		1 - AIMPLAS	4	Risk indicators for all WPs identified in order to start Risk Management procedures

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Failure to accomplish deadlines and EC procedures	WP1, WP10, WP2, WP3, WP4, WP5, WP6, WP7, WP8, WP9	Risk: Delays Mitigation: Experienced Coordinator that will keep continuously updated about new procedures. Risk Level/Impact = L (Low), M (Medium),
2	IPR/innovation conflicts	WP9	Risk: lack of cooperation, delays Mitigation: a solid Consortium Agreement will be signed before accessing to the G.A, which will include clear rules, Background included/excluded, clear IPR ownership & consensus and conflict resolution mechanisms through the IPREB, external arbitration if necessary. Risk Level/ Impact = L (Low), , H (High).
3	Partners not committed, under-performing, lack capabilities or leaving	WP1, WP10, WP2, WP3, WP4, WP5, WP6, WP7, WP8, WP9	Risk: Work not done, delays Mitigation: The Partners have a strong network of contacts that can replace such a partner and the RO have broad capabilities to fill some temporary gaps. Risk Level/Impact = L (Low), M (Medium),
4	Insufficient cooperation among Partners	WP10	Risk: Work not done, delays Mitigation: frequent meetings to keep partners connected. Strong and proven Organizational structure to ensure workflow.
5	High variability of feedstock (Medium , Likelihood 80%)	WP1	This variability has been taken into account in the strategies proposed in WP1 and mitigated already by collecting feedstock from diverse but related provenances (ex. MSW sourced in specific regions in Spain and MRRM from NUTRIMAR). Additionally, a representative feedstock with a reasonable content of all the fractions required for further upgrading can be selected.
6	Interferences of fractions for the recovery of others (Risk Lever Medium , Likelihood 30%)	WP1	To interact with WP2 to optimize the procedures.
7	Production of toxic compounds for further fermentation as a result of pretreatment (Risk Level Medium, Likelihood 30%)	WP1	To implement detoxify mechanisms, depending on the target compounds
8	Undesired product formation (Risk Level Medium, Likelihood 40%)	WP1	To do all operations with the highest cleaning and sterilization measures. To add industrial antibiotics if required (this is not possible if the objective is bacterial fermentation of the substrate, but for yeast transformation is recommended)

9	Low contents of nucleic acid in unsorted viscera (Risk Level Medium, Likelihood 30 %)	WP1, WP2	Sorting and selection of fractions with a higher content (eg milt or row) will be required

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
10	Insufficient purity of the nucleic acids for application as FR (Risk Level High , Likelihood 30 %)	WP2, WP4	Selection of RRM-fractions with higher nucleic acid content will also make it easier to obtain a high purity. If not sufficient, more expensive purification methods may be required
11	Insufficient amount or purity of extracted gelatine to provide for WP5 (Rosk Level Medium , Likelihood 40%).	WP2	Additional technologies steps such as fractionation and treatments of raw materials will be proposed to improve the amount or the purity of the gelatine extracted.
12	Inability of yeast strains to efficiently utilize all the components in MSW feedstock	WP3	Implement adaptive laboratory evolution on pure underutilized component. If not possible, then propose a process solution, in interface with the DSP development tasks, where a specific medium component can be utilized in other ways than conversion by the used microbe
13	Low titters and/or productivities of the fermentation products (diacids and/or diamine) on MSW (Risk Level Medium , and Likelihood 50%)	WP3	Iteration with WP1 to improve pre-treatment. If not sufficient, optimize the medium further, e.g., by supplementing the MSW feedstock with fractions rich in minerals and vitamins, for example nutrients from fish RRM. Consider other host strains.
14	Strain development is delayed (Risk Level Medium , and Likelihood 25%)	WP3	Coordination with partners working on the fermentation and optimization and downstream processing. Focusing the effort on the most promising product.
15	Developed DSP method does not yield diacid or diamine that meets desired specifications (Risk Level Medium , and Likelihood 25%)	WP3	Additional purification steps can be included to give higher purity. There will be also a close feedback loop with the metabolic engineering strategy to reduce the formation of by-products which would be more costly to remove during the DSP. Interfering medium components coming from the raw materials will be identified and the information fed to WP1 to achieve the production of a raw material stream allowing the facilitated recovery of the target fermentation product.
16	Thermal stability of the extracted and or modified biomacromolecules not suitable for their compounding in PA (Risk Level Medium , and Likelihood 30%)	WP4	Exploitation of other modification reactions for enhancing the thermal stability of the obtained products; possible utilization of micro encapsulation methods of the extracted biomacromolecules in order to increase their thermal stability

17	FR protection provided by the extracted/modified biomacromolecules may not be sufficient (Risk Level Medium , and Likelihood 25%)	WP4	Exploitation of additive or synergistic effects of the extracted/modified biomacromolecules with other FR compounds (Polyphosphates, Phosphinates etc.)
18	Insufficient purity of gelatine provided from WP2 to provide good barrier properties. (Risk Level High , and Likelihood 30%)	WP5	Iteration with WP2 to obtain more suited fraction. Enhancement through addition of bioactive substances.
19	Organoleptically unacceptable taste of the edible coatings. (Risk Level Medium , and Likelihood 20%)	WP5	Gelatine is generally very mild. The selection of the bioactive substances will be made in tune with the foodstuff to be coated or packed respectively. Fish is generally forgiving, making this risk relatively low.
20	Low market acceptability of the use of new biopolymers from rest fraction in food packaging (Risk Level High , and Likelihood 30%)	WP5, WP7	Gelatine is a well-known food additive as well as already approved to be used in food contact materials with FCM substance nº547 (without mention to origin) with no specific migration assigned. In addition, past development of biopolymers from animal by-products at IRIS, like whey protein, could benefit from exemption from being listed in Regulation 10/2011 under the scientific demonstration that the substance is safe (Article 19 says that for substances not listed in Annex I, the compliance with article 3 of the Regulation 1935/2004 is assessed in accordance with internationally recognized scientific principles on risk assessment) as long as it is pure and food grade. NUTRIMAR currently produces feed grade protein concentrates and their facility is being upgraded to comply with food grade standards. As such the partners see no barrier for the future use of the gelatine from NUTRIMAR in the planned edible coating and food packaging applications.
21	Low reactivity between diacids and diamines due to the presence of inhibitors (Risk Level High , and Likelihood 20%).	WP6, WP7	Alternative catalyst routes and processing parameters will be explored, based on the experience of the involved partners. Increase of residence time of reactive extrusion by combine more than one extruder in a tandem configuration. Remove impurities using the experience of BIOTREND & BBEU expertise.

22	Low yield of biopolyamide synthesis (Risk Level Medium , and Likelihood 30%)	WP6, WP7	Alternative catalyst routes and processing parameters will be explored, based on the experience of the involved partners. Increase of residence time of reactive extrusion by combine more than one extruder in a tandem configuration. Remove impurities using the experience of BIOTREND & BBEU expertise.
23	Thermo-mechanical resistance of polyamides is not enough for further processing Reduction in biopolymer properties, bad smell. (Risk Level High , and Likelihood 30 %).	WP6, WP7	Different compatible antioxidants, lubricants and plasticisers will be tested to reduce the action of compounding extruder over the material. Use of low shear screw elements and minimize residence time thanks to the modular configuration of multi screw extruders.
24	Low conversion rate of muconic acid with H2S (Risk	WP6, WP7	Conversion of Maleic acid/ester is well documented. Since both muconic and maleic acid are unsaturated diacids it is expected that

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
	Level High , and Likelihood 40%)		similar catalysts will be efficient. Addition of H2S on other unsaturated monoacids/esters is also documented, included in Arkema's patent applications, with different catalytic systems. So, patented ARKEMA catalysts will be also investigated.
25	Low yield of dinitrile in the conversion of the muconic acid (Risk Level Low , and Likelihood 20%)	WP6, WP7	The major risk is linked with the possible formation of Diels Alders products during the high temperature reaction. This should lead to polyfunctional dinitriles, which however might have valuable properties as polyamines precursors. Instead of a liquid phase process with long residence time at high temperature, a gas phase process with short residence time (seconds) and different catalysts will be tried. Arkema recently patented new catalysts for esters and acids conversion in such a process.
26	Fermentation scale-up from 100 L to 1000 L has lower productivity or titer (Risk Level Low, and Likelihood 30%)	WP7	Optimize fermentation reproducibility at 100 Lscale if applicable and study strain under different conditions to find causes of batch variability at 1-L scale. Altered oxygen transfer, bulk mixing, reactor geometry and CO2 accumulation are typical problems that result in different growth/ production profiles. BBEU's core business is scale-up up to 15.000 L and can anticipate these problems during tech transfer phase.
27	Applied DSP technology (developed at lab-scale) may not yield product within set specifications during pilotscale validation (Risk Level Medium , and Likelihood 20%)	WP7	Different DSP routes are developed in parallel. Additional process steps can be included to increase purity (e.g. recrystallization).

28	Final film properties is not enough to fulfil the selected food package requirements (Risk Level Medium , and Likelihood 20%).	WP7	Additional layers will be proposed and optimized using commercial biodegradable materials. At industrial level up to eleven layers can be used just for one films
29	DAFIA's products might be relatively expensive (Risk Level Medium , and Likelihood 40%)	WP8	To revaluate and improve the production process to minimize energy consume or look for alternative and cheaper energy source. Look for alternative catalysts that improve yield or reduce energy consume.
30	LCA or Energy consume is higher than current process Risk Level H,Likelihood 10%	WP8	To revaluate and improve the production process to minimize energy consume or look for alternative and cheaper energy source. Look for alternative catalysts that improve yield or reduce energy consume.

1.3.7. WT7 Tentative schedule of project reviews

Review number [19]	**Tentative timing**	**Planned venue of review**	**Comments, if any**
RV1	18	Brussels (Tentatively)	
RV2	36	Brussels (Tentatively)	
RV3	48	Brussels (Tentatively)	

Project Number [1]	720770	Project Acronym [2]	DAFIA

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS	AIMPLAS	Spain	1	48
2	POLITECNICO DI TORINO	POLITO	Italy	1	48
3	SINTEF FISKERI OG HAVBRUK AS	SINTEF-FA	Norway	1	48
4	STIFTELSEN SINTEF	SINTEF-MC	Norway	1	48
5	DANMARKS TEKNISKE UNIVERSITET	DTU	Denmark	1	48
6	CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS	CNRS	France	1	48
7	NUTRIMAR AS	NUTRIMAR	Norway	1	48
8	INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL	IRIS	Spain	1	48
9	BIOTREND - INOVACAO E ENGENHARIA EM BIOTECNOLOGIA SA	BIOTREND	Portugal	1	48
10	DAREN LABORATORIES & SCIENTIFIC CONSULTANTS LTD	DLABS	Israel	1	48
11	MINE PLASTIK ENDUSTRIYEL URUNLER VE SERT MADEN KIMYA LIMITED SIRKETI	MINE	Turkey	1	48
12	BIO BASE EUROPE PILOT PLANT VZW	BBEPP	Belgium	1	48
13	BIOPOLIS SL	BIOPOLIS	Spain	1	48
14	ARKEMA FRANCE	ARKEMA	France	1	48
15	The National Non-Food Crops Centre	NNFCC	United Kingdom	1	48

4. MEMBERS OF THE CONSORTIUM

4.1. PARTICIPANTS (APPLICANTS).

PARTNER 1 (Coordinator) **Full name**: AIMPLAS-Asociación de Investigación de Materiales Plásticos y Conexas (AIMPLAS) **Web page**: www.aimplas.es	**Country:** Spain **Type:** RO	AIMPLAS PLASTICS TECHNOLOGY CENTRE

Brief description of the organization:

AIMPLAS, Technological Institute of Plastics located in Valencia, is a private, non-profit Association with more than 500 associated companies created in 1990. AIMPLAS is formed by +120 highly skilled professional, more than 65% with a Masters, Engineering or equivalent degree in Chemistry, polymer engineering, materials engineering or equivalent, including 15 PhD.

AIMPLAS' fields of work are related to technological research and development on thermoplastic and thermosetting plastic materials & products, its transformation processes and their recyclability and sustainability. AIMPLAS generates new knowledge and technologies that can be transferred to companies in order to help them to increase their effectiveness and competitiveness.

AIMPLAS has a broad expertise in the fields of recycling, compatibilization, reactive extrusion, biopolymers and renewable source materials, special assisted processing technologies and nanocomposites, gained during the last years in the frame of EU and national projects. Nowadays, AIMPLAS is involved in more than 25 European projects and has participated in 77 projects in FP5, FP6, FP7, LIFE+, CIP-EcoInnovation EU Programmes, among others, coordinating 27 of them. At National and Regional levels AIMPLAS participates in around 100 projects yearly.

AIMPLAS, as RO is focused to help companies in the plastic sector to develop new products and increase their competitiveness through innovation, has more than 20 pilot plants representing the most relevant polymer/plastics/composites production technologies present in the industry nowadays and has state-of-the art test facilities for chemical, optical, morphological, mechanical and physical characterisation. These pilot lines and laboratories are used by many customer's every year allowing them to test new materials, optimize production processes and launch new products to the market, supported by AIMPLAS technical staff, resulting in more than 5000 assays, 170 technical assessments and 120 skills training actions to more than 1500 clients per year. AIMPLAS has state-of-the-art 8500 m^2 facilities, including thermoplastics and thermoset pilot plants, analysis and testing laboratories (physical-mechanical, chemical, packaging, automotive and construction) and training areas.

Role in the Project: Project coordination (WP10 leader). Production of bio-polyamides from aliphatic acid and 1.5pentanediamine. Develop of flame retardant polyamides compounds using the nucleic acid-flame retardant. LCA studies of the key develop products from MRRM &MSW bio-waste. Contribution to communication, dissemination, exploitation and other related activities.

Scientific team:

Dr. Amador García (M), PhD in Organic Chemistry by the University of Valencia, working in AIMPLAS as researcher in the chemical laboratory since 2009. He has been involved in several regional and FP7 European projects focused in nanotechnology areas, materials from renewable resources and coatings with enhanced properties, etc. (SUPIN, TRAINER, NMP INNOREX, RECAN, KBBE BRIGIT, KBBE BISIGODOS, BIOREFINE-2G). Before joining to AIMPLAS, he worked as researcher assistant in CIPF (Centro de Investigación Principe Felipe) and in the University of Valencia. During 2007 he was in the chemical group of Professor Dennis P. Curran (University of Pittsburgh) as visiting researcher. In 2010 he received the Plastic & Innovation award.

Ms. Raquel Llorens-Chiralt (F), Master in Material Engineering by the Polytechnic University of Valencia, working in AIMPLAS as Senior Researcher since 2005 in the Compounding Department, focused in new material formulations. She has been involved as main researcher in the technical execution of several FP7 Projects focused

on nano-materials, biopolymers, flame retardants, blends, mineral and natural fibres composites and PVC. Before joining AIMPLAS she worked in different private industries (paints, ceramics and metallurgical).

Dr. Ana Espert (F), PhD in Polymer Technology by the Royal Institute of Technology (KTH) in Stockholm (Sweden). Since 2005, she is working as a Senior Researcher in AIMPLAS. In this position, she is in charge of the technical coordination of several of the Projects in which AIMPLAS participates (+75 Projects since 2000, 36 of them FP7 in 18 as Coordinator). Dr Espert has been the technical coordinator of several FP7 Projects focused in biotechnologies, materials from renewable resources and sustainability (Collective FLOWFREE, R4SMEs ECOBIONET, NMP BUGWORKERS) and is participating in others such as NMP ECOPLAST and KBBE BRIGIT. Before joining AIMPLAS, she worked at the Royal Institute of Technology, where she was also involved in EU projects Technical and Administrative Management of FP6 projects (ECOSITES, PRONACOM). She is the main author of more than 10 peer-reviewed articles in Scientific Journals and Conference proceeding in the field of Polymer Technology

Mr. Santiago Gálvez (M). Degree in Agricultural Engineering by the Polytechnic University of Valencia, and working for AIMPLAS as Project Manager in the International Projects Department since 2010. Mr Gálvez has participated in many EU projects focused in biotechnology, materials from renewable resources and sustainability; expertise in plastic materials and plastic processing, and in management of EU funded projects. Involved in the project coordination and/or management activities of some FP7 (R4SME THERMALCOND, R4SME REFRESH, DEMO XXL-REFRESH, NMP COALINE, R4SME BIOFIBROCAR, KBBE FIBRAGEN, JTI WAVECOM), CIP-EcoInnovation (CAYLEY), LIFE+ (ECOFLEXOBAG, BREAD4PLA, PISA), Leonardo (ENERGYWISE PLASTICS, iPOLYCOND)and Interreg IVB (CARBONINSPIRED, CARBONINSPIRED 2.0) projects and in the preparation of some others.

Publications:

- Signori, F., Boggioni, A., Righetti, M. C., Rondán, C. E., Bronco, S. and Ciardelli, F. (2015), Evidences of Transesterification, Chain Branching and Cross-Linking in a Biopolyester Commercial Blend upon Reaction with Dicumyl Peroxide in the Melt. Macromol. Mater. Eng., 300: 153–160.
- Campanella, A., Wool, R. P., Bah, M., Fita, S. and Abuobaid, A. (2013), Composites from northern red oak (*Quercus robur*) leaves and plant oil-based resins. J. Appl. Polym. Sci., 127: 18–26. doi: 10.1002/app.36663.
- R.M. González, M.P. Villanueva19 – Starch-based polymers for food packaging, Multifunctional and Nanoreinforced Polymers for Food Packaging,2011, Pages 527–570
- Coltelli, M.-B., Maggiore, I. D., Bertoldo, M., Signori, F., Bronco, S. and Ciardelli, F. (2008), Poly(lactic acid) properties as a consequence of poly(butylene adipate-*co*-terephthalate) blending and acetyl tributyl citrate plasticization. J. Appl. Polym. Sci., 110: 1250–1262. doi: 10.1002/app.28512
- Martí-Ferrer, F., Vilaplana, F., Ribes-Greus, A., Benedito-Borrás, A. and Sanz-Box, C. (2006), Flour rice husk as filler in block copolymer polypropylene: Effect of different coupling agents. J. Appl. Polym. Sci., 99: 1823–1831
- Fang Jm, Fowler Pa, Escrig C, Gonzalez R, Costa Ja, Chamudis L: Development of Biodegradable Laminate Films Derived From Naturally Occurring Carbohydrate Polymers. *Carbohydrate Polymers* 2005, 60:39-42.

Projects:

Project acronym/description.	Scope	End Year	Website
BUGWORKERS: New tailor-made PHB-based nanocomposites for high performance applications produced from environmentally friendly production routes.	FP7-NMP - 2009-SME	2013	www.bugworkersproject.eu/.

BRIGIT: New tailor-made biopolymers produced from

FP7-KBBE- www.brigitsugars obtained from lignocellulosic residues for flame 2017 SME project.eu/ retardancy highly demanding applications.

INNOREX: Realising the continuous, highly precise, metal- FP7-NMPfree polymerisation of PLA using alternative energies for an 2016 www.innorex.eu/ 2012 in novative reactive extrusion.

BIOREFINE 2G : Development of 2nd Generation FP7-KBBE Biorefineries Production of Dicarboxylic Acids and Bio- 2017 www.biorefine2g.eu/ 2013 ba sed Polymers Derived Thereof

BISIGODOS: High value-added chemicals and BIoreSIns FP7 from alGae biorefineries produced from CO2 provided by 2017 www.bisigodos.eu industrial emissions. KBBE 2013

Technical equipment:

- Equipment for mechanical, thermal, rheological and physical tests accredited to perform plastic characterization using the EN/ISO standards
- Rheometer, Optical and electronic Microscopy, DMA, DSC, Thermogravimetry. Raman Spectrometer.
- Twin-screw co-rotating extruders with gravimetric feeders (to feed solid and liquid additives) and other multiscrew planetary extruder.
- Thermoplastic injection machines.
- Multilayer blown film co-extrusion line and a pilot plant lamination unit.

PARTNER 2

Country: Italy **Full name**: Politecnico di Torino (POLITO).

Type: HE **Web page**: www.polito.it

Brief description of the organization: Politecnico is a "Research University" acknowledged as a centre of excellence in all over the world for the qualifying research areas of Architecture and Engineering. Politecnico has participated and participates at the highest levels of international scientific research, including more than 210 projects in the 7th Framework Programme, with a total EU contribution of nearly 62 million Euros. The development of the fields of excellence, the investment on frontier subjects, and also the support to the curiosity driven research allow a virtuous mix of basis and applied research.

Within POLITO, the Department of Applied Sciences and Technology (DISAT) promotes the study of fundamental sciences and their technological applications to engineering as well as other emerging areas within the interdisciplinary sciences. DISAT staff counts on more than 120 faculties, 200 research assistants and collaborators, in the area of materials engineering, physics and chemical engineering. Lab facilities of DISAT are located either in the main campus in Torino, or in a second campus in Alessandria, in the eastern part of the region, where the polymer processing and characterization labs are installed. The laboratories involved in this project are mainly related to the characterization of the extracted and/or modified biomacromolecules and to the preparation of blends (at a lab scale) and their overall characterization.

Role in the Project: POLITO will be involved in the characterization of the extracted biomacromolecules and modified products. More specifically, POLITO will perform morphological tests (SEM-EDS), thermogravimetric analyses, LOI, cone calorimetry and UL94 tests. Furthermore, POLITO will be also involved in WP2 - Task 2.3

Isolation and purification of nucleic acids, for which POLITO will help in setting up a low impact extraction method to be used at large scale and will assess the molecular size and purity of the obtained extracted products. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Giulio Malucelli (M) got his Master Degree in Chemical Engineering in 1992. He received his Ph.D. in Chemistry in 1996. From 2003 he is Associate Professor of Materials Science and Technology at the Department of Applied Science and Technology of Politecnico di Torino, Italy. He has co-authored more than 170 peerreviewed papers; his main current research interests include the synthesis and characterization of nanocomposite structures (hybrid organic–inorganic coatings and polymeric nanocomposites), as well as the modification of plastics and textiles for improving their flame retardant features.

Dr. Francesca Bosco (F) got her Master Degree in Biological Science in 1987. She received her Ph.D. in Biology and biotechnology of fungi in 1995. She is Assistant Professor at the Department of Applied Science and Technology of Politecnico di Torino, Italy. She has co-authored more than 35 peer-reviewed papers. Her main research interests concern the production of microorganisms and biomolecules of industrial interest, valorisation processes of wastes and industrial by-products in the production of biomolecules.

Dr. Chiara Mollea (F) got her Master Degree in Biological Science in 1998. She received her Ph.D. in Biology and Biotecnology of Fungi in 2002. She is Laboratory Technician at the Department of Applied Science and Technology of Politecnico di Torino, Italy. Her main research interests concern the extraction of biomacromolecules from food industry wastes for their valorisation as well as the extraction of aromatic molecules for the improvement of food products.

Five relevant publications, patents or products:

- F. Bosco, A. Casale, C. Mollea, M.E. Terlizzi, G. Gribaudo, J. Alongi, DNA coatings on cotton fabrics: effect of molecular size and pH on flame retardancy. In: Surface & Coatings Technology, 272, 86-95 (2015)
- Alongi J; Cuttica F.; Bourbigot S.; Malucelli G., Thermal And Flame Retardant Properties of Ethylene Vinyl Acetate Copolymers Containing Deoxyribose Nucleic Acid or Ammonium Polyphosphate. In: Journal Of Thermal Analysis & Calorimetry, 122, 705-715 (2015)
- Giulio Malucelli, Francesca Bosco, Jenny Alongi, Federico Carosio, Alessandro Di Blasio, Chiara Mollea, Fabio Cuttica, Annalisa Casale, Biomacromolecules as novel green flame retardant systems for textiles: an overview. In: RSC Advances, 4 n. 86, 46024-46039 (2014)
- Alongi J.; Di Blasio A.; Cuttica F.; Carosio F; Malucelli G., Bulk or surface treatments of ethylene vinyl acetate copolymers with DNA: investigation on the flame retardant properties. In: European Polymer Journal, 51, 112-119 (2014)
- Alongi, Jenny; Carletto, Riccardo Andrea; Bosco, Francesca; Carosio, Federico; Di Blasio, Alessandro; Cuttica, Fabio; Antonucci, Vincenza; Giordano, Michele; Malucelli, Giulio, Caseins and hydrophobins as novel green flame retardants for cotton fabrics. In: Polymer Degradation & Stability, 99, 111-117 (2014).
- Alongi, Jenny; Carletto, Riccardo Andrea; Di Blasio, Alessandro; Cuttica, Fabio; Carosio, Federico; Bosco, Francesca; Malucelli, Giulio, Intrinsic intumescent-like flame retardant properties of DNA-treated cotton fabrics. In: Carbohydrate Polymers, 96, 296-304 (2013).

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
"HEFEST – Smart flame retardant coatings based on intumescent nanocomposites"	FP7 SME	2011	http://www.hefest.info

"FRONT - Flame Retardant ON Textile"	FP7 SME	2011	
"BIOMOLENER – Recovery of waste from agro-livestock and agro-industrial activities for high value biomolecules and biofuels"	National Italian Project	2015	http://rps.entecra.it/project_BIOMOLENER_79

Technical equipment:

- Polymer processing facilities, including compounding of thermoplastics by twin screw extrusion (different scales, from 15 g to tens of kg), injection molding (starting at 50 g, min batch), compression moulding.
- Characterization methods including thermal analysis (TGA, DSC, laser flash for thermal diffusivity, thermal conductivity), rheology, mechanical and dynamic-mechanical properties, HPLC, FTIR, UV-VIS, SEM, FE-SEM, flammability (UL94, LOI) and forced combustion (Cone calorimeter) tests and SDS-page and agarose gel electrophoresis

PARTNER 3

Full name: SINTEF Fiskeri og Havbruk AS (SFA).

Country: Norway
Type: RO

Web page: www.sintef.no/fish

Brief description of the organization: As one of the largest contract R&D organizations in Europe, the SINTEF Group has a long and successful track record both as participant and as coordinator in Norwegian and European projects. Annually, the SINTEF Group supports the development of approximately 2000 Norwegian and overseas companies and the group employs about 2150 persons. For instance, SINTEF Group has participated in 116 FP7projects and coordinating 33 of them. In Horizon2020, SINTEF Group is participating in 31 projects so far. The SINTEF Group consist of the SINTEF Foundation and four subsidiary research institutes, of which SINTEF Fisheries and Aquaculture is one.

SINTEF Fisheries and Aquaculture (SFA) represents technological expertise and industry knowledge in the utilization of renewable marine resources. Under the vision "Technology for a better society" we are working for a knowledge-based bio-marine industry. Our goal is to meet market demands for technological research and development on renewable marine resources. Our expertise, interdisciplinary approach and industrial knowledge provide our customers with added value in all the projects we undertake. SFA performs basic and applied research for commercial customers as well as governmental institutions. SFA with more than 120 employees is located in Trondheim at SINTEF SeaLab with excellent facilities for applied research. In these new facilities we have a range of different laboratories, direct supply of seawater and mobile oil- hydrolysates pilot plant. These excellent working facilities, our experienced and highly trained researchers and our partnership and close cooperation with the other SFA departments and the Norwegian University of Science and Technology (also located in Trondheim) provide a sound base for both basic and applied research – from very specialized scientific fields to multidisciplinary and applied solutions for our clients.
The Process Technology group delivers industrial solutions for the marine value chain, focusing on the interactions between raw materials, processes and products. Researchers have for more than 15 years been working with the technology development for better utilisation of marine raw materials. One of the focus areas is marine oils and hydrolysis of marine proteins. SFA has collaborated with companies producing marine oils in a number of projects to solve different industrial and technological challenges.
Role in the Project: SFA will lead WP2 and will be responsible for work related to utilization of fish rest raw materials with emphasis on processes for extraction of gelatine. Residual raw material remaining after extraction

of gelatine and nucleic acids will be characterised in order to define the possible use of valuable marine proteins/hydrolysates and marine oil. Much work will also be connected to determination of chemical composition, microbiological quality and technological properties of the residual fractions in relation to their use as food/feed/fertiliser products. SFA by integration of different technological solutions will upgrade the obtained products by increasing the yield and quality protein and lipid fractions. Methods for preservation and

stabilisation of residual fractions as well as concentration and removal of undesired components will be evaluated. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Rasa Slizyte (F) is a biotechnologist with the main focus on utilization of marine rest raw materials with emphasis on use of hydrolysis technology and extraction and separation of proteins. Her PhD work is titled “Hydrolysis of cod (Gadus morhua) by-products: Influence of raw material composition and process conditions" and was defended at Norwegian University of Science and Technology (NTNU) in 2004. She was experience in leading international and industrial projects, has published 18 peer-reviewed papers and is co- author of several book chapters.

Dr. Ana Carvajal (F) a research manager at SINTEF Fisheries and Aquaculture. She is responsible for the research group working raw material quality and processing and utilization of rest raw material and new sources for food and feed. Her PhD thesis ‘Utilization of Norwegian spring-spawning herring by- products for human consumption’ was defended in Trondheim in 2013 at the Norwegian University of Science and Technology. From 2007 and up till now her main research activities have been within the field of utilization and valorization of rest raw material with focus on production and quality of oil and protein for food and feed. She has been project manager for several research projects and worked closely together with the Norwegian industry. She is author and co- author of several peer-reviewed publication, book chapters and several SINTEF reports.

Dr. Revilija Mozuraityte (F). She received PhD in Biotechnology in 2007 from NTNU, Trondheim, Norway. Since 2007, she is a researcher with main expertise in marine lipids, antioxidants and prooxidants: extraction technologies, characterisation methods, and stabilisation. Revilija Mozuraityte is involved in the project activities related to processing and utilisation of marine oils with the focus on extraction of lipids and antioxidative phenolic compounds.

Dr. Inger Beate Standal (F) received her PhD in Biotechnology in 2009 from NTNU, Trondheim, Norway. Her main fields of expertise are lipid technology, Nuclear magnetic resonance (NMR) spectroscopy, metabonomics, compositional and quality analyses of marine raw materials and products. She is involved in activities related to processing of marine raw materials and characterization of raw material and extracts/products obtained by HR NMR.

Five relevant publications, patents or products:

- Rasa Šližytė, Revilija Mozuraitytė, Oscar Martínez-Alvarez, Eva Falch, Martine Fouchereau-Peron and Turid Rustad, (2009), Functional, bioactive and antioxidative properties of hydrolysates obtained from cod (Gadus morhua) backbones, Process Biochemistry, 44, 668–677.
- Turid Rustad, Ivar Storrø & Rasa Slizyte (2011), Possibilities for the utilisation of marine by-products, Invited review, International Journal of Food Science and Technology, 46, 2001-2014, Process Biochemistry, 44, 668–677.
- Rasa Šližytė, Ana Karina Carvajal, Revilija Mozuraityte, Marit Aursand and Ivar Storrø, (2014), Nutritionally rich marine proteins from fresh herring by-products for human consumption, Process Biochemistry, 49 (7), 1205–1215.
- Margareth Opheim, Rasa Šližytė, Hallgeir Sterten, Fiona Provan, Eivind Larssen, Nils Petter Kjos, (2015), Hydrolysis of Atlantic salmon (Salmo salar) rest raw materials – Effect of raw material and processing on

composition, nutritional value, and potential bioactive peptides in the hydrolysates, Process Biochemistry, Volume 50, Issue 8, 1247–1257.

- Mozuraityte R., Kristinova V., Rustad T. (2016) Oxidation of Food Components. In: Caballero, B., Finglas, P., and Toldrá, F. (eds.) The Encyclopedia of Food and Health vol. 4, pp. 186-190. Oxford: Academic Press.

Five Relevant Projects:

Project acronym/description.	**Scope**	**End Year**	**Website**
Maximal resource utilisation superior utilisation of fish byg/Publications/maximumproducts. Financed by the Nordisk Innovation Center (Nr. NICe 2009 resource-utilisation-value04252). added-fish-by-products/			www.nordicinnovation.or
APROPOS *(Added value from high oil and high protein industrial co-streams KBBE*	*FP7*	2014	www.euapropos.eu/
CYCLE (Total utilization of raw materials in the supply chain for food with a bio-economical perspective), NRC 2016 http://cycleweb.no/ financed by the Norwegian Research Council).			
Processing of macroalgae for food and feed (**PROMAC**), financed by The Research Council of Norway. Objective: Utilization of harvested and cultivated macroalgae in production of ingredients for food and feed.	NRC	2018	http://promac.no/
Technological solutions for production of safe and high quality proteins from salmon rest raw materials for dietary applications in poultry and pet food, financed by the Norwegian Research Council (No. 256439).	NRC	2018	No available

Technical equipment:

SFA's facilities can be used for lab scale tests, industrial research and process optimisation. SFA was selected by the Research Council of Norway to be the national laboratory for marine oils with laboratory facilities for research, education and process optimisation. SFA has designed and constructed both a hydrolysis-protein and a lipid platform. The creation of these platforms has resulted in a mobile pilot plant for production of protein hydrolysates and lipids. All this allows SFA to perform both biotechnological processing studies for finding the optimal and sustainable technologies for valorisation of marine rest raw materials into marketable products.

PARTNER 4

Full name: Stiftelsen SINTEF (SMC). **Country:** Norway **Type:** RO

Web page: www.sintef.no

Brief description of the organization: SINTEF Materials and Chemistry belongs to the SINTEF Foundation ("Stiftelsen SINTEF"), which is part of the SINTEF Group ("SINTEF"). Being the largest non-profit independent research organization in Scandinavia and one of the largest in Europe, SINTEF performs contract R&D for a wide range of both Norwegian and international industry customers and has more than 2100 employees, see Partner 3, SINTEF Fisheries and Aquaculture.

SINTEF Materials and Chemistry (SMC) is the largest SINTEF-institute, with 450 employees and expertise in materials- and nano-technology, oil and gas, applied and analytical chemistry, chemical engineering, **biotechnology** and environmental sciences. SMC has a strong track record in participating and managing EU projects, coordinating 19 projects in FP7 and the related programmes.

The Biotechnology and Nanomedicine department at SMC has 60 employees in three research groups (Biotechnology, Mass Spectrometry and Polymer Particles and Surface Chemistry). Key expertise and technologies of relevance for DAFIA include microbial strain improvement by metabolic engineering/ synthetic biology, development, optimization and scale-up of microbial fermentation processes, biomass processing, and isolation and purification of biological compounds. Furthermore, the biotechnology group has expertise in

enzyme development, the discovery of anti-microbial and anti-cancer compounds, recombinant protein expression, functional metagenomics and more. The biobased economy and biorefineries in general, and more specifically, the conversion of higher generation biomass feedstocks, including waste streams, into value-added products using microbial cell factories, are prioritized research areas at SMC.

Role in the Project: SMC will 1) contribute to the design of MSW and MRRM pre-processing strategies critical for the subsequent application as fermentation substrates and source of nucleic acids, 2) Take a leading role in extraction of nucleic acids from MRRM, and 3) develop microbial cell factories for conversion of MSW and MRRM substrates into the diamine, and evaluate selected strains during controlled conditions in lab-scale fermentations. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Ms Inga Marie Aasen (F) is a Senior Research Scientist in the Biotechnology group. She has a broad experience in biochemical engineering, fermentation and bioprocess development, downstream processing and technoeconomic analyses. She is manager of several projects related to biomass processing, biorefineries and fermentation process development. Current projects include utilisation of seaweed as an alternative carbon source for biofuel production and as a source of valuable chemicals, utilisation of MRRM as fermentation substrates, and microbial production of omega3 fatty acids.

Dr. Alexander Wentzel (M) is a Senior Research Scientist in the Biotechnology group. He has expertise in microbial strain development for production of fuels and chemicals from biomass, metagenome-based discovery of enzymes, and directed molecular evolution of peptides and proteins.

Mr Ingemar Nærdal (M) is a Research Scientist in the Biotechnology group and holds a Master of Science degree in biotechnology from the Norwegian University of Science and Technology (2007). He has broad experience within molecular biology and microbiology and is specialized in metabolic engineering of microorganisms, recombinant protein expression, cultivation of microorganisms at controlled conditions in fermentors and isolation and purification of biomacromolecules, including proteins and nucleic acids. Nærdal is author and coauthor of totally four peer-reviewed scientific publications and several confidential SINTEF reports.

Five relevant publications, patents or products:

- Nærdal I, Pfeifenschneider J, Brautaset T, Wendisch VF. (2015). Methanol-based cadaverine production by genetically engineered *Bacillus methanolicus* strains. *Microb Biotechnol.* 8:342-350.
- Nærdal I, Netzer R, Ellingsen TE, Brautaset T. (2011). Analysis and manipulation of aspartate pathway genes for L-lysine overproduction from methanol by *Bacillus methanolicus*. *Appl Environ Microbiol.* 77:6020-6026.
- Aasen IM, Ertesvåg H, Heggeset TMB, Liu B, Brautaset T, Vadstein O, Ellingsen TE (2016) Thraustochytrids as production organisms for docosahexaenoic acid (DHA), squalene and carotenoids. Appl. Microbiol. Biotechnol. In press
- Jakobsen AN, Aasen IM, Josefsen KD, Strøm AR (2008) Accumulation of docosahexaenoic acid-rich lipid in thraustochytrid Aurantiochytrium sp. strain T66: effects of N and P starvation and O_2 limitation. Appl. Microbiol. Biotechnol., 80: 297-306
- Horn SJ, Aasen IM, Østgaard K (2000) Ethanol production from seaweed extract. J. Ind. Microbiol. Biotechnol., 25: 249-254

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website

PROMYSE/Products from methanol by synthetic cell factories.	FP7 KBBE	2014	https://www.sintef.no/projectweb/promyse/

MetApp/Systems biology of bacterial methylotrophy for products from methanol — FP7 KBBE ERA-NET — 2017 — http://www.sintef.no/projectweb/metapp/ — biotechnological

ThermoFactories/Thermophilic cell factories for efficient conversion of brown algae biomass to high-value chemicals — H2020 ERA-NET — 2018 — http://www.marinebiotech.eu/thermofactories

Promac/ Processing of macroalgae for food and feed, — National funded — 2019 — http://promac.no/

AERTOs Biobased Economy/ Development of processes for modern biorefineries and the extraction of algae. — ERA / Own funding — 2016 — http://www.aertos.eu/ — lignocellulosic biomass

Mar3Bio/ Biorefinery and biotechnological exploitation of marine biomasses — H2020 ERA-NET — 2019 — http://www.marinebiotech.eu/mar3bio

Technical equipment:

Relevant infrastructure for SMC's contribution to DAFIA includes;

- State-of-the-art molecular biology laboratories, including sequencing facilities
- Fermentation laboratories (32x1-3L fermenters, full process control, used for process development), 1x 50L fermenter (up-scaling, if requested), pilot fermenter (200 L).
- Downstream processing equipment from laboratory to pilot scale, including centrifuges, membrane filtration, high-pressure homogenizers and chromatography.
- High-throughput screening laboratory, to be applied for screening of growth substrates, evaluation of mutants, sample preparation and assays.
- Laboratories for routine HPLC analysis and a laboratory for mass spectrometry, including LC-MS-QTOF, LCMS/MS-QQQ, FFF/ICP-MS/MS and Ft-ICR.

PARTNER 5 **Full name**: Danmarks Tekniske Universitet (DTU). **Web page**: www.dtu.dk	**Country:** Denmark **Type:** SME	DTU Technical University of Denmark

Brief description of the organization:

Danmarks Tekniske Universitet (DTU) is recognized internationally as a leading university in the areas of technology and the natural sciences. DTU is known for its business-oriented approach, its focus on sustainability and its appealing study environment. In 2012, DTU had a total of 5227 employees and 7843 students, of which 1136 were Master students. Among Master students, 412 were international.

The Novo Nordisk Foundation Center for Biosustainability (CfB) is a research and innovation center at DTU focusing on products that can be produced using bacteria, yeasts, and mammalian cells. CfB has about two thirds of its approx. 240 employees at the Center premises in Hørsholm, and the rest at related universities in Denmark, the US, South Korea and Sweden. CfB is organized as a hybrid research institute with eight research sections that perform basic research and one application-oriented research section. The latter includes a team that gives priority to commercial considerations and addresses the potential for spin-outs from CfB. CfB offers excellent opportunities for research programmes in collaboration with leading universities and industrial partners in the development of cell factories.

Role in the Project: DTU will have the primary responsibility for development of microbial cell factories for conversion of waste streams into dicarboxylic acids via fermentation (WP3). Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Irina Borodina (F), Chemical Engineer (2001) from Kaunas University of Technology and PhD in Biotechnology (2007) from the Technical University of Denmark. Dr. Borodina is Senior Researcher and Group Leader at the Novo Nordisk Foundation Center for Biosustainability, DTU. Her research group focuses on metabolic engineering of yeast cell factories for biosustainable production of chemicals from renewable feedstocks (link to Yeast Metabolic Engineering Group). Dr. Borodina's group currently includes 9 employees: 2 Research Scientists, 2 PostDocs, 2 PhD students, 2 Research Assistants, 1 Lab Technician and a business development manager, all financed by national and international grants. Additionally the group includes several Danish and international master students and hosts international guest researchers on a regular basis. Dr. Borodina is the coordinator of the EU FP7 project BioREFINE-2G (www.biorefine2g.eu). She has published 24 peer-reviewed articles, which have been cited over 700 times, and she is co-inventor of 6 patent applications.

Mrs. Birte Kastrup Rasmussen (F), Senior Advisor. She provides legal and administrative support for the externally financed projects at the Center for Biosustainability. Mrs. Rasmussen will be the contact concerning legal and financial issues.

Five relevant publications, patents or products:

- **Borodina I** & Nielsen J (2014) "Advances in metabolic engineering of yeast *Saccharomyces cerevisiae* for production of chemicals". *Biotechnol J,* 9(5):609-620. doi: 10.1002/biot.201300445.
- Kildegaard KR, Hallström BM, Blicher TH, Sonnenschein N, Jensen NB, Sherstyk S, Harrison SJ, Maury J, Herrgård MJ, Juncker AS, Forster J, Nielsen J, & **Borodina I** (2014) "Evolution reveals a cyclic glutathionedependent mechanism of 3-hydroxypropionic acid tolerance". *Metab Eng,* 26C:57-66. doi: 10.1016/j.ymben.2014.09.004.
- **Borodina I**, Kildegaard KR, Jensen NB, Blicher TH, Maury J, Sherstyk S, Schneider K, Lamosa P, Herrgård MJ, Rosenstand I, Öberg F, Forster J, & Nielsen J (2015) "Establishing a synthetic pathway for high-level production of 3-hydroxypropionic acid in *Saccharomyces cerevisiae* via -alanine". *Metab Eng,* 27:57-64. doi: 10.1016/j.ymben.2014.10.003.
- Kildegaard KR, Wang Z, Chen Y, Nielsen J, **Borodina I** (2015) "Production of 3-hydroxypropionic acid from glucose and xylose by metabolically engineered *Saccharomyces cerevisiae*". *Metab Eng Commun*, 2:132-136. doi: 10.1016/j.meteno.2015.10.001.
- Stovicek V, Borja GM, Forster J, & **Borodina I** (2015) "EasyClone 2.0: Expanded toolkit of integrative vectors for stable gene expression in industrial *Saccharomyces cerevisiae* strains". *J Ind Microbiol Biotechnol,* 42(11):1519-1531. doi: 10.1007/s10295-015-1684-8.

Five Relevant Projects:

Project acronym/description	Scope	End Year	Website
BioREFINE-2G: Development of 2nd Generation Biorefineries Production of Dicarboxylic Acids and Biobased Polymers Derived Thereof	FP7-KBBE 2013	2017	www.biorefine2g.eu/
LipoFINE: *Yarrowia lipolytica* as a Platform Cell Factory for Fine Chemicals	NNF – national grant	2019	
BioProtect: Cell Factories for Production of Biocides by Fermentation	Pre-seed grant - Novo Seeds A/S	2018	

Technical equipment:

State-of-the-art facilities for molecular biology, fermentation, high -throughput screening, next generation sequencing, advanced analytics laboratory (HPLCs, GC-MSs, LC-MSs, MALDI-TOF, Orbitrap Fusion). Robotics for adaptive laboratory evolution and for strain construction and fermentation.

PARTNER 6 **Full name**: Institut de recherches sur la catalyse et l'environnement de Lyon (CNRS) **Web page**: http://www.CNRS.univ-lyon1.fr/	**Country:** France **Type:** HE	

Brief description of the organization: The Centre National de la Recherche Scientifique (**CNRS**) (National Centre for Scientific Research) is a Government -funded research organization under the administrative authority of French Ministry in charge of research. As the largest fundamental research organization in Europe, CNRS is involved in all fields of knowledge. Interdisciplinary programs and actions offer a gateway into new domains of scientific investigation and enable CNRS to address the needs of society and industry. CNRS is organized in 1121 research units spread throughout France. These units are either intramural or in partnership with universities, other research organizations, or industry.
L'Institut de Recherches sur la Catalyse et l'Environnement de Lyon **(RCELYON**) is the largest CNRS department devoted for fundamental and applied research in Catalysis and Environment in France. It offers the Top World infrastructure and state of the art facilities as well as cutting-edge for the characterization of solid catalysts.
Role in the Project: CNRS will be involved in the heterogeneous catalytic sulphuration of extracted muconic acid or its esters. More specifically CNRS will design and optimize efficient catalyst for the sulphuration reaction. The reaction will first be conducted in batch processing and then the shift of batch process to a continuous "assembly line" chemical process will be studied. The purification of the sulphurated product will be taken up for the delivery of a first lab scale batch. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Jean-Marc M. MILLET (M), PhD in inorganic chemistry (1983), PhD in catalysis (1990), coordinator of the project is senior researcher at CNRS. He was project leader of the group "Refinery and innovative valorization of hydrocarbons" at IRCELYON and in charge of activities concerning the development of new catalysts for alkanes oxidation and valorization of biomass for the synthesis of basic chemicals. He has been in charge of numerous industrial contracts and ANR projects in the field over the past fifteen years. He authored more than 150 papers published in international journals and is co-inventor of 14 patents on light alkanes oxidation and biomass valorization.

Dr. Gilles BERHAULT (M) is CNRS researcher at IRCELYON, University of Lyon since 2005. He got his diploma of "Grandes Ecoles chemical engineer" in 1992 from CPE. He received his Ph.D. in Physical Chemistry in 1997 working on acid-base properties of sulfide catalysts. After a post-doctorate at the University of Texas with former Exxon researcher R.R. Chianelli, he joined CNRS in 2000 as Chargé de Recherches, first at the University of Poitiers then at IRCELYON. He has co-authored 90 peer-reviewed papers and 4 book chapters; his main current research interests include the activation (sulfidation, dopants) and characterization of sulfide catalysts, the use of phosphide catalysts for biomass upgrading applications and the synthesis of shape- and composition-control (bi)metallic and oxide nanoparticles for catalytic and photocatalytic applications.

Dr. Dorothée LAURENTI (F) got is PhD in Organic Chemistry in 1999. She is CNRS researcher at IRCELYON from the end of 2002. She has co-authored more than 40 papers in various fields as organic synthesis, homogeneous catalysis, heterogeneous catalysis, hydrotreating, biomass conversion. Her main research interest is now

directed to catalysis for lignocellulosic or algal biomass hydroconversion and characterization tools for complex mixtures. She is the leader of Lignarocat project funded by ANR on lignin catalytic hydroconversion to aromatics.

Dr. Pavel AFANASIEV (M) is senior researcher at IRCELYON, CNRS. Master degree in physical chemistry (1986) PhD. in catalysis (1989). Author of more than 150 peer-reviewed publications. Since nomination as CNRS staff in 1994 he coordinated many projects with industry and academia. The main axis of research is synthesis of catalytic materials including nanoparticles of oxides, sulfides, nitrides and other inorganics. In parallel to the materials synthesis, PA is actively involved in the application of advanced physico-chemical methods of materials characterization such as in situ X-ray absorption and emission, as well as DFT calculations.

Dr. Françoise Bosselet (**F**) obtained her PhD in chemistry at the University of Saint-Etienne, France (1984) on the Application of DSC to study diesel engine fuels and their improvement by doping. She integrated the CNRS in 1983 and worked into two teams: on the characterization of zeolites and on the study of reinforcement-matrix interactions in metal matrix composites, ceramic or intermetallic. Since 2008, she is engineer at IRCELYON and at the head of XRD Service. She is co-author of near than 70 publications, more than 60 communications with proceedings and one patent.

Five relevant publications, patents or products:

- Method for trapping sulphurous impurities using specific trapping masses Girard, V.; Bazar-Bachi, D.; Chiche, D.; Baudot, A.; Geantet, C. Eur. Pat. Appl. (2013), EP 2591842 A1 20130515
- Schoumacker, K.; Cattenot, M.; Geantet, C.; Puzenat, E.; Lacroix, M.; Herrmann, J. M., Catal. Comm.(2010), 11(14), 1116-1119.
- Catalytic process for manufacturing of mercaptans by hydrogen sulfide-cleavage of thioether in the presence of H2 Fremy, Georges; Essayem, Nadine; Lacroix, Michel; Zausa, Elodie, Fr. (2004), FR 2844726 A1 20040326. |
- Characterization of acid-base paired sites on silica-supported RuS2 by infrared spectroscopy and methyl mercaptan condensation reaction Berhault, Gilles; Lacroix, Michel; Breysse, Michele; Mauge, Francoise; Lavalley, Jean-Claude; Qu, Lianglong J. Catal. (1997), 170(1), 37-45.
- Lignin hydroconversion on MoS2-based supported catalyst: Comprehensive analysis of products and reaction scheme Joffres, B.; Nguyen, M. T.; Laurenti, D.; Lorentz, C.; Souchon, V.; Charon, N.; Daudin, A.; Quignard, A.; Geantet, C. Appl. Catal., B: Environmental (2016), 184, 153-162.

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website	
BIOCOUP: Co-processing of upgraded bio-Liquids in refinery units –co processing with HDT	FP6	2011	www.biocoup.com	standard

Technical equipment:

The team has access to all the technical equipment from IRCELYON (http://www.ircelyon.univ-lyon1.fr/) with access in particular to various characterization techniques (TEM, SEM, XRD, NMR, XPS, Raman and Mössbauer spectrometer, Calorimeter, and has specific technical equipment by itself: zeta-meter, ⍰⍰GS -MS, DTA-DTG analyser, TPD CHSN analyser, IRTF spectrometer, DRIFT spectrometer, Raman spectrometer, UV-Vis

spectrometer, densitometer, 2D chromatograph, and several devices to test catalysts (7 atmospheric pressure testing apparatus, 4 high pressure testing apparatus, 6 batch testing apparatus).

PARTNER 7 **Full name**: Nutrimar AS (NUTRIMAR).	**Country:** Norway **Type:** SME	nutrimar

Web page: http://nutrimar.no/

Brief description of the organization: Nutrimar is a Norwegian bio marine company that produces high quality oil, protein hydrolysate and meal from salmon. Nutrimar was established in 2007 next door to InnovoMar facility of SalMar AS at Frøya. The purpose of creation was for better utilization of the growing amount of residual materials from SalMar and surrounding slaughterhouses and processing of salmon. Facilities at Nutrimar are built to handle to have higher level of processing than traditional silage. Separate fractions of raw materials from InnovoMar is received is separate lines to give a possibility of recipe for unique combinations in the final product. Nutrimar is expanding to a new food grade production plant with vision to produce novel products from rest raw materials. Nutrimar aims to utilize the rest raw material from the salmon farming industry to ensure the optimal use of the natural resource.

Role in the Project: NUTRIMAR will contribute its knowledge to describe and define possible technical process solutions for the products and results acquired during the research. NUTRIMAR will work together in direct cooperation with SINTEF to meet the goals assigned in the work packages 1 and 2 of the project. NUTRIMAR will work with IRIS to scale-up the gelatin based coating and their validation over a fish slice. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Raja M. Rathore (M) is the leader for quality and research at Nutrimar. He has a Ph.D. in fish physiology and nutrition (2007). He has experience in utilization of raw materials in fish hydrolysates as a research scientist at Nofima, NIFES and University of Bergen. He has more than 15 peer reviewed scientific publications to his name.

Mr. Kristbjørn Bjarnason (M) is the General Manager of Nutrimar. He has more than 10 years of experience in industrial management, administration and market.

Mr. Tor Egil Eide (M) is the production manager at Nutrimar. He has more than 8 years of experience in process optimization and industrial production.

Five relevant publications, patents or products: No relevant publications.

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
STABIL: Proteins for human consumption from marine residual raw materials - mathematical model for optimal product preservation.	Norwegian Regional funding	2016	http://www.nfr.no

Technical equipment: The facility is equipped for processing fish oil, fish meal and protein hydrolysates. The daily intake capacity of raw material is about 180 tons and a daily production capacity of oil, meal and hydrolysate about 40-, 20- and 3 tons respectively. In addition basic oil testing instrument is available at the production unit.

PARTNER 8 **Full Name:** Innovacio i Recerca Industrial i Sostenible (IRIS) **Web page**: http://www.iris.cat	**Country:** Spain **Type:** SME	

Organization description: IRIS is an advanced engineering company that specialises in process monitoring and control, and process optimisation. Our commercial division offers advanced monitoring and control solutions based on: NIR chemical monitoring (handheld device and in-line solutions); hyperspectral imaging solutions for foreign body detection and chemical composition analysis; data management solutions for data processing, decision support, automation and data analysis (big production data).

Our research division has 4 core research lines:

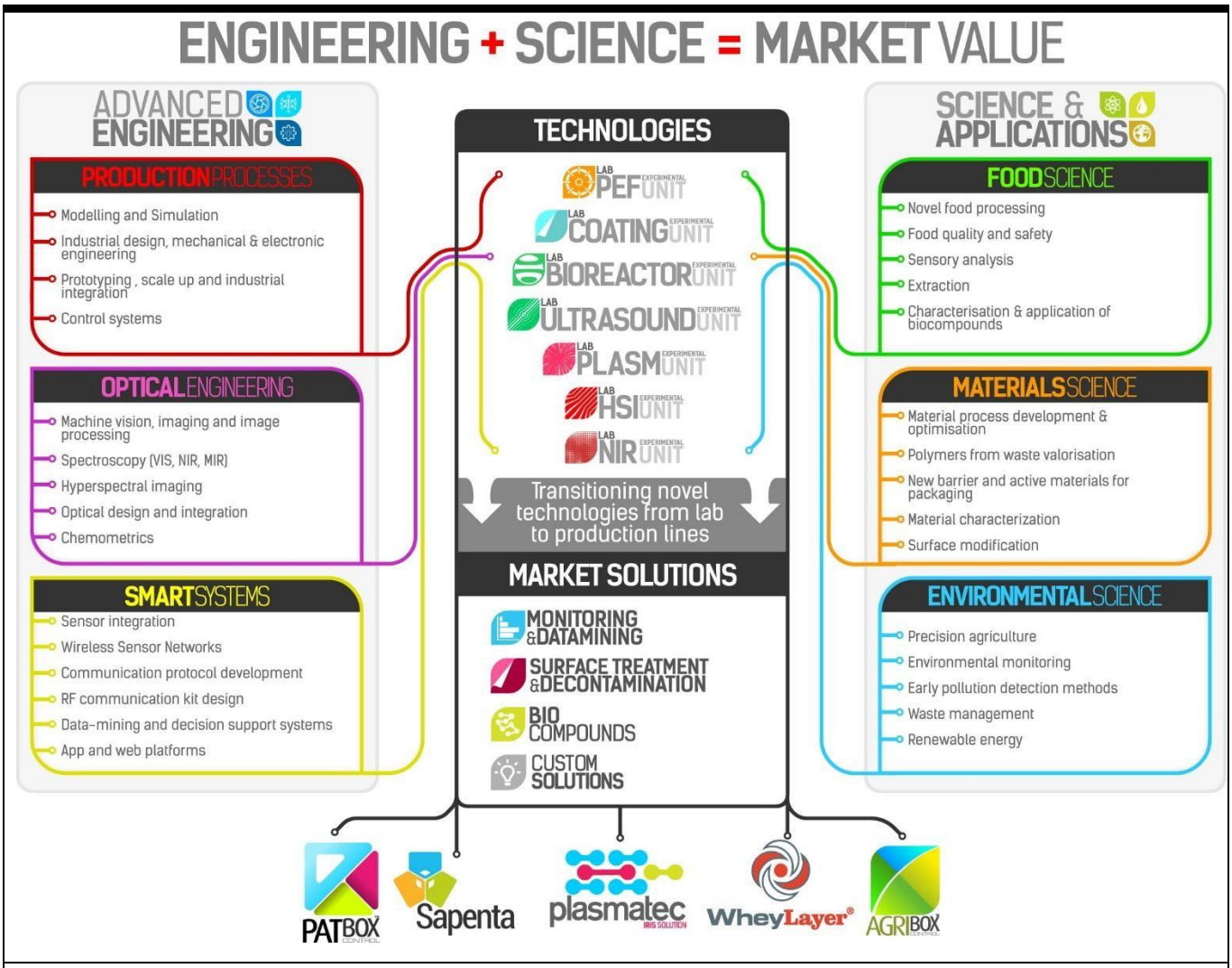
1. Optical and spectroscopy technologies (e.g. MEMs spectrophotometers, fluorescence and Raman) for process monitoring for on-line and PAT applications in food, pharmaceuticals and the process industries
2. *Data Processing & Information Systems (using eg. Cloud technologies, Predictive modelling, Data mining, Deep Learning) for the development of intelligent data management and decision support systems for smart manufacturing, processing and agriculture.*
3. Novel processing technologies (ultrasound, plasma, microwave…) for process intensification and nonthermal decontamination
4. Sustainable processing technologies: waste valorisation strategies, biocompounds and biopackaging for the circular economy

The company was established in 2007 and currently employs 48 staff with a turnover of €3.8 M. We are based over two sites (Barcelona and Dublin) that are equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab.

IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. IRIS is an industry member if the following sectoral networks: SPIRE, SPIE and SECPHO.

IRIS has a matrix of multidisciplinary knowledge groups in the company. The main Groups that will be involved in this project are: food science and materials science (who have expertise in developing bio-based coatings and testing resulting packaging as well as active packaging).

IRIS' research team consists of biologists, food and material scientists, physicists, among other, who have the expertise to validate the technologies developed with support in terms of process from its broad crossdisciplinary engineering team, including electronics engineers, mechatronics, mechanical, industrial design engineers, system engineers, application programers, web developers, etc.



Role in the Project: The main role in the project will be as a leader of WP5, where they will develop different fist-waste based active packaging and edible coatings solutions that will be tested for the optimal conservation of fish fillets. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Ing. Elodie Bugnicourt. *(female)* Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Leader of the Eco-Materials business unit in IRIS since 2009. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: R&D in chemical industry especially related to reactive polymer formulations: i) Arkema (ex. Cray Valley, 2002, Spain) in the field of polyester powder coating; Wacker Chemie (2002-2005, Germany and France), Silicone division that funded her PhD thesis: development of nanocomposites based on an epoxy matrix and silica; iii) Ashland (20062009, located sequentially in US, Spain and France) in the Composite Polymers division, focussing on bio-sourced resins, flame retardant resins, SMC resins, PU and epoxy structural adhesives. Led and participated in many R&D projects aimed at developing new formulations, understanding mechanisms and structure-properties relationships, developing new analytical methods and modelling (for improved quality management); as well as collaborations with academic labs and other players in the supply and value chain. Author of number of papers in peer-reviewed and technical journals, and has presented her work in numerous international conferences.

Dr. Cristina Serrano *(female)* is currently Project Manager in the Eco-Materials business unit in IRIS where she coordinates projects in the fields of sustainable packaging and surface treatment. She received her PhD (2011) at the University Complutense of Madrid (Spain) working at the Institute of Science and Technology of Polymers (2006-2011) on the development and characterization of new polymeric nanocomposites for self-sterilization of food packaging materials, based on the incorporation of non-doped and doped-TiO_2. She benefited from short pre-doctoral stays in PLAPIQUI (Argentina, 2008) to study the rheological behaviour of thermoplastics and composites, and in the Max Planck Institute for Polymer Research (Germany, 2011) where her studies were focused on obtaining nanostructured surfaces by plasma etching. After her PhD degree she worked as a postDoc (2011-2013) at the Max Planck Institute for Polymer Research, Germany, on several R&D projects in the field of bio-inspired adhesives and stimulus sensitive polymers involving synthesis and modification of polymers and polymer´s characterization. She is co-author of fourteen scientific articles and has presented her work in national and international conferences.

Dr. Laura Rodríguez Turienzo. *(female)* B.S in Food Science & Technology (2008) Masters in Innovation in Safety & Food Technologies (2010), PhD in Biological Sciences (2011), all from University of Santiago de Compostela (USC), Spain. She has participated in a number of R&D regionally and nationally funded projects aimed at developing and validating new bio-based active edible packaging, physico-chemical characterization of novel foods and valorisation of food industry wastes. She is author of several papers in peer-reviewed journals and has participated in different national and international conferences.

Mr. David Montes Monteserin. *(male) BSc* in Mechanical Engineering (2011), Polytechnic University of Catalonia (UPC), Barcelona. Prior to joining the Production Processes group at IRIS as Mechanical Design Engineer, he gained wide experience in product design and development, including dosing systems for the chemical industry and both industrial and domestic appliances for the water treatment sector. He has also held positions as Engineering Team Manager, applying renewable energies to industrial-sized drinking water plants among other projects. He is specialised in design for manufacturing and is proficient in SolidWorks, design software which has taught to several companies as a freelancer.

Dr. Edurne Gaston. *(female)* PhD in Food Technology (Dublin Inst. of Tech.). Thesis: Visible and hyperspectral imaging systems for the detection and discrimination of mechanical and microbiological damage of mushrooms. Postgrad. Dipl. in Food Quality and Safety, BSc in Food Science and Technology (Uni. of the Basque Country).
Leader of Food Tech Group in IRIS. Expertise in R&D projects, sensorial analysis. Industry expertise (Aguas Municipales de Vitoria) performing physical, chemical, and microbiological analysis of crude and drinking water, sensory analysis protocol for routine quality control. Author of numerous publications. Project management expertise.

Dr. Sara Peiró (female) PhD student performing her thesis at IRIS. Research topic: Natural antioxidants from plants: extraction, characterization and incorporation to food models and products. Co-directed by UPC and IRIS, Food Tech Group. She has the degree in Biology specialize in Vegetal Biology and Ecology by UAB, 2006.

Ms. Marisa Erencia *(female)* Technical Industrial Engineer, specialized in Industrial Chemistry, from Polytechnic University of Catalonia (UPC; Spain, 2007) and Chemical Engineer from Autonomous University of Barcelona (UAB; Spain, 2009). She benefited from pre-doctoral grant in the Polymer Materials and Textile Chemistry research group (POLQUITEX) of UPC (Spain, 2011) to develop her research in the field of polymers and nanomaterials focused on the control and optimization in manufacturing protein nanofibers (gelatin and collagen) using the electrospinning technique in order to obtain scaffolds available for tissue engineering applications. From this research, she has published three scientific papers as a main author and has presented her work in national conferences. She is currently finishing the drafting of her PhD thesis and working as Polymer Chemist in IRIS since March 2015, mainly involved in projects related with new bio-based packaging materials.

Dr. Ana García Sanz *(female)* PhD degree in Biophysics from the Autónoma University de Madrid (PhD thesis: Electron and positron interactions with molecules of biological interest"). Masters in Biophysics from the Autónoma University in Madrid and Master-degree in Physics from the Complutense University de Madrid. Prior to joining IRIS, she has worked at the National Research Council Centre (CSIC) and has been invited as a research intern in Universitá La Sapieza di Roma, The Open University and Sherbrooke University. Her research has been focused on the study of the damage induced by radiation (electrons, positrons and X-rays) in biological systems at molecular level, by means of both theoretical (quantum chemistry calculations) and experimental (spectroscopy, HPLC, LC-MS/MS) techniques. Author of 17 scientific papers in peer reviewed Journals.

Five relevant publications, patents or products:

- "Whey protein coated films"; **world patent** IB2011/053271; Inventors: M. Schmid, K. Noller, F. Wild, **E. Bugnicourt**; positive PCT examination; Application date: July 22, 2011.

- 'Method to prepare a coating made of whey-proteins treated with ultrasounds. **L. Rodríguez Turienzo** et al. **Patent** application 201130582
- Effects of edible coatings based on ultrasound treated whey proteins in quality attributes of frozen Atlantic salmon (Salmo salar), **Rodríguez-Turienzo, L**., Cobos, A., & Díaz, O., (2012), Innovative Food Science & Emerging Technologies, 14, pp. 92-98.
- Processing and Validation of Whey-Protein-Coated Films and Laminates at Semi-Industrial Scale as Novel Recyclable Food Packaging Materials with Excellent Barrier Properties. **Bugnicourt, E,** Schmid, M., Mc. Nerney, O., Wildner, J., Smykala, L., Lazzeri, A., Cinelli, P. (2013) Adv in Materials Sci Eng, 10 p.
- State of the art in the development and properties of protein-based films and coatings and their applicability to cellulose based products: an extensive review; M.B. Coltelli*, F. Wild*, **E. Bugnicourt***, P. Cinelli, M. Lindner, M. Schmid, V. Weckel, K. Müller, P. Rodriguez, A. Staebler, L. Rodriguez, A. Lazzeri; Coatings 2016, 6, 1. (*equal contribution of corresponding authors)

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
WHEYLAYER Whey protein-coated plastic films to replace expensive polymers and increase recyclability	FP7-SME	2011	NA
WHEYLAYER2 Barrier biopolymers for sustainable packaging.	FP7-DA	2014	www.wheylayer.eu
THERMOWHEY Whey protein based barrier coatings with enhanced thermoformability and agglomerated formulations for a range of packaging applications	FP7-MANUNET	2016	www.thermowhey.eu
OLI-PHA A novel and efficient method for the production of PHA polymer-based packaging from olive oil waste water	FP7-NMP	2015	www.olipha.eu
SPECTRAFISH Hyperspectral imaging technology for the quality inspection of fish products	FP7-SME	2016	www.spectrafish.eu

Technical equipment:

Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc noncommercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.

IRIS' optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

Food and wet chemistry lab equipped with a DSC, a Surface Energy analyser, an HPLC, a Cobb tester, microbiological equipments, microspcope and microtome for structural analysis, migration cells, colourimeter (for colour and opacity measurement), Ford cup viscometers, water vapour permeation cells, turbidimeter,

climatic chambers, manual and automatic co aters, electrospray, different drying technologies, plasma surface activation, etc.

PARTNER 9 **Full name**: Biotrend - Inovação e Engenharia em Biotecnologia, S.A. (BIOTREND) **Web page**: www.biotrend.biz	**Country:** Portugal **Type:** SME	biotrend Inovação e engenharia em biotecnologia

Brief description of the organization:

Biotrend is a research-based company focused on the development, optimization, de-risking and scale-up of bioprocesses for industrial biotechnology. Clients range from start-up companies without bioprocessing knowhow and infrastructure to large chemical to pulp and paper companies exploring the opportunities of industrial biotechnology and of the use of renewable raw materials, including industrial by-products, to create value. Additionally, Biotrend invests in internal projects to create its own IP portfolio. Biotrend carries out research aiming at the production of bio-based chemicals, materials and fuels from renewable raw materials. Further, we actively participate in collaborative projects in the fields of the bioeconomy, nanobiotechnology and marine biotechnology. Our team's expertise includes biotechnology, chemical engineering, organic chemistry, and analytical chemistry.
The core activity of Biotrend is the development of novel bioprocesses and their transition from lab scale to bioprocesses that can be implemented in production scale, process integration and early evaluation of the technical and economical feasibility of processes before investment in higher scale trials are performed. We are experienced in the development of processes based on a wide range of biological systems, from naturally occurring bacterial strains and yeasts to recombinant microbial cells, for the production of biomass, primary and secondary metabolites, biopolymers and recombinant proteins.
Role in the Project: BIOTREND will participate in WP1, extracting physiological data from strains producing chemical intermediates in iteration with the partners developing strain design activities. Further, it will contribute to bioreactor design using gas-liquid membrane contactors in WP2. Leader of WP8 (techno-economic assessment) and contributions to WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Bruno Sommer Ferreira (M): CEO, MSc Chemical Engineering and PhD Biotechnology (Technical University of Lisbon, Portugal). Responsible for the company strategy and management. More than 15 years' experience in bioprocess development and technology transfer. Author of 30 international scientific publications. Advisor of public and private entities, including the Portuguese Government, the EU and the OECD on the topic of the bioeconomy.

Dr. Frederik van Keulen (M): CSO, MSc Molecular Sciences Engineering (Univ. Wageningen, The Netherlands), PhD Biotechnology (Technical University of Lisbon, Portugal). Coordinator of all scientific activities at Biotrend. 20 years' experience in bioprocesses, biocatalysis and organic and analytical chemistry. Author of 25 scientific publications.

Dr. João Cavalheiro (M): Pilot plant manager, MSc Biological Eng. and PhD Biotechnology (Technical University of Lisbon, Portugal). Performs the scale-up experiments and oversees the development of downstream processes. Has 10 years research experience in bioprocessing and authors 5 scientific publications.

Ms. Ana Carolas (F): Lab leader, MSc Biological Engineering (Technical University of Lisbon, Portugal).

Five relevant publications, patents or products:

- Fava, F., Diels, L., Reis, M., Duarte, J., Carioca, O.B., Poggi-Varaldo, H.M., Ferreira, B.S. (2015) Biowaste biorefiney in Europe: opportunities and R&D needs. *New Biotechnology*, 32: 100-108.
- Cesário, M.T, Raposo, R.S., de Almeida, C.D., van Keulen, F., Ferreira, B.S., da Fonseca, M.M.R. (2014) Production of poly (3-hydroxybutyrate-4-hydroxybutyrate) by *Burkholderia sacchari* using wheat straw hydrolysates. *International Journal of Biological Macromolecules*, 71: 59-67.
- Cesário, M.T, Raposo, R.S., de Almeida, C.D., van Keulen, F., Ferreira, B.S., da Fonseca, M.M.R. (2014) Biological conversion of wheat straw lignocellulosic hydrolysates to polyhydroxyalkanoates. *New Biotechnology*, 13: 104-113.
- Cavalheiro, J.M.B.T, Raposo, R.S., de Almeida, M.C.M.D., Cesário, M.T., Sevrin, C., Grandfils, C. and da Fonseca, M.M.R. (2012) Effect of cultivation parameters on the production of poly(3- hydroxybutyrate-co-4hydroxybutyrate) and poly(3- hydroxybutyrate-4-hydroxybutyrate-3-hydroxyvalerate) by *Cupriavidus necator* using waste glycerol, *Bioresource Technol*, 111, 391-397.
- Ferreira, B.S. (2011)Ten-year forecast of disruptive technologies in industrial biotechnology, Pira Internat.

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
TRANSBIO: Biotransformation of by-products from fruit and vegetable processing industry into valuable	FP7-KBBE- bioproducts.	2015	http://www.transbio.eu
O4S: Sustainable surfactant production from renewable resources through natural fermentation for applications in natural, organically-certified products	FP7-SME-2011-2	2014	http://www.organic4surfactants.eu
BLUEGENICS: From gene to bioactive product: Exploiting marine genomics for an innovative and sustainable biotechnology industry.	FP7-KBBE-2012-6	2016	http://www.bluegenics.eu
BRIGIT: New tailor-made biopolymers produced from sugars obtained from lignocellulosic residues for flame retardancy highly demanding applications.	FP7-KBBE-SME	2016	www.brigit-project.eu/
BIOREFINE 2G : Development of 2nd Generation Biorefineries Production of Dicarboxylic Acids and Bio-based Polymers Derived Thereof	FP7-KBBE 2013	2018	www.biorefine2g.eu/

Technical equipment:

- Bioreaction equipment for physiological studies of microbial strains and process development from 1 mL to 250L scale.
- Lab and pilot scale downstream equipment (tangential flow filtration, centrifugation, homogenization, solvent extraction)
- Multiplexed UV/VIS spectrophotometry and HPLC, optical microscopy, on-line gas sensors.

PARTNER 10 **Full name**: Daren Laboratories & Scientific Consultants Ltd (DLab). **Web page**: www.darenlabs.co.il/	**Country:** Israel **Type:** SME	Daren Labs

Brief description of the organization: DLab is a private company that promotes its own research and development as well as solving problems brought by external clients. Its expertise is in the fields of plastics, polymers, materials and organic chemistry. It synthesise new materials, tailor made to meet specific applications. Its expertise includes polymerizations, photochemistry, catalysis, optical properties of polymers, flame-retardants, halogenations, hydrogels and expert opinions to governmental authorities and courts of law. The company is also involved in small volume high value production of specialty products such as optical adhesives and filters.

Solutions to technological problems always need to include economic and market parameters. It has proven experience in evaluations of chemical products and processes. It utilizes this experience to perform marketing and cost estimates for High-Tec start-up companies who are seeking funding such as technological incubators. It continues to provide technical support to such companies after the funding has been obtained.

Its fields of activities are as broad as its clients' needs. During past years, it has, among others, been involved in the process development of cholesterol-reducing drug, drug delivery systems, and polymeric electrolytes for ultra-thin batteries and new applications for bromine-containing molecules. All of these projects were performed by its professional staff in its own laboratories.

Role in the Project: Synthesis of polymers and oligomers which will contain flame retardant components derived from protein and nucleic acids. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Steve Daren-Managing Director and Founder (M). Graduate of the University of London and the Weizmann Institute of Science. Dr. Daren has more than thirty years of experience in the chemical industry and academia. For more than twenty years he has been a consultant providing solutions to numerous problems in the chemical, pharmaceutical, polymer and plastic industries. Dr. Daren is a recognized authority on plastics, polymers, adhesives and flame retardants. He holds several patents in the fields of agriculture, battery-electrolytes, plastics and flame retardants.

Vlada Nikolayevsky M.Sc - Chemist, Laboratory Manager (F) Graduate of the Donetzk University, Vlada has over 16 years of experience in R&D at Daren Labs. Vlada progressed from a purely technical role and today performs independent research and project management. In addition to managing the R&D laboratory, Vlada is responsible for purchasing and client relations at the Daren Innovation Centre.

Pavel Basin, M.Sc – **Chemist (M)** Graduate of Tel-Aviv University ,Pavel is a very knowledgeable chemist and has a vast range of knowledge in organic ,inorganic and analytic chemistry. He is experienced in polymer research for industry and novel materials development. At Daren Laboratories, Pavel performs advanced chemical

syntheses including the FR materials that are presented here. He is also an expert in metal corrosion. Pavel does all of this despite his congenital hearing difficulties.

Five relevant publications, patents or products:

Publications

- A Stone Lamp with Seven Nozzles from the Late Second Temple Period. Rosenfeld, S. Ilani, W. Krumbein, S. Daren, H. Feldman. Liber Annus 60 (2010), 377 - 391
- One-step Holographic Grating Inscription in Polymers S. Kibben; M. Koerdt; J. Garcia; S. A. Kolpakov; S. Daren; F. Solomon-Tsvetkov; Photonics West 2009-OPTO: Practical Holography XXIII: Materials and Applications, eds: H. I. Bjelkhagen, R. K. Kostuk. Society of Photo-Optical Instrumentation Engineers (SPIE) Bellingham 7233 72330M-1-72330M-10, 2009.
- Single Step Holographic Grating Inscription in a PMMA-based Copolymer. T. Seefeld; F. Vollertsen SPIE: 3 February 2009; 10 pages; DOI: 10.1117/12.808752

Patents

- Israel Patent 100540 Flame Retarded Polymethylmethacrylate
- Israel Application 148578 Flame Retardant Grafted Rubber **Five Relevant Projects:**

Project acronym/description.	Scope	End Year	Website
POLYMICRO Polymer Micro-optics	Euro-practice IST 2001	2002	Discontinued.
BRIGHTLIGHT Laser Diode Radiation	FP 6	2009	www.bias.de
EPHOCELL Smart light collecting system for the efficiency enhancement of solar cells.	FP7 ENERGY	2013	//cordis.europa.eu/project/rcn/90161_en.html
INFORM: Integrating Nanomaterials in Formulations	FP7 NMP	2012	//cordis.europa.eu/project/rcn/91315_en.html
BRIGIT: New tailor-made biopolymers produced from sugars obtained from lignocellulosic residues for flame retardancy highly demanding applications.	FP7-KBBE-SME	2016	www.brigit-project.eu/

Technical equipment:

Full laboratory equipment: Drying Ovens, Hot press, 1-20L scale glass-ware. Daren Laboratories has 500m^2 of laboratory space including its Innovation Centre. In addition to well-equipped laboratories for organic and polymer chemistry the following analytical instruments are available in-house: HPLC, FTIR, UV-Visible Spectrophotometer, DSC and TGA.

PARTNER 11 **Full name**: Mine Plastik Endüstriyel Ürünler Kimya Ltd Sti. (MINE). **Web page**: www.minecolours.com	**Country:** Turkey **Type:** SME	Mine Colours

Brief description of the organization: Mine Colours is a family company with large experience in production of petrol-based plastic compounds for several industrial sectors. Its business is to develop new plastic grades suitable for specific needs. Established in 2010.Mine Colours is an innovative and technology oriented supplier of colouring solutions and speciality additives master batch form .Our long lasting cooperation with leading producers of pigment and speciality chemicals results in advanced formulation tailor made to meet the most critical requirements of our customers. Mine Colours has a holistic approach to Environmental are addressing continuous improvements within total enterprise all way from product planning & development, manufacturing & supply product use, recyclability end –of –life for the products. Specific measures are initiated with in the implemented management system ISO:14001.

Role in the Project: MINE will design and optimization of twin screw parameters to produce thermoplastic compounds standard polymers and bio polyamides and DAFIA Flame Retardant. Mine Colours will also take part to the dissemination the project. Contributions to WP8 (LCA and techno-economic assessment) and WP9 leader (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Nalan Onar (F) The founder of the company. She used to be teacher and had experience on training for 25 years. She was a projects coordinator ona European Nation Comenius Project (TR1-COM06-36346-Cultural Heritage Reflected in Our Hometown) Responsible for management and reporting the project.

Emin Semih Onar (M) He is the general manager of Mine Colours. He has a master degree on Human Resources and Management. He has experience in Industrial Management and administration and market.

Gülce Öklem (F) Graduated Chemistry Department of Middle East Technical University in 2014. She has master degree on Polymer Science and Technology. She will work as lab leader during the project.

Five relevant publications, patents or products: No previous relevant publications or patents.

Five Relevant Projects: No previous relevant projects.

Technical equipment: Co-rotating twin screw extruder, auxiliary (dosing system, side feeders, screw elements) and laboratory equipment to carry out the work proposed in WP6.

PARTNER 12 **Full Name**: Bio Base Europe Pilot Plant vzw (BBEU) **Web site:** www.bbeu.org	**Country:** Belgium **Type:** non-profit SME	Bio Base Europe Pilot Plant

Organization description:

BBEU (www.bbeu.org) is a flexible and diversified pilot plant for the development and scale up of new, bio-based and sustainable processes. It is capable of development of new bioprocesses, optimization of existing processes and scale up of a broad variety of bio-based processes up to an industrial level (from 5L to 50m3 scale, depending on the process). It can perform the entire value chain, from the green resources up to the final product. BBEU intends to close the gap in the innovation chain of the bio-based economy, bridging science and industrial production. BBEU was selected by the European Commission as a multi-KET Pilot Lines demonstrator (www.mkpl.eu). It is located in Ghent, Belgium.

The activities of BBEU can be categorized in:- development of bio-based and sustainable processes (TRL 2-4); scale up (TRL 5-6) and pilot and demo production to allow market introduction (TRL 7-8)

The BBEU team combines over 50 years of experience in fermentation development and scale-up, and also has extensive expertise in biocatalysis, green chemistry, and the development of industrial purification techniques. It is responsible for these tasks in several publicly funded projects of which 13 are ongoing. In addition, BBEU is a service provider to many companies in the biotech domain and bio-based economy. In the period 2013-2015 it served 129 customers for development, scale up or pilot production.

Role in the Project: BBEU will be mainly involved in the following tasks:

WP3: Microbial conversion of feedstocks into value chemicals: BBEU together with BIOTREND will develop costeffective DSP routes for the recovery of diacids and diamines from fermentation broth. Also, BBEPP will focus on easy implementation of the developed DSP methods on larger scale (WP7).

WP7: Product validation by industrial partners: BBEU will scale up the fermentative production of 2 compounds (diacids or diamines) to 150 L in cooperation with BIOTREND. The most promising compound will be further scaled up to 1500 L, together with a large scale validation of the developed DSP process.

Scientific team:

Frederik De Bruyn, PhD (M) is Business Development Manager at the Bio Base Europe Pilot Plant (BBEPP). He received his Master's degree in Bioscience Engineering in 2010, at Ghent University, and his PhD in 2014 at the Centre for Industrial Biotechnology and Biocatalysis (InBio.be). He has been active as program manager at the Belgian Science Policy Office (BELSPO), where he was involved in various national and European research projects. He currently focuses on enhancing bilateral/public collaborations between industrial partners/research institutes and BBEPP to enable the transition to a sustainable bio-based economy.

Lieve Hoflack, PhD (F) is project manager at BBEPP. She holds a PhD in biotechnology (Ghent University, 1998) and has been working as a researcher in academia (University of TX Health Science Center, Ghent University), as well as in biotech SMEs (Plant Genetic Systems, Dyax, Vivactis), and later as project manager in SMEs (MintEurope, BBEPP).

Ir. Helena Moens (F) is Process Engineer. She holds a master degree in bioengineering sciences (Ghent University, 2009). She has extensive experience in equipment design, scale-up of fermentations and downstream processing as well as European project experience (Biosurfing and IB2Market).

Ir. Reinout de Boeck (M) is proces engineer. He holds a master degree in applied engineering sciences, discipline chemistry (Hogeschool Gent, 2008). He is an expert in working under ATEX environment conditions and has experience with various DSP technologies such as liquid/liquid extraction, distillation, crystallization, absorption.

Five publications/patents related to the project:

- Hamer S, Cord-Landwehr S, Biarnés X, Planas A., Waegeman H, Moerschbacher B, Stephan Kolkenbrock (2015). Enzymatic production of defined chitosan oligomers with a specific pattern of acetylation using a combination of chitin oligosaccharide deacetylases. Nature Scientific Reports 5, 8716.

- Roelants S, Cielsielska K, De Maeseneire S, Moens H, Everaert B, Verweire S, Denon Q, Vanlerberghe B, Van Bogaert I, Van der Meeren P, Devreese B, Soetaert W (2015). Towards the industrialization of new biosurfactants: Biotechnological opportunites for the lactone esterase gene from *Starmerella bombicola*. Biotechnology and Bioengineering 8.

Five Related Projects:

Description of experience/project	Scope (European or national)	End Year	Website (for further information)
DEMOPROBIO: Demonstration of In Situ Product Recovery (ISPR) to improve the productivity in fermentation processes (responsible for scale-up).	National, Agentschap ondernemen	2016	Not available
IB2Market: Bringing innovative industrial biotechnology research to the market (coordinator and responsible for scale-up)	European, FP7	2016	www.ib2market.eu
Nano3Bio: NanoBioEngineering of BioInspired BioPolymers (responsible for scale-up)	European, FP7	2017	www.nano3bio.eu
CARBOSURF: New processes for the fermentative production of biosurfactants and specialty carbohydrates (responsible for scale-up and downstream process development)	European, H2020	2018	www.biosurfing.ugent.be/
PULP2VALUE: Processing Underutilised Low value added products (responsible for scale-up)	European, H2020	2019	//pulp2value.eu/

Technical equipment.

Bio Base Europe Pilot Plant has 3 process halls, and a supporting lab for analysis: a biomass pre-treatment hall of 907m^2, a fermentation hall of 500 m^2 for industrial biotechnology processes and product purification and an ATEX-proof 'green process hall' of 285 m^2 for green chemistry. A full list of the equipment at BBEU can be found at http://www.bbeu.org/pilotplant/expertise/. For the scale up of fermentatively produced diamines or diacids, BBEPP has fermentors ranging from 7L up to 15.000L available. For this project, the fermentors used will range

from 150L to 1500L. For the development and scale-up of the downstream processing, BBEU has a wide variety of state-of-the-art unit operations for recovery of compounds from complex broths on both lab- and industrial scale.

PARTNER 13 **Full name**: BIOPOLIS, S.L. (BIOPOLIS). **Web page**: www.biopolis.es	**Country:** Spain **Type:** SME	

Brief description of the organization:

Biopolis, S.L. is a leading biotechnology company based in Valencia (Spain) offering contract research and manufacturing. The company is classified as an SME. Our core competences are the isolation, selection, design, characterization, validation and production of microorganisms and cellular metabolites with applications in agrifood, pharmaceutical and chemical industries. This results into the development of microbial strains and metabolites (most of them biopharmaceuticals) by strain and bioprocess design followed by fermentation and downstream process development and the production at an industrial scale. Prototype projects are (i) the selection, validation and production of starters and probiotics, (ii) the selection and production of enzymes, (iii) the identification, characterization and process development of functional ingredients, (iv) the overexpression of recombinant protein in microbial systems, (v) the overproduction of synthesis intermediates and active pharmaceutical ingredients by fermentation, (vi) the design of novel processes based on microbial and enzymatic catalysis.

Role in the Project: BIOPOLIS will participate in tasks of extraction of lipids and protein from MSW for further chemical conversions, and develop appropriate separation methods; pre-treatment, hydrolysis and fractionation of MSW, for use as a fermentation substrate for chemical building blocks; design microbial hosts for production of adipic diacids, from the carbohydrates of MSW, and protein and lipid fractions of MSW or RRM; design of strategies for standardizing the feedstock composition (e.g. by mixing) as a fermentation substrate. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr Daniel Ramón Vidal (M) CEO and CSO of Biopolis S.L, with more than 160 publications and patents in microbial biotechnology (selection of microbial starters including probiotics, fermentation development and scale-up, strain improvement) and biotechnological production of microbial metabolites

Dr Marta Tortajada (F) Microbial Biotechnology Manager at Biopolis (Chemical Engineer, PhD in Automatics and System Engineering). She has 12 years' experience in protein overexpression in microbial systems and strain and bioprocess design for biochemical and bulk chemicals overproduction. From 2005 to the present she has been active in the preparation, management, coordination and / or development of several European R & D projects in the areas of bio-based industries, sustainability and revalorization of urban and agricultural wastes/residues (i.e. Integrated project FP6, KBBE-FP7, ERA-IB, EUROSTARS, NMP-FP7, CENIT, CIEN, etc.).

Dr Antonia Rojas (F) (Chemical Engineer, PhD in Biotechnology), head of Bioenergy Laboratory at Biopolis. She has large experience in microbial systems and strain selection, validation and bioprocess design for bulk chemicals overproduction.

From 2009 to the present she has been active in the preparation, management, coordination and / or development of several European R & D projects in the areas of bio-based industries, sustainability and revalorization of urban and agricultural wastes/residues (i.e. Integrated project FP6, KBBE-FP7, ERA-IB, EUROSTARS, NMP-FP7, CENIT, CIEN, etc.).

Dr Silvia Segarra (F) (Organic Chemist, PhD in Biotechnology and Molecular Biology of Plants), head of Biocatalysis Laboratory at Biopolis.

Mr. Diego Rodriguez (M) Biochemist, MSc Biochemistry and Molecular Biology), responsible of bench-top fermentation and scale-up, both experienced in metabolic engineering and whole cell biocatalysis.

Five relevant publications, patents or products:

- Rojas A, Segarra S, Montesinos A, Tortajada M, Ramón D, Santos VE, Ladero M, García-Ochoa F, Ripoll V, Rodríguez A (2012) P201231119. Process for the production of 2,3-butanediol using improved strains of *Raoultellaplanticola*. BIOPOLIS SL – Universidad Complutense de Madrid. Priority date 17/07/2012.
- Del Pozo MV, Fernández-Arroyo L, Gil-Martínez J, Montesinos A, Chernikova T, Nechitaylo T, Waliszek A, Tortajada M, Rojas A, Huws S, Gloyshina OV, Newbold CJ, Polaina J, Ferrer M, Golyshin P. (2012) Microbialbglucosidasesfromcow rumen metagenomeenhancethesaccharification of lignocellulose in combinationwith comercial cellulasecocktail. *Biotechnol. Biofuels*, 5, 73-85.
- Tortajada M, Llaneras F, Ramón D, Picó J. (2012) Estimation of recombinant protein production in *Pichiapastoris*based on a constraint-based model. *J. Process Control*, 22, 1139-1151.
- Matencio E, Navarro V, Enrique M, Tortajada M, Lukovic D, Rodríguez D, Ramón D, Álvarez B. (2011) PCT/ES2011/070403. Obtention of oligosaccharides through a biotechnological procedure. Priority date 07/07/2011
- 5. M. Tortajada, F. Llaneras, J.Picó. (2010). Validation of a constraint-based model of *Pichia pastoris* metabolism under data scarcity. BMC Systems Biology 4: 115.

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
SYNPOL. Biopolymers from Syngas Fermentation	FP7	2016	http://www.synpol.org/

Optimized oxidoreductases for medium and large scale biotransformations (INDOX)	FP7	2017	http://www.indoxproject.eu/ industrial
Production of Organic Acids for Polyester Synthesis" (POAP)	VII FP. ERA-IB	2016	http://www.era-ib.net/poap
Production and Upgrading of 2.3-Butanediol from Biomass (PUBB)	6FP ERA-IB	2012	http://www.era-ib.net/pubb
3R2020+: From waste to resource by recycling	Spain	2019	//3r2020.com/

Technical equipment:

9 specialized research laboratories (Biochemistry, Molecular Biology, Cellular Biology, Microbiology, Filamentous Fungi, Cell Culture, Chemistry, Analytics, and Bench-top fermentation) and an industrial plant with a total capacity of 9,000 L fermentation.

PARTNER 14 **Full name**: Arkema France S.A. (ARKEMA). **Web page**: www.arkema.com	**Country:** France **Type:** SME	ARKEMA INNOVATIVE CHEMISTRY

Brief description of the organization: A global chemical company and France's leading chemicals producer, Arkema is building the future of the chemical industry every day. Deploying a responsible, innovation-based approach, we produce state-of-the-art specialty chemicals that provide customers with practical solutions to such challenges as climate change, access to drinking water, the future of energy, fossil fuel preservation and the need for renewable source materials.

ARKEMA is a French chemical company with about 19,000 collaborators in the world, and 7.5 G€ of sales. Arkema generates about 1 G€ of sales from products derived from renewable resources, including specialty polyamides; epoxidized vegetable oils, terpenes and limonenes; fatty amines; coating resins and activated carbons and holds leadership positions in all its markets with a portfolio of internationally recognized brands. Arkema has 13 research centers in the world; 7 or them being located in France. Arkema is active in producing and selling industrial chemicals with worldwide leading positions in fluorine, sulfur and acrylate based chemistries as well as in chemical specialties and polymer materials such as its range of bio-based long chain polyamides (Rilsan®), PMMA(Altuglas®), PVDF (Kynar®) and PEKK (Kepstan®).

The research center located near Lyon (CRRA, Pierre-Bénite) is working mostly on Fluorine chemistry, H2O2, monomer synthesis, Fatty nitriles and fatty amines. This centre hosts the corporate research activities dealing with catalytic processes. The research center located near Pau (GRL, Lacq) is mostly working on zeolite synthesis, polymers and sulphur chemistry. The research centre located near Rouen (CERDATO, Serquigny) is mostly working on specialty polyamides.

Role in the Project: Arkema will investigate the Flame retardants produced in the project in its polymer applications, and will support the investigations on monomer synthesis for polyamides and polyesters. More particularly in this later case, Arkema will support the investigations on the use of muconic acid for specialty

applications. Contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr Jean-Luc Dubois (M), Scientific director for Catalysis, Processes and Renewables. JL Dubois supervises long term research projects related to development of monomers and other chemicals. He has been involved in the EU projects EuroBioRef (as WP leader, IP manager, Executive Board Chair, and Exploitation manager), and is currently involved in Cosmos (H2020) with similar responsibilities. He is the author of more than 100 publications and 150 patent applications, h-index=31, i-10 index=59.
(//scholar.google.fr/citations?user=Fv2D9LkAAAAJ&hl=fr)

Dr Michel Glotin (M), received his PH.D (Physical Chemistry) at the University of Southampton (UK). After 2 years as a Research Assistant at Florida State University (USA), he joined Ato Chemistry as a researcher on polymers in 1982. After holding various positions in Serquigny research center, he joined the management of research as Atofina delegate search Technical Polymers, PMMA and PVC in 2000. He is currently Scientific Director of Arkema materials. Michel Glotin was a member of the Board of the Chemical Society of France and GFP (French Association of study and application of polymers) which he was president from 2009 to 2012.

Dr Sébastien Taillemite (M), Chemical Engineer of Chimie ParisTe ch and Ph.D. Organic Chemistry from the University Pierre & Marie Curie, Paris. He joined Arkema (formerly Cray Valley) in 2007 to be in charge of Marketing and New Markets for the Business Unit Composites, then he was managing the Coating Additives Business and recently took over the Development of the Elium Composites Business.

Five relevant publications, patents or products:

- Jean-Luc Dubois et al., **EuroBioRef Final booklet**. (available for download) https://www.researchgate.net/publication/285054983_EuroBioRef_Public_Final_Booklet_April_2014
- JL Dubois, Chapter 2: **Refinery of the future: feedstock, processes, products**, (2012), in "*Biorefinery: From Biomass to Chemicals and Fuels*", Ed. by Aresta, Michele / Dibenedetto, Angela / Dumeignil, Franck, http://www.degruyter.com/view/product/177487
- JL Dubois and JL Couturier, Chapter 10: **Oil chemistry: chemicals, polymers, and fuels**, (2015), in "*Biorefineries, An Introduction*", Ed. by Aresta, Michele / Dibenedetto, Angela / Dumeignil, Franck, http://www.degruyter.com/view/product/210639
- JL Dubois, Chapter: **Arkema's Integrated Plant-based Factories**, (2016), in "*Chemicals and Fuels from Bio-based Building Blocks*"**,** Fabrizio Cavani, Stefania Albonetti, Francesco Basile,

http://eu.wiley.com/WileyCDA/WileyTitle/productCd-3527338977.html

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
Eurobioref – Biorefinery related project	FP7	2014	www.eurobioref.org/
Biocore – Lignocellulosic Biorefinery project	FP7	2014	www.biocore-europe.org
Cosmos – Camelina & crambe Oil crops as Sources for Medium-chain Oils for Specialty oleochemicals	H2020-Biotec	2019	Cosmos-h2020.eu
OPTICO Model-based optimization & control for process-intensification in chemical and biopharmaceutical systems	FP7 NMP	2015	www.opticoproject.eu/
STEELCOAT Development of green anticorrosion coatings for steel protection based on environmentally friendly nanoparticles and conducting polymers	FP7 NMP	2014	http://steelcoatproject.com

Technical equipment:

Arkema research centre in Pierre Bénite is working on nitriles synthesis with equipment for conversion of acids to nitriles, and has been working on conversion of Diacids (sebacic acid) to the dinitrile, and corresponding diamine. Equipment is available at several kg scale for validation at TRL5 of the involved chemistries. Characterization techniques (TEM, SEM, XRD, Calorimeter, DTA-DTG analyser, IRTF spectrometer, DRIFT spectrometer, Raman spectrometer, UV-Vis spectrometer, and several reactors of different sizes to test different types of catalysts.

PARTNER 15 **Full name**: The National Non-Food Crops Centre (NNFCC). **Web page**: www.nnfcc.co.uk/	**Country:** United Kingdom **Type:** SME	NNFCC

Brief description of the organization:

The National Non-Food Crops Centre (NNFCC) is a non-profit Research and Technology Organisation (RTO). NNFCC was originally established by the UK Government in 2003 to provide expert and impartial **advice to business and the public sector** on the development of the bioeconomy. The company mission is to provide clients with a holistic view of feedstock, technology, policy and market development across the bioeconomy, enabling them to make informed business decisions and develop sustainable business strategies. NNFCC provides a range of facilitation, advisory and consultancy services. We act as a **valued programme coordinator and evaluator** for UK research and innovation funders, which makes us able to **validate ideas** and as an advisor to the UK's Department of Energy and Climate Change. Our consultancy services include; Market analysis, feedstock and logistics planning, sustainability strategy development, initial technology and economic appraisal, project feasibility assessment and policy and regulatory support. NNFCC has over 12 years of experience in supporting the development of the bioeconomy.

The organisation has worked extensively in the areas of supply chain building, network facilitation, technology development, providing evidence to inform policy, and market analysis and development. NNFCC has worked internationally to identify opportunities for transnational working (IEA Bioenergy Task 42 member, Interreg IVB NWE Bio Base NWE project.) and to identify bilateral collaboration opportunities (UK-Norway).

Role in the Project: NNFCC will develop business models to enable technology exploitation and support the communication and dissemination of project outputs. Networking with other EU related projects.

Scientific team:

Adrian Higson PhD (M). Goal focused Business Strategist – Over 12 years' experience in coordinating large multi-stakeholder projects and 10 years specifically working in the area of bio-based products. Excellent track record of delivering complex projects to quality targets, on budget and within deadlines. Adrian holds a PhD from the University of Liverpool and has held research positions at the University of Colorado (USA) and University of Dundee (UK). Prior to joining NNFCC he was a project manager at SAFC Pharma (Sigma Aldrich) supervising the manufacturing of pharmaceutical ingredients for clinical trials. Adrian is a member of the Scottish Industrial Biotechnology Development Group, the European Commission Expert Group for Bio-Based Products (UK representative), UK Bioeconomy public sector working group and a bioeconomy advisor to Innovation Norway.

David Turley (M) is NNFCC's Lead Consultant for bio-based feedstock with over 20 years' experience in provision of research, techno-economic and strategic market analysis services in the area of biomaterials, bioenergy and bio-based feedstock. Broad experience in biorefinery applications and experience in LCA analysis and interpretation.

Dr Davide Di Maio PhD (M). Consultant and chartered engineer, with expertise in biopolymers, materials science and engineering. Davide has experience in sustainable materials selection, data analysis, and modelling. He has experience in managing technical projects through working in the private sector and as a scientist at the National Physical Laboratory. Davide obtained his DPhil from the University of Oxford (UK) and qualified as Materials Engineer from the University of Trieste, (Italy).

Five relevant publications, patents or products:

- Five recommendations to kick-start bioeconomy innovation in the UK, Burns, C., Higson, A. and Hodgson, E. (2016), Biofuels, Bioprod. Bioref., 10: 12–16. doi:10.1002/bbb.1633
- How will the changing biofuels market affect the development of bio-based chemicals? Aylott, M. and Higson, A. (2013), Biofuels, Bioprod. Bioref., 7: 9–11. doi:10.1002/bbb.1384
- Product developments in the bio-based chemicals arena. de Jong, E., Higson, A., Walsh, P. and Wellisch, M. (2012), Biofuels, Bioprod. Bioref., 6: 606–624. doi:10.1002/bbb.1360
- What is the most appropriate LCA method for measuring greenhouse gas emissions from bioenergy? Aylott, M., Higson, A., Evans, G., Hamer, A. and Mortimer, N. (2011), Biofuels, Bioprod. Bioref., 5: 122–124. doi:10.1002/bbb.282
- IEA Bioenergy - Task 42 Biorefinery: Biobased Chemicals - value added products from biorefineries (http://www.ieabioenergy.com/publications/bio-based-chemicals-value-added-products-frombiorefineries/)

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
BioBaseNWE: Providing advice and business support to SMEs in the bio-based products sector across North West Europe. The development of recommendation to guide regional policy development to stimulate the bio-based economy.	Interreg North West Europe	2015	www.biobasenwe.org
Energetic Algae (EnAlgae) was a four-year Strategic Initiative with the aim of developing sustainable technologies for algal biomass production.	Interreg North West Europe	2015	www.enalgae.eu

UK Industrial Biotechnology Catalyst: NNFCC are programme
co-ordinators for UK's leading Industrial Biotechnology UK Active www.bbsrc.ac.uk funding programme.

UK Integrated Biorefining Research and Technology Club
(Biotechnology and Biological Sciences Research Council): UK Active www.bbsrc.ac.uk

http://www.nnfcc.co.
Capabilities and Opportunities for Joint Working on Biorefining uk/tools/capabilitiesand Industrial Biotechnology. NNFCC were commissioned to and-opportunities-foridentify high level opportunities and synergies for Norway and joint-working-onthe UK in the area of industrial biotechnology and biorefining. biorefining-andNNFCC undertook an assessment of the market pull for UK-

industrialdifferent
products, the availability of raw materials, expertise Norway
and infrastructure in the UK and Norway. The assessment bilateral 2011 biotechnology-mayresulted in the identification of a number of areas where the initiative 2011-dr.-adriancountries have mutual or synergistic interests and expertise. higson-and-dr.-claireThese areas were consolidated and refined into a series of key smith.-a-project-forshort term and longer term opportunities for collaboration innovation-norway-

between the countries. and-technologystrategy-board

Technical equipment:---

4.2 THIRD PARTIES INVOLVED IN THE PROJECT (INCLUDING USE OF THIRD PARTY RESOURCES) No third parties involved

5. ETHICS AND SECURITY

5.1 ETHICS

Pre-Grant Agreement requirement:

The ethical standards and guidelines of Horizon2020 will be rigorously applied, regardless of the country in which the research is carried out.

The materials to be imported to/exported from the EU are shown below:

SINTEF	Proteins →	POLITO (Italy)
	+ Nucleic Acid →	DLABS (Israel) (500g)
BIOPOLIS	Proteins →	DLABS (Israel) (500g)
IRIS	Gelatine →	AIMPLAS (Spain) (200g-300g)
DLABS	Flame Retardant →	MINE (Turkey) (2kg)
		AIMPLAS (Spain) (2-3kg)
		ARKEMA (France) (2-3 kg)

Following REACH regulations.

The specific authorisations for the import/export of materials will be provided in due time after being provided by the relevant authorities.

Companies have the responsibility of collecting information on the properties and the uses of substances that they manufacture or import at or above one tonne per year. They also have to make an assessment of the hazards and potential risks presented by the substance. However, this tonnage is far from being achieved in the project.

The substances involved in the project can be considered products oriented to research and development, whose definition in REACH regulation is any scientific development related to product development or the further development of a substance, on its own, in mixtures or in articles in the course of which pilot plant or production trials are used to develop the production process and/or to test the fields of application of the substance; scientific research and development meaning any scientific experimentation, analysis or chemical research carried out under controlled conditions in a volume less than one tonne per year.

Therefore, this project will apply the exemption from the general obligation to register for products and process oriented to research and development (PPORD):

- Articles 5, 6, 7, 17, 18 and 21 of REACH Regulation shall not apply for a period of five years to a substance manufactured in the Community or imported for the purposes of product and process orientated research and development by a manufacturer or importer or producer of articles, by himself or in cooperation with listed customers and in a quantity which is limited to the purpose of product and process orientated research and development.

Regarding the protection of human health and the environment from the risks that can be posed by chemicals, all the substances used in the project will be evaluated to ensure that no risks are involved in their use. And in case anyone could have any hazard property, the specific substance will be sent to the destinatary accompanied by a document describing the risk management measures to be used to make safe use of it. For partners established outside the EU this requirement will apply as for the rest of partners, in this particular project.

Detailed information will be provided, when applicable, to confirm that fair benefit-sharing arrangements with stakeholders from low and/or lower-middle income countries are ensured during the project.

Environmental protection and safety EPQ - Requirement No. 2

Further information will be provided about the possible harm to the environment caused by the research and state the measures will be taken to mitigate the risks. As relevant, copies of facilities authorisations will be provided (e.g. security classification of laboratory, GMO authorisation). It will be ensured that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

Protection of personal data POPD - Requirement No. 3

Detailed information will be provided on the procedures that will be implemented for data collection, storage, protection, retention and destruction and confirmation that they comply with national and EU legislation. It will be explicitly confirmed that the data used is publicly available. In case data is not publicly available, relevant authorisations will be provided.

There are no other ethical issues associated with this project. The work conforms to all relevant national or international regulations and EC legislation. There are no foreseen negative ethical issues associated with the implementation of the project results. **There will be no experimentation on animals, with human cloning, genetic modification or human embryos or stems cells as a part of the project**.

5.2 SECURITY.

Please indicate if your project will involve:

- activities or results raising security issues:
- 'EU-classified information' as background or results:

There are no security issues associated with this project.

ANNEX I. Standardization committees related to biobased products.

ISO/TC 276 Biotechnology Standardization in the field of biotechnology processes that includes the following topics: Terms and definitions; biobanks and bioresources; analytical methods; bioprocessing; data processing including annotation, analysis, validation, comparability and integration and metrology.

CEN/TC 411 – Biobased products. The Technical Committee CEN/TC 411's main task is to develop standards for bio-based products covering horizontal aspects, including consistent terminology, sampling, certification tools, biobased content, application of and correlation towards life cycle analysis, among others.

CEN/TC 183 - Waste management .Standardization in the field waste management including public cleaning, taking into particular account technical and logistical aspects. Drafting of Standards for products and procedures as well as safety requirements for the collection, transport, storage and transfer of solid and liquid waste. The characterization of waste and sludge as well as functional aspects for snow clearing equipment and road service area maintenance equipment is not covered by the scope of CEN/TC 183.

CEN/TC 249/WG 17 for Biopolymers. The CEN Technical Committee 249 is in charge of all standards for plastics. One of its working groups, the Working Group 17, is responsible for the development of standards for biopolymers. WG17 was established in October 2008 and has been active since January 2009. As of October 2012, CEN/TC 249 has published one technical report and three technical specifications relevant to the biobased industry.